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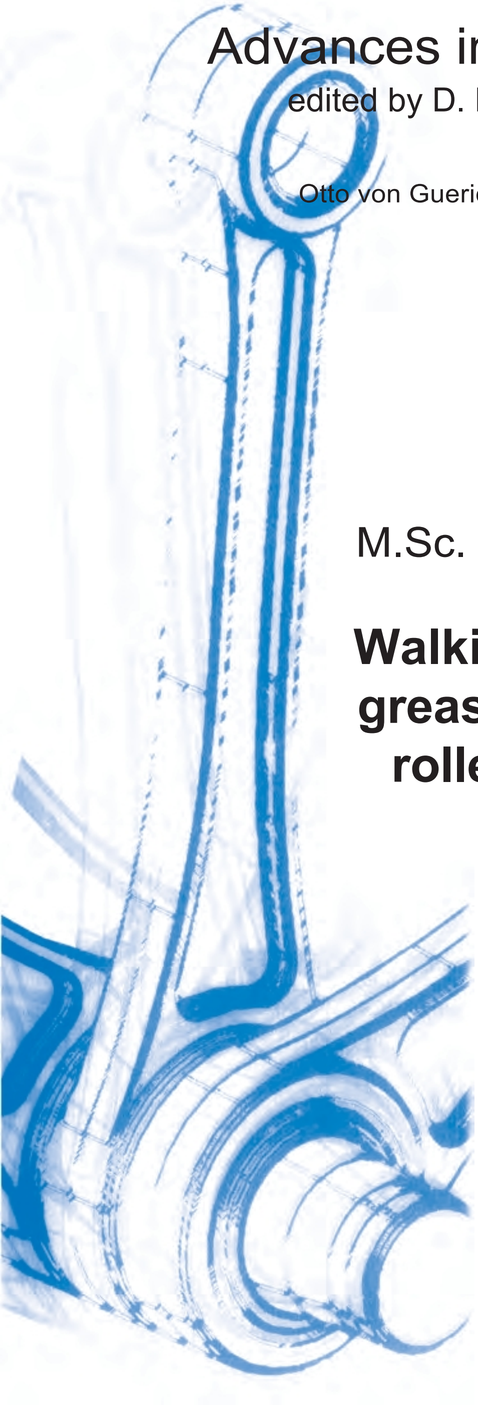

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Walking losses in grease-lubricated roller bearings

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Walking losses in grease-lubricated rolling bearings

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Foreword

This work is the result of my activities at the Chair of Machine Elements and Tribology at the Institute of Machine Design at Otto von Guericke University Magdeburg.

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Ricardo Lühe

Magdeburg, May 2024

Abstract

One of the key challenges facing modern drive systems, such as those used in electric mobility, is to reduce bearing losses [ZSCH21]. This includes reducing friction losses in rolling bearings. Friction occurs at rolling contacts, i.e., between rolling elements and bearing rings. There are also sliding contacts, such as the contact between the rolling elements and the cage. However, friction also occurs independently of contacts, such as during displacement and shearing of the lubricant. In grease lubrication, this results in kneading work. These friction losses can be quantified, for example, by the friction torque. In rotating roller bearings, the friction torque is calculated with the speed to determine the friction power. If the speed increases, the friction power also increases, which leads to a rise in temperature. High temperatures, in turn, have an impact on the service life of grease lubrication. In addition, heat from bearing losses cannot be dissipated as effectively in grease-lubricated systems as in oil-lubricated systems. In high-speed applications, it is particularly important to understand the system in order to optimize it.

Standard methods for calculating friction torque are based on catalog procedures provided by roller bearing manufacturers. However, in certain configurations, these procedures can lead to an underestimation of the friction torque. This is also known to occur with grease-lubricated roller bearings. Kneading losses play a particularly important role here. In oil lubrication, the so-called splash losses are well known. For grease-lubricated rolling bearings, there is currently no way to calculate rolling losses.

This paper focuses on grease distribution in rolling bearings and kneading losses. To this end, a test methodology for rolling bearing test benches is presented, which allows kneading losses to be quantified by evaluating metrological variables. Variations in speed, grease rheology, and grease quantity, as well as rolling bearing kinematics and shaft position, are used to investigate the dependencies of kneading losses on three different rolling bearing test benches, with speeds of up to 18,000 rpm being achieved. On the other hand, a lubricating grease model is presented, which is used in a 3D CFD model and is based on the rheology of the lithium complex greases considered here. The fluid mechanics simulation of the rolling bearing provides the most detailed insight possible into the grease-lubricated rolling bearing and is intended to help understand the distribution mechanism of different bearings. Rolling bearing tests with transparent bearing components are used for validation. This allows imaging technology, such as high-speed cameras, to provide interesting insights into a rotating rolling bearing.

Table of contents

Nomenclature

1 Introduction 1
1.1 Problem 1
1.2 State of research. 3
1.3 Aim and content of the thesis 11

2 Fundamentals 13
2.1 Investigation of rolling work with roller bearings of different designs 13
2.2 Basic properties of test greases 15
2.3 Grease quantity in roller bearings 16
2.4 Minimum load of rolling bearings 19
2.5 Rheological properties 20
2.6 Models for describing flow behavior 21

3 Testing and measurement technology 25
3.1 Rolling bearing test benches 25
3.1.1 Rotational tribometers 25
3.1.2 High-speed test bench 27
3.1.3 Special tribometer 30
3.2 High-speed recordings 32
3.2.1 High-speed camera 32
3.2.2 Thermal imaging camera 32
3.2.3 Measurement setup with high-speed camera 32
3.2.4 Measurement setup with thermal imaging camera 33
3.3 Rheometer 35
3.3.1 MCR 301 35
3.3.2 MCR 702e 36
3.3.3 Comparison of rheometers 37

4 Experimental investigations 38
4.1 Rheological investigations 38
4.1.1 Apparent viscosity according to DIN 51810-1 38
4.1.2 Storage and loss modulus according to DIN 51810-2 42
4.1.3 Oil separation according to DIN 51817 43
4.1.4 Further rheological investigations 45
4.2 Basic test methodology for roller bearing tests 46
4.3 Rotational tribometer 50
4.3.1 Tests without temperature control 50
4.3.1.1 Repeat tests 52
4.3.1.2 Determination of the fulling work 54

4.3.1.3 Lubricant quantity variation 56
4.3.1.4 Increase in friction torque with the RRKL 6007 61
4.3.2 Tests with temperature control (at 80 °C) 62
4.3.2.1 Repeat tests 63
4.3.2.2 Lubricant quantity variation 64
4.3.3 Walking work from all tests 66
4.4 High-speed rolling bearing test bench 72
4.4.1 Test methodology for high speeds 72
4.4.2 Lubricant quantity variation at 80 °C 76
4.5 Special tribometer 80
4.5.1 Tests with the AZRL 81212 80
4.5.2 Tests with the SKL 7007 84
4.5.3 Measurements using a thermal imaging camera 87
4.6 Comparison of roller bearing test benches 90
4.7 Summary of the roller bearing tests 94

5 Numerical simulation 97
5.1 Analytical grease model 97
5.1.1 Determination of the Herschel-Bulkley coefficients 98
5.1.2 Special features of regression and model modification 100
5.1.3 Extension of the model to include temperature dependence 104
5.1.4 3D representation of the developed lubricating grease model 107
5.1.5 Extension of the model to include base oil viscosity 109
5.1.6 Extrapolation with regard to the shear rate 109
5.2 CFD model for rheometer 111
5.2.1 Boundary conditions 111
5.2.2 Calculation of the friction torque in the CFD 112
5.2.3 Network study 115
5.3 CFD model for roller bearings 117
5.3.1 Boundary conditions 117
5.3.2 Networking 119
5.3.3 Multiphase flow 121
5.3.4 Initial greasing 122
5.3.5 Convergence 123
5.3.6 Stationary simulation with RRKL and SKL 124
5.3.7 Transient simulation of AZRL 128

6 Discussion of the results 130
6.1 Comparison between experiment and simulation 130
6.1.1 Model validation with rheometer experiments 130
6.1.2 Grease distribution and friction torque in rolling bearings 132
6.2 Final considerations on flow behavior 136

6.2.1 Change in flow behavior as a result of stress 136
6.2.2 Shear tests at maximum shear rates 140
6.3 Classification of results 144

7 Summary 150

8 Outlook 152

9 References 156

Nomenclature

Latin capital letters

Symbols	**Unit**	**Designation**
A_E	[m²]	Area of an element in considered calculation area
B	[mm]	Width of the roller bearing
C_0	[kN]	Static load rating
C_1	[-]	Vogel equation coefficient
C_2	[-]	Coefficient of the Vogel equation
C_r	[kN]	Dynamic load rating of a rolling bearing
C_0	[kN]	Static load rating of a roller bearing
D	[mm]	Outer diameter of the rolling bearing
$E_{Borosilikatglas}$	[Pa]	Modulus of elasticity for borosilicate glass
E_A	[kJ/mol]	Activation energy
$E_{Saphirglas}$	[Pa]	Modulus of elasticity for sapphire glass
E_{Stahl}	[Pa]	Modulus of elasticity for steel
F_a	[N]	Axial bearing load
$F_{a,min}$	[N]	Minimum axial load for roller bearings
F_r	[N]	Radial bearing load
M	[Nmm]	Torque measured on cone in shear test
M_{Y-Int}	[Nm]	Integral moment around the y-axis
G	[kg]	Weight of the roller bearing according to GfT worksheet 3
G'	[Pa]	Storage module DIN 51810-2 (elastic portion of viscoelasticity)
G''	[Pa]	Loss modulus DIN 51810-2 (viscous component of viscoelasticity)
M	[Nm]	Moment
P_{Lager}	[W]	Performance in storage
P	[kN]	Dynamic equivalent bearing load
R	[J/mol·K]	Universal gas constant
T_1	[°C]	Temperature 1 in Vogel equation

T_2	[°C]	Temperature 2 in Vogel equation
V_E	[m^3]	Volume of an element in the calculation area under consideration
V	[m^3]	Free storage volume according to GfT worksheet 3
X	-	Bearing coefficient for the applied radial load in the calculation of the dynamically equivalent bearing load
Y	-	Bearing factor for the applied axial load when calculating the dynamically equivalent bearing load

Lowercase Latin letters

Symbol	**Unit**	**Designation**
d	[mm]	Bore diameter of the roller bearing
d_k	[µm]	Cone tip reduction in cone-plate measuring system
h_K	[m]	Height of the cone at a specific cone radius
n	[-]	Flow index
n_E	[No.]	Number of elements in the calculation area under consideration
r_k	[mm]	Cone radius in the cone-plate measuring system
u_k	[m/s]	Circular path speed
ƌ`	[m/s]	Resulting velocity
u	[m/s]	Speed in x-direction
v	[m/s]	Velocity in the y-direction
w	[m/s]	Speed in the z direction

Lowercase Greek letters

Symbol	**Unit**	**Name**
α_{100Cr6}	[1/K]	Coefficient of linear expansion for roller bearing steel 100Cr6
α_{Saphir}	[1/K]	Coefficient of linear expansion for sapphire glass
α_k	[deg]	Cone angle in cone-plate measuring system
$\dot{\gamma}$	[s^{-1}]	Shear velocity or shear rate
$\dot{\gamma}_c$	[s^{-1}]	Critical yield point for critical flow in numerical fluid mechanics (c for "critical")
κ	[Pa·s]	consistency coefficient

η	[Pa·s]	Dynamic viscosity
η_0	[Pa·s]	Dynamic viscosity at a shear rate of zero
η_A	[Pa·s]	Initial viscosity (apparent dynamic initial viscosity) of the measurement phase according to DIN 51810-1
η_b	[Pa·s]	Dynamic viscosity at a shear rate of zero
η_c	[Pa·s]	Limit value of dynamic viscosity at very high shear rate
η_E	[Pa·s]	Final viscosity (apparent dynamic final viscosity) of the measurement phase according to DIN 51810-1
η_{eff}	[Pa·s]	Effective viscosity
η_{Rel}	[Pa·s]	Relative change in dynamic viscosity according to DIN 51810-1
η_∞	[Pa·s]	Limit value of dynamic viscosity at maximum shear rate
ϕ	[-]	dissipation function
τ	[Pa]	Shear stress
τ''	[Pa]	Total shear stress tensor
τ_0	[Pa]	Yield strength determined by regression of the shear test according to DIN 51810-1
τ_f	[Pa]	Yield strength determined via the intersection of Speier and loss modulus according to DIN 51810-2
τ_x	[Pa]	Shear stress in the x-direction
τ_z	[Pa]	Shear stress in z-direction
ϑ	[°C]	Temperature
ν	[cSt]	Kinematic viscosity
ω	[rad/s]	Angular velocity
ω_{AR}	[rad/s]	Angular velocity of the outer ring
ω_{IR}	[rad/s]	Angular velocity of the inner ring
ω_K	[rad/s]	Angular velocity of the cage
ω_{WK}	[rad/s]	Angular velocity of the rolling element

Abbreviations

AFM	Atomic force microscope
AZRL	Axial cylindrical roller bearing, in this work the AZRL 81212
Example	Example
BT	Persistence temperature, corresponding to the friction and rotational speed at the bearing back
CFD	Computational Fluid Dynamics
EHD	Elasto-hydrodynamics
Grease 1	Test grease 1
Grease 2	Test grease 2
HB	Herschel-Bulkley
IMK	Institute for Machine Design at the University of Magdeburg Oil
1	Base oil from test grease 1
Oil 2	Base oil from test grease 2
PAO	Polyalphaolefin, a synthetic base oil SEM
	Scanning electron microscope
RRKL	Radial groove ball bearing, in this work the RRKL 6007-TVP
SKL	Angular contact ball bearing, in this work the SKL 7007 B-XL-2RS-TVP
UDF	User Defined Functions in the commercial software ANSYS Fluent
VF	Volume fraction in two-phase flow in computational fluid dynamics
VOF	Volume of Fluid, calculation approach in CFD for mapping multiple phases

1 Introduction

1.1 Problem

Grease lubrication of rolling bearings offers many advantages over oil lubrication, such as e.g. low construction and maintenance costs. For this reason, grease-lubricated roller bearings are very commonly used in all areas of machine and vehicle construction. Predicting the grease service life, which is largely determined by the bearing temperature, is an important parameter for the service life of the entire bearing. Roughly speaking, a temperature increase of 15 K halves the grease service life [SCHAE13]. The temperature development in the bearing is primarily influenced by the friction in the bearing itself. Bearing friction consists of various components: friction at the contacts between the rolling elements and either the bearing rings or the cage and, if applicable, the flange. Furthermore, contact seals can be used, which usually seal on the inner ring. The lubricant is involved in all these contacts. In addition to these friction contacts, there is friction caused purely by the lubricant due to displacement and shearing. In oil lubrication, this can lead to splashing. In grease lubrication, the process is called "walking." The friction in the bearing leads to a friction torque in the drive train. The friction power of the bearing is derived from the friction torque and the existing speed. This friction power leads to heating in the rolling bearing. Finally, the temperature development in the bearing also depends on the ambient conditions that influence heat dissipation. Thus, a reduction in the friction power in the bearing means a reduction in bearing losses. This in turn leads directly to a reduction in the temperature in the entire bearing. This reduces the thermal stress on the lubricating grease, which increases the service life of the grease. At the same time, the efficiency of the bearing is increased, reducing energy consumption and CO2 emissions [WOYD21].

This work focuses on the analysis of lubricant work, which has not been sufficiently researched to date. The ability to calculate the required amount of grease, ensuring that all contacts are sufficiently supplied with grease and that the work and bearing temperature are minimized, would eliminate the need for a series of tests. A more in-depth understanding of grease lubrication as an interplay between grease properties, grease quantity, bearing design, grease distribution, and operating point-dependent temperature development would also make it possible to adjust the rheological properties of the greases more specifically to the respective application in order to shorten the development process for the greases. Predicting and reducing work of rolling in high-speed applications is becoming particularly important. In the field of electromobility, electric motors with speeds of up to 20,000 rpm are increasingly being used. In the field of machine tools, (n·dm) values of 1 million are currently being achieved, with values of 1.2 million being targeted in the future using angular contact ball bearings or spindle bearings. Due to the increased work of wear in high-speed applications, the speed limits of the rolling bearings used are determined by the maximum permissible temperature. High temperatures often arise here, which can lead to serious damage. The viscosity in the lubrication gap and its effect on the contact temperatures in the rolling bearing can already be taken into account in parts by calculation. However, this does not make it possible to predict the heating of the lubricant in all areas of the rolling bearing and, from this, the temperature development in the entire bearing.

There are many known cases in which certain rolling bearing greases cause permanently high kneading work during operation. It is often unclear which test or system parameters lead to increased heating. Lubricating greases always fulfill several tasks in a rolling bearing, for which they are specifically equipped with rheological properties. It is not known which rheological properties are directly related to bearing losses due to kneading work. In addition, very specific lubricant distribution processes take place in roller bearing types of different designs. It is therefore crucial to know and understand the distribution processes in the roller bearing. Furthermore, when planned relubrication is carried out, the problem arises that the roller bearings of the drive train are then supplied with very different quantities of lubricant. The fill level can also influence the work of the rolling elements. Currently, the amount of lubricant required can only be estimated using calculation methods. It is not currently possible to calculate how sensitively a rolling bearing reacts to a variation in quantity. All in all, there are many questions for users that can often only be answered through costly experiments and empirical values.

1.2 Research Status

Losses in lubricated rolling bearings are composed of various components. For many users, the measured variables "friction torque" and
"Temperature" suitable reference values. The various contacts in the rolling bearing can be dominant for the friction torque. These include rolling element-raceway, rolling element-cage, and, if applicable, rolling element-flange and inner ring seal contacts. Normally, the lubricant is involved in all contacts. In addition to the friction from the lubricated contacts, there are also losses based purely on the lubricant. In the case of oil, the amount of oil present in the rolling bearing, together with the operating viscosity, plays a central role. This can result, for example, in slick losses, which are particularly common in gear applications. The lubricant is displaced and sheared. Similarly, kneading can occur in roller bearings with grease lubrication. In this context, well-known methods for calculating the friction torque and publications from research are presented below.

Fundamental investigations to describe the individual components of friction were carried out empirically by PALMGREN [PALM57]. Based on his rolling bearing experiments, he developed one of the best-known equations for calculating the friction torque of rolling bearings [PALM64]. PALMGREN uses the term
"total friction torque," although in this paper the term "friction torque" is used synonymously.

$$M_R = M(\ 0_) + M_1 \qquad (1)$$

The friction torque$M_R$is composed of the speed-dependent torqueM_0 and the load-dependent torqueM_1. This calculation method also allows the type of lubrication to be taken into account. A number of factorsare included in the two main componentsM_0 and$M_{(1)}$.
are based on extensive empirical testing. On this basis, the following influences can be taken into account using Eq. *(1)*:

- Rolling bearing type and size,
- type and condition of lubrication,
- Operating conditions such as load, temperature, and speed.

In further work, the determination of the two components from Eq. *(1)* was modified. Among the best-known publications are the modifications by HOLLATZ, JEDRZEYEWSKI, and STEINERT [HOLL84, JEDR89, STEIN95]. In the investigations
JEDRZEYEWSKI demonstrated that equation *(1)* according to PALMGREN

calculates significantly higher friction torques than those measured in experiments, especially with grease lubrication. However, due to its ease of application, PALMGREN's preliminary work still forms the basis for the calculation of friction moments by various bearing manufacturers. Since the internal geometry of rolling bearings has changed over the years, individual factors are still determined empirically through rolling bearing tests and thus reflect the current state of rolling bearing design.

Until the 2000s, roller bearing manufacturer SKF offered a catalog calculation based on the same principle. However, this concept was replaced by a new method [SKF14]. SKF's current catalog method takes into account the lubrication condition, which in turn is influenced by the lubricant viscosity and the speed.

$$M = M_{rr} + M_{sl} + M_{seal} + M_{drag} \quad (2)$$

The "total friction torque" M is composed of the rolling friction torque M_{rr} , the sliding friction torque M_{sl} , the friction torque from contacting seals M_{seal} , and the friction torque from flow, splashing, and spray losses M_{drag} . Here, M_{rr} has a

Significant dependence on the formation of an EHD contact, as this includes the lubricating film thickness and lubricant displacement. M_{sl} takes into account the lubrication condition, which corresponds to the principles of the Stribeck curve

[STRI02]. This results in a lower friction torque in many areas than that calculated using similar methods according to PALMGREN.

Fig. 1 compares the catalog methods described by Schaeffler and SKF. Both methods take temperature and flow behavior into account via the base oil viscosity. Grease lubrication is equated with oil mist lubrication.

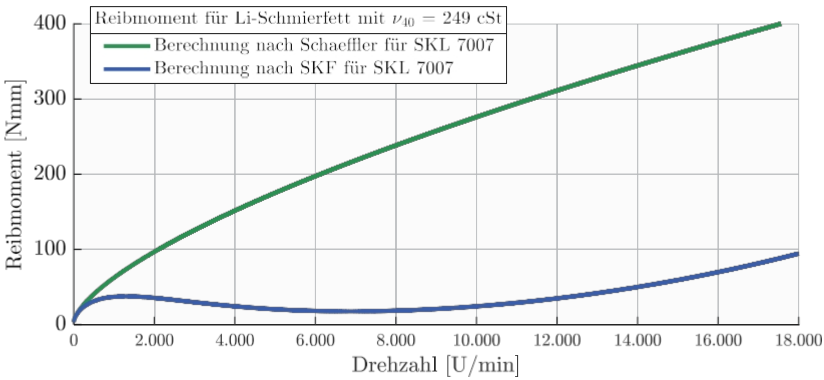


Fig. 1: Friction torque calculated using the catalog method according to Schaeffler Eq. *(1)* and SKF Eq. *(2)* for the angular contact ball bearing 7007 and a Li grease with a base oil viscosity of 249 mm²/s at 40 °C

Despite many similarities, the sometimes striking differences between the two catalog methods are clearly illustrated in Fig. 1. Neither the specific amount of grease in the rolling bearing nor the grease viscosity are taken into account.

This can also lead to so-called walking in grease-lubricated roller bearings. Similar to splashing losses, losses also occur here as a result of the displacement of the lubricant. Due to the high consistency of greases compared to oils, this is referred to as walking or kneading rather than splashing. In English-speaking countries, the term "churning loss" is used for both churning losses and splashing losses [LUGT13]. It is well known from gear applications that splashing losses can be a persistent effect during operation with oil lubrication. Whether this is also the case with kneading and to what extent it changes the rheology of the lubricating grease has been the subject of frequent research in recent years.

LUGT describes that the life cycle of a rolling bearing grease begins with the "churning phase" [LUGT13, 2.4]. This phase, in turn, begins with what is known as "channeling" – the formation of a reservoir. This reservoir formation is associated with a significant increase in friction torque and thus also in temperature, which in radial bearings is usually measured at the outer ring. In studies by GERSTENBERGER, the position of the grease collar next to the raceway was recorded optically. These results suggest that, macroscopically, no significant changes occur after the initial grease distribution [GERST00]. Following grease distribution, the temperature usually drops as the churning decreases. However, there are greases whose flow behavior prevents them from leaving the churning phase. HORTH already showed, using two fats with very different rheological properties, that the measured bearing back temperature remains permanently high in one case and drops in the other [HORTH71]. The grease with the permanently high temperature is described as "poor channeling." The grease that leaves the high temperature level again after a short time is described as "channeling."

Based on this categorization, CHATRA and LUGT published several findings between 2020 and 2023. Initially, tests were conducted using a roll stability tester, whereby various fats were examined in terms of their walking behavior and divided into the two groups mentioned above. Fats with a long fat distribution phase were able to maintain a high temperature for three to five hours or longer until a drop of more than 50 % was finally observed. These were classified as

"poor" in terms of the creeping behavior . Fats with a short

Grease distribution phases that are considered "good" exhibit a comparatively high friction torque for less than one hour. The special behavior of the greases was also confirmed in the R0F roller bearing test bench [CHATR20]. AFM images of tested greases provided qualitative evidence that "poor" greases exhibit greater thickener change (degradation) [CHATR21]. Similarly, following tests with "poor" greases, a more pronounced change in oil release behavior (increase) and yield point (reduction) was observed [CHATR22]. The yield point of greases is usually derived from the intersection of the storage and loss modulus [DIN 51810-2]. This represents the resistance to flow. The authors postulate an energy density law that is intended to predict the change in the yield point due to the energy introduced [CHATR23-1]. Finally, the aspect of uniform grease distribution on the bearing cover was investigated, following the hypothesis that a successfully completed kneading phase should be characterized by uniform grease distribution on the bearing cover [CHATR23-2].

In addition to aspects of mechanical aging, the aspect of thermally induced aging of rolling bearing greases was also investigated. After all, heat input into the grease always leads to a reduction in service life. A well-known rule of thumb postulates that a temperature increase of 15 Kelvin halves the service life of the grease [SCHAE13]. Based on the first-order reaction (kinetics), the Arrhenius equation (*3)* describes the thermal aging of materials as a function of time. The

equation includes the reaction rate constant k , the so-called frequency factor A, the activation energy E_A , the universal gas constant R and the absolute thermodynamic temperature T .

$$k = A \cdot e^{\frac{E_A}{R \cdot T}} \qquad (3)$$

The equation has been applied several times for roller bearing greases, whereby a thermogravimeter [RHEE09, DGMK 788, SMOOK22], a differential calorimeter [REZ16-2, REICH22], a rheometer [DORN16, FRANK23], or an autoclave [MATZ19, FRANK23] can be used to determine the coefficients. The latter method is currently being qualified to supplement DIN 51808 [DIN 51808]. When determining the coefficients, the experiment is terminated at the same temperature due to thermal conditions. By taking the logarithm, the equation can be represented as a straight line. To determine the coefficients, the downtimes of at least two different test temperatures are required, with the above-mentioned standard recommending three different temperatures. To apply this theory to roller bearings, In the lubricant and rolling bearing industry, rolling bearing tests are often performed on an FE9. The procedure for determining the coefficients according to Eq. *(3)* is identical [REICH22, FRANK23].

ZABEL describes how oxidation products increasingly form during the thermal stressing of greases. The most common oxidation product that is formed is carboxylic acid. This acid leads to decomposition in most types of thickeners. As a result, the thickener is increasingly less able to bind the base oil. Oil depletion is therefore often a cause of grease lubricant failure. Compared to oils, heat dissipation from the bearing is poorer with greases. Thermal aging is therefore an important aspect of greases. To prevent this, greases contain various oxidation inhibitors that slow down this process [ZABE06].

Starvation effects can also occur with grease lubrication. In addition to the frequently used translation "Mangelschmierung" (insufficient lubrication), the terms "Mindermengen- oder Minimalmengenschmierung" (reduced quantity or minimum quantity lubrication) are also used in German-speaking countries. This is because starvation does not always lead to undesirable rolling bearing damage, but can even have a positive effect on the friction torque behavior and thus contribute to reducing rolling bearing losses. This topic was comprehensively investigated in the FVA 388 I and II research projects. An important component here is the capacitive measurement of lubricant film thicknesses in the rolling bearing. This made it possible to determine that the actual lubricant films in grease-lubricated rolling bearings can in some cases be significantly below the lubricant film height predicted by elasto-hydrodynamics under full lubrication [FVA 388]. Some of the observations on bearing friction were incorporated into the FVA Workbench software tool within the Low Friction Powertrain research cluster [FVA 1112]. Building on numerous findings from these projects, grease-lubricated rolling bearings up to 30,000 rpm were investigated in project 914 I. Analytical considerations were made for the angular contact ball bearing used. The considerations include the speed-dependent contact angle, the bore slip, and the gyroscopic moment from the rotation of the rolling element. Together with the measured lubricating film height, an extension of the SKF catalog calculation according to Eq. (*2)* could thus be proposed.

In order to clarify the conditions prevailing in a grease-lubricated rolling contact, numerous investigations of ball-disc contact were carried out at Imperial College. LAURENTIS investigated the dependencies of lubricant film formation in grease-lubricated point contact and varied the speeds for this purpose. At some operating points, it was observed that thickener particles also migrate through the concentrated contact, which means that the fluid properties can vary greatly [LAUR16].

In addition, the FVA Project 580 II "Thin Lubricating Films" investigated starvation effects in grease-lubricated rolling element-raceway contact [FVA 580]. Starvation can cause the lubricating film thickness to fall below a critical value of 0.07 µm, which can lead to undesirable boundary layer effects. Extensive investigations were carried out on the EHD tribometer to examine the limits of starvation as a function of different test greases. Lubricating film thickness measurements were also taken during the tests. The evaluation of the tests showed that the thickener adheres to the contact surfaces and changes their properties. In certain cases, this even leads to so-called "expansion" in the lubrication gap. To classify the effect of expansion, it should be noted that the lubrication gap height under full lubrication with the base oil initially serves as the reference height. Two different cases can occur with grease lubrication. If the lubrication gap height is greater than in the reference case, this is referred to as expansion [COUSS12]. If the lubrication gap height is less than in the reference case, this is referred to as starvation [WILS79]. Both cases can occur with grease lubrication. In the case of expansion, it is assumed that thickener particles migrate through the contact. LAURENTIS observed that this is possible [LAUR16]. Thus, empirical values from the calculation of the EHD contact of oil-lubricated systems may lose their validity when using grease lubrication. Furthermore, the experiments from FVA 580 show that characteristic grease displacement occurs at the start of the experiment. After that, the grease reservoir is present in the form of grease lanes, which confine the oil located in the concentrated contact area. Based on this assumption, a CFD model was developed that takes two phases into account: oil and air. The oil has the properties of a Newtonian fluid. The simulation reveals how the so-called replenishment (in German:
"Replenishment" or "reflow") occurs between two contacts. This effect describes the migration of the lubricant in the raceway area. To avoid harmful lubrication deficiencies, the oil that is displaced from the contact during rollover must flow back into the contact area before another rollover occurs.

LIEBRECHT investigated the splashing losses for a fully lubricated tapered roller bearing. For this purpose, the test chamber and the test bearing were examined in a CFD simulation. With the help of the CFD model, it was possible to determine that the flow losses observed were only a relatively small proportion of the splashing losses.

[LIEBR15]. FELDERMANN also used CFD simulation to investigate the sliding losses in a radial cylindrical roller bearing [FELD17]. Assuming closed gaps between a rolling element and its raceways, a fluid space was described that is based exclusively on the space between two rolling elements. This automatically displaces lubricant that is transported in the direction of the rolling element-raceway contact to the side. The model was used to simulate three speed stages up to 3800 rpm. A calculation based on the SKF model was also performed and compared with existing test results. The comparison showed that both the analytical equation and the CFD simulation exceed the losses in the test at increasing speeds.

CONCLI also explained a meshing concept for radial cylindrical roller bearings for roller bearing simulation under oil lubrication [CONCLI20]. In it, the author discusses in detail the so-called blocking – a method that enables a fully structured mesh. The rolling bearing is supplied radially with ISO VG 320 at 95 °C. The flow behavior of the lubricant is assumed to be Newtonian and incompressible. In the simulation, the authors achieve 4,500 rpm.

MACCIONI presents a test bench that includes an optically accessible tapered roller bearing with an outer ring made of sapphire glass [MACC22-1]. The aim of the test setup is to investigate the axial flow through the tapered roller bearing up to a speed of 600 rpm. Particle image velocimetry (PIV) is used for this purpose. This determines the particle velocity. Later, the influence of ventilation on the tapered roller bearing was investigated [MACC22-2]. A CFD model was also presented, which could be used to describe the ventilation in a first approximation.

GONDA presented a test methodology that can be used to investigate lubricant-based losses. This was used to demonstrate the slick losses in the tapered roller bearing as a function of the fill level [GONDA19]. Tests were carried out with either full fill, half fill, or minimum quantity lubrication. The sliding losses could be determined by subtracting the measured friction torques from full filling or half filling on the one hand and minimum quantity lubrication on the other. In addition, a CFD simulation of the test chamber under full lubrication was carried out, in which the axial position of the tapered roller bearing was also varied. The CFD simulation considered the oil as a Newtonian fluid.

A Newtonian fluid exhibits proportionality between shear stress and shear rate. This results in constant dynamic viscosity as a function of shear rate. Greases often behave non-Newtonian. Their flow behavior is generally described as plastic-structural viscous. There are several models for this in the literature, which are explained separately and distinguished from each other in this paper due to their scope (see section 2.6). To describe the plastic-structural viscous flow behavior, SARKAR used the Herschel-Bulkley model [HERS26] for rolling bearing greases of NLGI classes 00 to 2 [SARK17]. Based on this analytical model, CFD simulations were used to investigate flow through pipes of various cross-sections. This resulted in plug flow. The plastic-structural viscous flow behavior of greases means that shear only occurs in the area of the pipe wall, whereas a large area in the middle—a plug—moves through the pipe without being sheared. The Herschel-Bulkley model thus enabled an essential property of lubricating greases to be implemented in a CFD model.

The DGMK Project 810 investigated the use of roller bearing greases in robotics. The operating conditions in robotics require special properties from a lubricant. Unlike in permanently rotating systems, rapid changes of direction and high accelerations represent the normal operating conditions. This quickly leads to a problem with the presence of grease. This leads to undesirable wear, which should be avoided by choosing the right lubricant. In addition to extensive rolling bearing tests, shear tests were therefore also carried out to understand the flow properties of the lubricating greases. Shear rates of up to 17,500 s$^{(-1)}$were achieved, which in some cases led to gap emptying. Based on this data, the flow behavior was modeled using the CROSS model [CROSS65]. This describes a structurally viscous flow behavior. CFD simulations of shear tests were performed using this model approach [DGMK 810].

Other studies have also dealt with 3D CFD simulation of rolling bearings. NODA initially presented the grease distribution in real rolling bearings using the X-ray CT method [NODA16]. Later, in addition to the visualization method, a CFD model with a two-phase simulation was published [NODA20]. The BINGHAM flow model serves as the basis for describing the flow behavior. This model describes plastic flow behavior [BING16]. Together with the X-ray CT method, NODA's work focuses on grease distribution at the cage.

1.3 's goal and content of his work

An analysis of the literature shows that both measurement-based and simulation-based studies on roller bearing lubrication have already been conducted. These studies often focused on investigating the contact between the rolling elements and the raceway under oil lubrication. The amount of grease in the roller bearing is an important factor that can influence the work done by the rolling elements. The optimum amount of grease for reducing the heat input from the kneading work has not been specifically investigated. Most rolling bearing simulations were carried out under oil lubrication with roller bearings. In the simulation examples with greases, no plastic-structural-viscous flow behavior was taken into account, although this behavior is known to occur in greases.

Similarly, common catalog methods include the property of base oil viscosity
– but not the viscosity of the grease, the grease rheology, the fill level, or the installation situation. It has been shown for wide speed ranges that the analytical approaches sometimes calculate friction torques that differ from those measured in the corresponding rolling bearing test. These approaches neglect both work losses and starvation effects.

Despite the significant advantages of grease lubrication over oil lubrication, there is currently no technical method for quantifying work losses. In the context of high speeds, however, work losses mean both efficiency losses and a reduction in the service life of the grease. Since it is not clear which rheological properties determine the suitability of a grease for high-speed applications, the optimization of these lubricants is based on trial and error. In the future, it should be possible to better understand the conditions under which increased work is required in order to move from an empirically based approach to a physically based method. The objectives of this work are to investigate and vary the following parameters:

- Lubricating grease properties,
- amount of lubricating grease used,
- bearing design,
- grease distribution, and
- operating point, consisting of speed and temperature.

The aim is to determine the effects of the above parameters on bearing losses during grease lubrication and thus on the temperature development in the bearing. These objectives will be pursued through experimentation and simulation. The focus will be on experimental investigations. For this purpose, measurement setups and a method will be developed to determine the influencing factors.

A measurement method is to be developed using a rotational tribometer. This widely used test system allows rolling bearings to be tested under grease lubrication at speeds of up to 3,000 rpm using a standardized test head.

In order to examine grease-lubricated roller bearings even at high speeds, a test system is to be set up that allows speeds of 18,000 rpm. This high-speed test bench will be constructed according to the FE8 principle with a fixed test head, whereby the friction torque measurement is ensured via a torque measuring shaft.

Distribution aspects are of central importance for kneading work. For this purpose, glass rolling bearing components are used, which allow insight into a rotating rolling bearing. These investigations are to be carried out on a special tribometer at speeds of up to 2,000 rpm. A high-speed camera will record the distribution conditions.

The kneading behavior of greases is also determined by rheology. In order to provoke different kneading behaviors, test greases with different rheological properties are to be produced. This requires a thorough understanding of the flow behavior. Rheological investigations are used to record properties that are decisive for distribution characteristics. On this basis, a lubricating grease model is to be developed and integrated into commercial CFD software.

In the 3D CFD simulation, the standardized shear test for determining the apparent dynamic viscosity according to DIN 51810-1 is to be used for validation. The flow conditions can be well described here. Building on this, CFD models of roller bearings of different designs are planned, which are to be investigated using the test setups mentioned above. These will be used to perform distribution simulations. Finally, the distribution simulation will be compared with the optical distributions from the special tribometer.

2 Fundamentals

2.1 Investigation of rolling work with roller bearings of different designs

Various types of friction can occur in a roller bearing. The significance of these types of friction in the overall system depends on the respective operating parameters and the properties of the tribological subsystems. Many types of friction are caused by contact, while others are caused by the respective lubricant:

- Friction from the displacement and shearing of the lubricant.
- Friction from rolling element-raceway contact
- Friction from rolling element-cage contact
- Friction from rolling element-flange contact
- Friction from the inner ring-seal contact

Since only the friction torque resulting from all contacts can be measured in tests, other friction components must be completely eliminated, reduced, or differentiated by means of comparative measurements wherever possible.

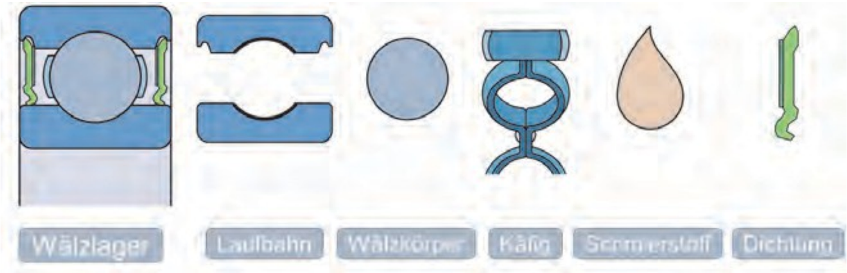


Fig. 2: Components of a rolling bearing and associated friction components [SKF14]

Various specifications were made to ensure this in a measurement setup. Contact seals, as shown in Fig. 2 on the right, generally dominate the overall friction in the rolling bearing [GERH95]. Non-contact seals are therefore used. In addition, the minimum load is required, which ensures a defined slip during operation and minimizes friction in the rolling element-raceway contact. Furthermore, support bearings are largely dispensed with. This is to ensure that only the torque from the test bearing is measured as far as possible. Furthermore, the choice of rolling bearings eliminates edge friction. The use of rolling bearings from the same production batch minimizes minor geometric deviations and the associated friction differences.

Based on these conditions, it is interesting to see how much friction a particular rolling bearing generates at minimum load. If this value can be determined experimentally, other aspects of grease-related bearing losses can be investigated. The differences include various lubricant rheologies and different fill levels. To determine the minimum friction, the smallest possible amount of oil or grease can be used. With grease lubrication, there is initially a strong redistribution. Subsequently, the lubricant is subject to continuous shearing in certain areas, whereas other areas are not subject to shearing. The resistance that the respective grease opposes to this quasi-stationary distribution state determines the extent of the work losses due to kneading.

The test bearings examined on all three test benches are the deep groove ball bearing 6007 (RRKL) and the angular contact ball bearing 7007 (SKL) (see Table 1). The axial cylindrical roller bearing 81212 (AZRL) is also being examined on one test bench. All bearings have a cage made of glass fiber-reinforced plastic (TVP, see [FAG00]).

Table 1: Test bearings and their specifications

Property	Detail	Unit	Test bearing		
			RRKL	SKL	AZRL
Designation	DIN 623-1	-	6007	7007	81212
Specification	Manufacturer	-	TVP	B-XL-2RS-TVP	TVP
Performance data	Dynamic load rating	kN	17	24.3	480
	Static load rating	kN	10.3	17.2	173
	Limit speed	rpm	16,100	14,700	3,700
Weight	gravimetric	kg	0.144	0.140	0.250
Rolling elements	Number	Pcs	10	15	15
Printing angle	static	°	9	40	-
Dimensions	D	mm	62		60
	d	mm	35		95
	W/D	mm	14		26
	DWK	mm	9,000	9.525	11,000
Cage	Construction	-	Snap cage	Window cage	
	Material	-	PA66 GF 25		
	Guide	-	Rolling element guided		

The angular contact ball bearing corresponds to the sealed version (2RS – abbreviation for "two radial sealings"). The elastomer seal and the initial grease lubrication have been properly removed. The internal geometry of the outer and inner rings of the angular contact ball bearing is completely identical in the versions with and without seals.

2.2 Basic properties e of the test grease

Rolling bearing greases consist of a base oil, a thickener, and various additives. The chemical composition and structure of the components sometimes differ greatly from one another. The test greases are not practical greases, which is why they can be classified as model greases. The test greases were defined so that they have the same chemical base and the same consistency class. However, they should differ in their rheology. This was achieved by using different base oil viscosities and different thickener proportions. Both test greases have the same thickener type and the same base oil type. For short, the two model greases are referred to as Grease 1 and Grease 2, and the respective base oils of the model greases are referred to as Oil 1 (for Grease 1) and Oil 2 (for Grease 1).

Table 2: Basic properties of test greases Grease 1 and Grease 2

Grease	Thickener	Base oil	Kin. Viscosity of base oil [mm²/s]		Grease density	Proportion of thickener	NLGI class	Walk-pene-tration
	Type	Type	ν_{40}	ν_{100}	[kg/m³]	[%]		[0.1 mm]
Grease 1	Li borate complex	PAO	46	8	874	30	I	310
Fat 2	Li borate complex	PAO	249	33	889	18	I	316

The density of the greases was determined by the lubricant manufacturer. The density of the respective base oil mixtures is estimated at 900 kg/m³ in each case. In order to rule out any influence from different additives, the same additive package was mixed into both model greases in equal quantities. The additives in their respective concentrations can be found in the following table.

Table 3: Additive package for both greases from Table 2

Additive type	Chemical basis	Designation	Concentration
Antioxidant	Aminic	Dilithium sebacate	~1
High pressure	Triphenyl	Triaryl thiophosphate	~1
Metal deactivator	Aromatic	Alkylated amine	~1
Antioxidant	phenolic	Phenol derivative	~1
Wear protection	Aminic	Phosphate ester derivative	~1

With the aim of entering high-speed applications, both greases were designed as NLGI Class I, giving them the consistency of a spindle bearing grease.

2.3 e grease quantity in roller bearings

There are several ways to estimate the correct amount of grease in a rolling bearing, some of which differ considerably in their results, which is why a few methods are explained below. These are explained using the ball bearings used (Table 1).

Rolling bearing manufacturers offer catalogs on the subject of lubrication [SCHAE13] and rolling bearings [SKF14, SCHAE18, NSK18]. However, the catalogs do not contain precise information on the amount of grease for every rolling bearing, every speed, and every cage design. Nor do they take into account the installation situation, such as the cover geometry or the shaft position. Schaeffler distinguishes between the free bearing volume and the undisturbed free bearing volume. The free bearing volume corresponds to the total free space in the rolling bearing. In contrast, the undisturbed free bearing volume corresponds to the free bearing volume without the volume between the rolling elements. Initial lubrication is generally between 30 and 100% of the free bearing volume [SCHAE18, p.77]. Normally, lubrication should be 90% of the undisturbed free bearing volume. For high-speed applications, especially spindle bearings, however, 60% of the undisturbed free bearing volume or 30% of the total free bearing volume is recommended [SCHAE13, p. 93]. Many users are familiar with this rule of thumb. It coincides with the basic statements of other manufacturers.

SKF does not differentiate between different types of clearance, generally referring to the clearance in the rolling bearing. At low speeds, 100% is normally used, and at medium to high speeds, 30-50% [SKF14,
p. 242]. KLÜBER also recommends a grease quantity of 25-35% of the bearing clearance for high-speed applications [KLÜB20, p. 13]. NSK recommends a fill level of 50-66% up to half the limit speed. Beyond that, a fill level of 30-50% is specified. However, the clearance in a specific rolling bearing cannot currently be researched using the rolling bearing catalogs [TIMK14, SKF14, SCHAE18, NSK18].

The Society for Tribology (GfT) has published GfT Worksheet No. 3 – Rolling Bearing Lubrication, which contains a calculation equation for estimating the free bearing space [GFT06].

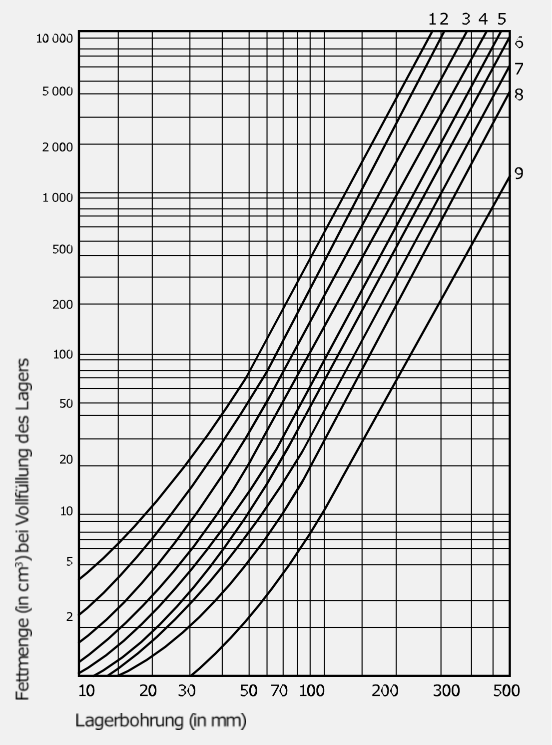
$$V=\pi / 4 \cdot B \cdot (D^{2}-d^{2}) \cdot 10^{-9} - G / 7800 \qquad (4)$$

The free bearing volume V is expressed in m³, whereby the dimensions width B, outer diameter D and bore diameter d are to be entered in mm. The weight G of the respective rolling bearing is to be specified in kg. The cage design is not taken into account. With a weight of 0.144 kg (RRKL), a free volume of

6.22 g/cm³ results. With an average density of the test grease of 882 kg/m³ (see Table 2) and a filling degree of 30%, this results in a grease quantity of 1.6 g. For the SKL, a grease quantity of 1.8 g would result in accordance with the lower bearing mass.

KLÜBER provides a chart for determining the grease quantity for Schaeffler rolling bearings when filled to capacity [KLÜB11, p. 11]. All roller bearing types and sizes are divided into nine lines. If the type is known, the corresponding free space in cm³ can be determined from the bore diameter. As a general rule, line 9 represents the smallest amount of grease and line 1 the largest. The RRKL 6007 and SKL 7007 are both assigned to line 6.

Fig. 3: Grease quantity for full lubrication of Schaeffler rolling bearings [KLÜB11]



With a bore diameter of 35 mm, a grease quantity of approximately 4.8 cm³ can be read for full lubrication. For an exemplary density of 882 kg/m³ and a fill level of 30%, this results in a grease quantity of 1.3 g.

If a clearance is known, a speed-dependent fill level in percent can be determined using the graph in Fig. 4 [KLÜB20].

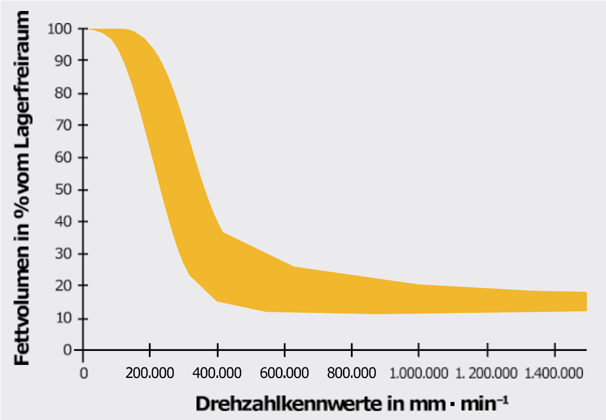


Fig. 4: Derivation of the bearing clearance for Schaeffler rolling bearings according to [KLÜB20]

Based on a specific inquiry to Schaeffler, ranges were specified for each of the sealed ball bearings that can be used as standard. This results in a grease quantity of 2.10–2.45 g for the SKL and a grease quantity of 1.87–2.20 g for the RRKL.

The Grease App from Schaeffler also provides an estimate of the amount of grease required in rolling bearings. The resulting amount depends in particular on the maximum speed, but also on the density of the grease. There are three limitations to using the app in this case. Firstly, a limit speed for greases is stored that is lower than the limit speed of the rolling bearings (see Table 1). No result can be derived above this speed. Specifically, this means a speed limit of 13,700 rpm for the RRKL and 11,000 rpm for the SKL. Furthermore, only greases from Schaeffler can be used, whereby the result of a weight specification corresponds to the fill level without a percentage specification. Finally, the variant with the TVP cage cannot be selected for the RRKL. Therefore, the following result applies to a larger internal volume corresponding to the RRKL 6007 variant with steel sheet cage (see Table 1). For all greases with a comparable density of around 882 kg/m³, the grease quantity range for the SKL is between 2.00 and 2.18 g and for the RRKL between 2.07 and 2.28 g.

Based on the recommendations outlined here, very different results can be determined for the continuous operation of the two ball bearings. Since the sealing concept also plays a role also plays a role, the amount of grease and the variation in grease quantity are determined at the beginning of the experimental investigations.

2.4 minimum load for rolling bearings

The equation for calculating the minimum load for ball bearings to ensure slip-free operation is [SCHAE18]:

$$P > \frac{C_0}{100} \qquad (5)$$

The dynamic equivalent bearing load P must be greater than one hundredth of the static load rating C_0.

$$P = 0.35 \cdot F_r + 0.57 \cdot F_a \qquad (6)$$

$$P = X \cdot F_r + Y \cdot F_a \qquad (7)$$

For SKL, this results in P according to Eq. *(6)* and for RRKL according to Eq. *(7)*. Both equations include bearing coefficients for weighting the radial force F_r and axial force F_a. Since the load conditions in the RRKL change significantly depending on the axial load, X and Y are linearly interpolated according to the catalog procedure if necessary. In the case of the minimum load from a pure axial load, no interpolation is necessary and a value of 2 is obtained for Y. At the same time, the radial load is neglected in both cases.

Combining the two equations with the condition from Eq. *(5)*, the following result is obtained:

$$F_{a,min} > \frac{C_0}{100 \cdot 0.57} \qquad \textit{(angular contact ball bearing)}$$

$$F_{a,min} > \frac{C_0}{100 \cdot Y} \qquad \textit{(radial ball bearings)}$$

2.5 Rheological properties

Thickeners, base oils, and additives influence the rheological properties of lubricating greases and thus their flow behavior. Some of the rheological properties are directly related to physical properties. Often, there is a strong dependence on individual variables, such as temperature. Standard tests are available in some cases to determine these grease properties. Furthermore, there are established models for certain dependencies. The dependencies are listed below:

- Viscosity
    - Shear rate dependence
    - Temperature dependence
    - Pressure dependence
    - Thixotropy
    - Viscoelasticity
- Consistency
    - Proportion of thickener and base oil
    - Cooking process (fat production)
    - Type and condition of thickener
    - Density ratio between thickener and base oil
    - Mobility of the oil
    - Oil separation
- Additive package
- Polarity of the base oil
- Specific heat capacity and thermal conductivity

Since the properties listed above interact with each other in many ways, numerous rheological properties of the two test greases were recorded. To simulate the grease distribution in the rolling bearing, the focus is on describing the flow behavior. This includes, in particular, shear rate dependence. Temperature also has a major influence on viscosity. This dependence is also integrated, as different temperatures are reached in the experimental investigations. The properties that can be determined for greases are listed below:

- Apparent viscosity according to DIN 51810-1
- Storage and loss modulus as well as yield point according to DIN 51810-2
- Walk penetration according to DIN ISO 2137
- Consistency class according to DIN 51818
- Flow pressure according to DIN 51805
- Oil separation according to DIN 51817
- Dropping point according to DIN ISO 2176
- Thickener content according to manufacturer's specifications

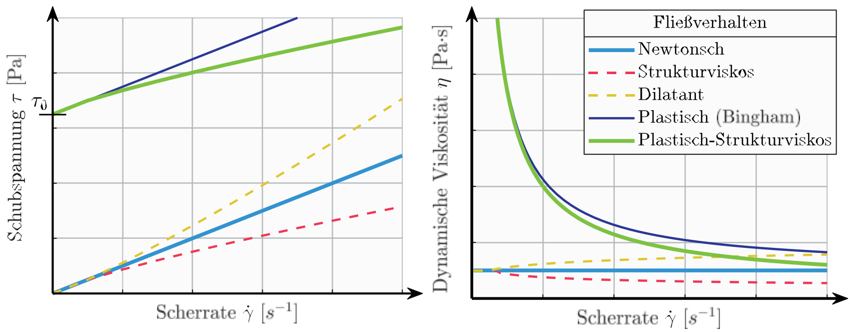
2.6 Models for describing flow behavior

There are various models for describing flow behavior (FB). Dynamic viscosity η is generally defined as the dependence of shear stress τ on shear rate $\dot{\gamma}$:

$$\eta = \frac{\tau}{\dot{\gamma}} \tag{8}$$

Equation *(8)* applies to a Newtonian fluid. Figure 5 shows different flow behaviors. On the left, you can see the dependence of the shear stress on the shear rate. On the right, the dependence of dynamic viscosity on shear rate is shown. In the case of Newtonian fluids (light blue curve), there is a proportionality between shear rate and shear stress (Fig. 5, left), so that the dynamic viscosity is constant regardless of the shear rate (Fig. 5, right). Non-Newtonian fluids behave differently. Either shear thinning occurs, which is referred to as structurally viscous FV (red dashed line), or shear thickening occurs, which is referred to as dilatant FV (yellow dashed line). Oils often exhibit structurally viscous behavior.

Fig. 5: Flow behavior of fluids. Left: shear stress τ versus shear rate $\dot{\gamma}$ Right: dynamic viscosity η versus shear rate $\dot{\gamma}$



Fats, on the other hand, have a yield point τ_0 , which must be overcome before the actual flow process. In the case of Bingham fluid (dark blue line), a yield point and thus plastic flow behavior is taken into account (cf. Eq. *(9)* and [BING16]). For this reason, the viscosity at low shear rates is very high compared to oils (Fig. 5, right).

$$\eta = \frac{\tau}{\dot{\gamma}} + \frac{\tau_0}{\dot{\gamma}} \tag{9}$$

However, greases generally exhibit plastic-structural viscous flow behavior (Fig. 5, green). There are numerous models for this flow behavior, which can be established using Eq. *(8)* based on the dynamic viscosity η . One of the best known

One such model is the Herschel-Bulkley model, which is described by Gl *(10)* [HERS26].

$$\eta = \frac{\tau_0}{\dot{\gamma}} + \kappa \dot{\gamma}^{n-1} \qquad (10)$$

Furthermore, Fig. 5 on the right shows that the dynamic viscosity of plastically structure-viscous flow behavior approaches structure-viscous flow behavior with increasing shear rate. This approximation occurs equally between fats and oils. Therefore, the greatest differences between fats and oils are in the low shear rate range.

Another model by SISKO shows great similarity to the model by HERSCHEL-BULKLEY [SISK58]. Here, η_b represents the viscosity value at a shear rate of zero. This is described by a constant value. It has no yield point and is therefore also not plastically viscous by definition, but only structurally viscous.

$$\eta = \eta_b + \kappa \dot{\gamma}^{n-1} \qquad (11)$$

In addition, there is also the PALACIOS model, which is shown in Eq. *(12)*. This includes both a yield point τ_0 and a base oil viscosity η_c and thus has parallels to Eqs. *(10)* and *(11)* [PALA84].

$$\eta = \frac{\tau_0}{\dot{\gamma}} + \kappa \cdot \dot{\gamma}^{n-1} + \eta_c \qquad (12)$$

CASSON developed another model for plastic-structural viscous flow behavior [CASS59]. Here, η_c represents the so-called viscosity coefficient. The flow index n is included here as a root function, which significantly reduces the resulting viscosity

.

$$\eta = \left[\left(\frac{\tau_0}{\dot{\gamma}}\right)^{\frac{1}{n}} + \eta_c^{\frac{1}{n}}\right]^n \qquad (13)$$

There is also the Cross model *(14)* [CROSS65] and the Carreau model *(15)* [CARR72]. Both equations are frequently used in oil-lubricated systems. They are similar in their basic structure and represent structurally viscous models.

$$\eta = \eta_\infty + \frac{\eta_0 - \eta_\infty}{\left[1 + \left(\frac{\eta_0 \cdot \dot{\gamma}}{\tau_0}\right)^{n-1}\right]} \quad (14)$$

$$\eta = \eta_\infty + \frac{\eta_0 - \eta_\infty}{\left[1 + \left(\frac{\eta_0 \cdot \dot{\gamma}}{\tau_0}\right)^{\frac{n-1}{2}}\right]} \quad (15)$$

The limit value for high shear rates η_∞ corresponds to the viscosity of the base oil in fats. At a shear rate of zero, the viscosity corresponds to the value η_0.

The models described here were developed for a range of very different substances in the field of plastic-structural viscous FV and are highly relevant to your application. For example, SISKO developed an approach for color-pigmented oil suspensions for the technical application of a printing machine. HERSCHEL and BULKLEY investigated the consistency of rubber-benzene solutions. The coefficients of the various equations were determined on the basis of data from a viscometer using a mathematical adjustment calculation (regression).

In addition to shear rate dependence, temperature dependence is also a relevant influencing factor. The Vogel equation [VOGE21] is often used for oils. The equation exists in two versions. Either two or three equation coefficients must be determined, which is why either two or three temperatures must be known

. For the base oils of the greases, the kinematic viscosity ꓒ at 40 and 100 °° °C
known. The following equation can be used to convert to dynamic viscosity.

$$= \frac{\eta}{\rho} \quad (16)$$

According to DIN 31652, the temperature-dependent dynamic base oil viscosity η_{eff} can be determined at two known temperatures using the following Vogel equation [DIN 31652, DUBB23]:

$$\eta_{eff} = (10)^{\frac{\eta C_1}{T_{eff} + 95°C} + C_2} \quad (17)$$

When determining the components of the equations C_1 and C_2, the following applies $T_1 < T_2$.

$$C_{(1)} = \frac{(T_1 + 95°C) \cdot \left\{\frac{ln\,[\eta(T_1)] - ln\,[\eta(T_2)]}{2.3026}\right\}}{\frac{T_1 + 95°C}{T_2 + 95°C} - 1} \quad (18)$$

$$C_1 = log[\eta(T_1)] - \frac{C_1}{T_2 + 95°C} \quad (19)$$

There are also models that describe the dependence of viscosity on pressure [BARUS93, ROEDE75]. These are taken into account in particular when considering isolated, highly loaded contacts such as those between rolling elements and raceways [TERW20, FVV 1277, TERW20].

Finally, there is the change in flow properties over the course of the stress duration (thixotropic flow behavior) and the regenerative capacity of greases. Both thixotropy and regenerative capacity have been observed in lithium greases, for example [PASZ14-1, PASZ14-2]. PASKOWSKI had already observed an increased (regenerated) shear stress in shear tests after short shear pauses. To date, no model has been established that reliably describes these properties for any grease.

The dependencies of pressure and thixotropy are not taken into account in the further course.

3 Test and e measurement technology

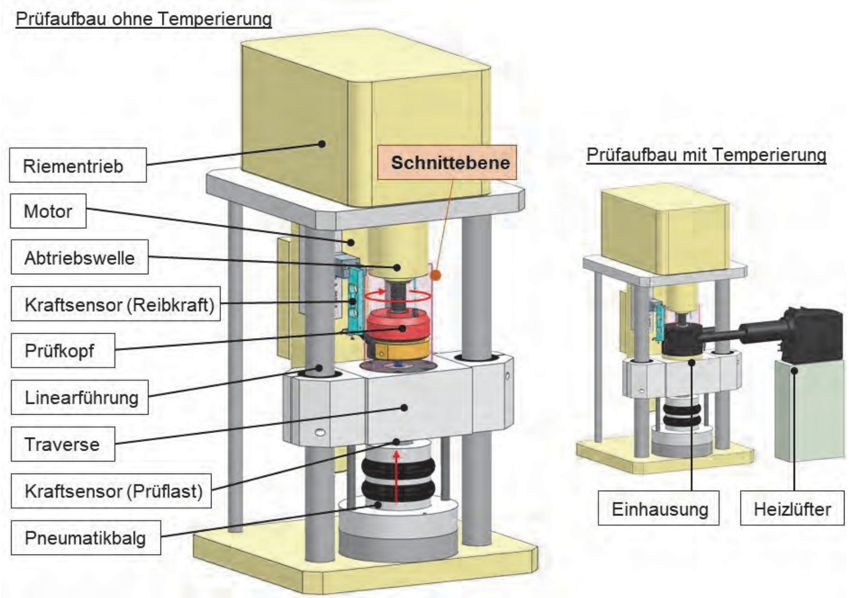
Before presenting the experimental results, the test systems and measurement technology used will be introduced. This concerns the roller bearing test benches, the rheometers, and the imaging technology used to record the grease distribution.

3.1 Rolling bearing test benches

3.1.1 Rotational tribometer

The rotational tribometer is shown with a standardized test head (tapered roller holder) in Fig. 6 in two CAD views. The view on the left shows the design without temperature control. The load is applied via a pneumatic bellows below the crosshead, where a load cell for measuring the test load is also located. When the load is applied, the crosshead moves along the linear guides toward the output shaft. The motor drives the output shaft via a belt drive with a fixed transmission ratio. A force sensor is connected to the test head to measure the friction force. Since the lever arm is known, the friction torque can be calculated.

Fig. 6: Rotational tribometer with standardized test head (tapered roller holder) and



adaptation for temperature control using a hot air gun

The test bench was adapted for the temperature control tests (Fig. 6, right). In principle, existing cooling channels in the test head can be used for liquid cooling.

This proved to be ineffective in preliminary tests, as the connection to the test head with temperature control hoses influences the friction torque measurement. This influence is too great when measured against the friction torques, which are often in the range of less than 100 Nmm. For this reason, air temperature control was implemented for the test bench when developing a test methodology. This is done using a powerful fan heater. Tests were carried out with and without temperature control. Without temperature control, the test showed a persistence in terms of friction torque (persistent friction torque) and temperature (persistent temperature). With temperature control, tests were carried out at a practical test temperature of 80 °C.

Fig. 7 shows sectional views from the sectional plane shown in Fig. 6. These reveal design details that are explained in more detail below.

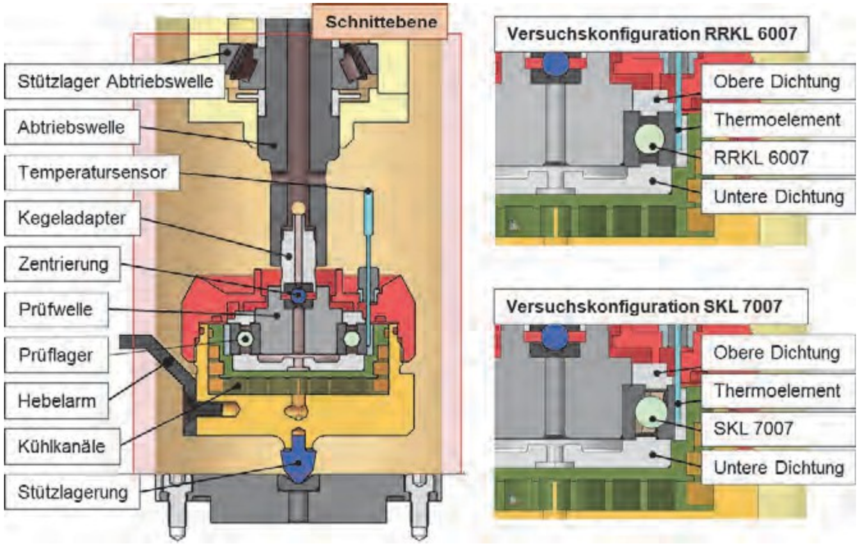


Fig. 7: Cross-sectional views of the rotational tribometer with the adaptation for the two test bearings RRKL 6007 and SKL 7007 ($_{nmax}$ = 3,000 rpm, $_{Fmax}$ = 10 kN)

The test head has the decisive advantage that the support bearing is designed as a cone. This means that only a small area of the cone tip generates friction force. According to the friction torque from the friction force multiplied by the lever arm, the very small radius at the tip of the cone results in a small friction torque. This leads to a negligible parasitic friction influence from the support bearing. Nevertheless, this must be taken into account when comparing results from different test benches, as measuring the friction torque, both the friction from the test bearing and from the support bearing is recorded. The friction torque is measured using a lever arm attached to the test head. Taking into account the length of the lever arm, the friction torque is recorded using a force sensor of the "double bending beam" type.

The test shaft of the test head is connected to the output shaft via a taper adapter (see Fig. 7, left). The test shaft is very short compared to the output shaft. The test bearing is mounted on it. The temperature is always measured on the bearing back in the area of the bearing raceway. Each rolling bearing that is tested has its own bearing cover, which acts as a contact-free seal (see Fig. 7, right).

The roller bearings are tested at minimum load. This raised the question of whether this minimum load is sufficient to preload the support bearing of the output shaft. For this purpose, the shaft was gradually loaded at a standstill. The axial displacement of the output shaft was recorded with a dial gauge. This preliminary test showed that the preload of the support bearing had to be increased with additional disc springs in order to ensure good guidance of the output shaft for all planned tests at minimum load.

3.1.2 High-speed test bench

As part of this work, the high-speed test bench for rolling bearings was constructed and modified according to the FE8 principle. The rolling work was also investigated at higher speeds on the high-speed test bench, whereby the maximum speed of 18,000 rpm for the rolling bearing size used (see Table 1) resulted in a speed characteristic value of 873,000 $n \cdot dm$. The test bench has two special features: The first special feature is the adjustability of the position of the entire drive train. This is mounted on a swiveling support plate and can be swiveled between 0 and 90° in just a few simple steps, allowing two shaft positions (horizontal/vertical) to be examined. Fig. 8 shows CAD views of the shaft positions at the top. The real test bench with the drive train on the swiveling support plate is shown at the bottom.

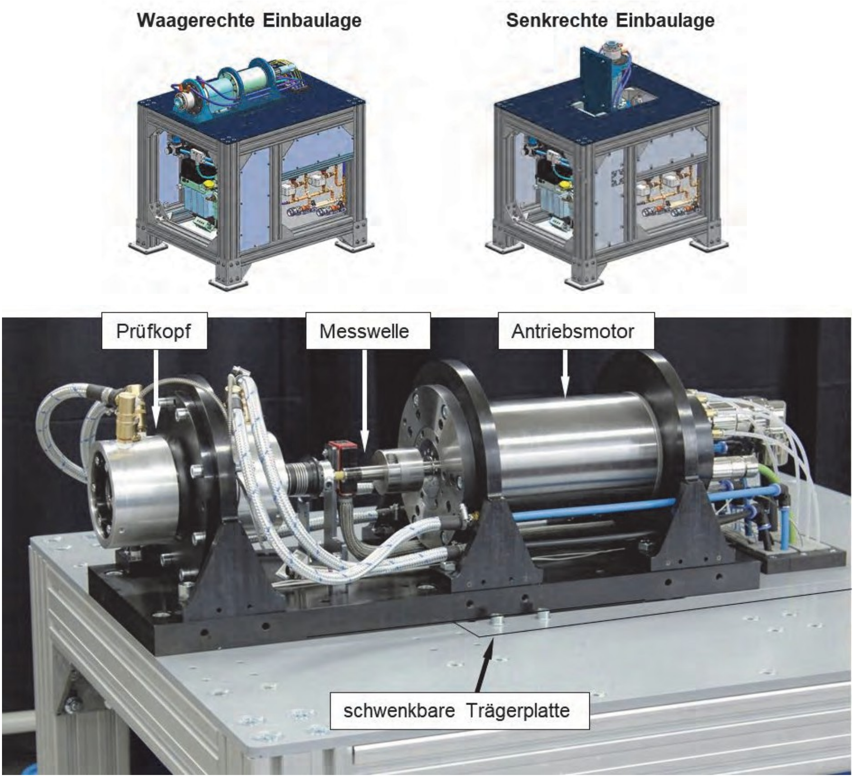


Fig. 8: Rolling bearing test bench based on the FE8 principle with media temperature control and a maximum speed of 18,000 rpm, drive train on swiveling support plate for two positions: horizontal/vertical shaft

The second special feature is the measuring principle, which, thanks to the design of the test head, eliminates the need for support bearings. This means that only the friction torque from the test bearings is measured.

Fig. 9 shows cross-sectional views of the test head. The top image shows the entire test head with test bearings. The bottom image shows an enlarged view of the left test bearing (here SKL 7007) with the bearing covers. The test bearings are always loaded in an X arrangement with the respective minimum load.

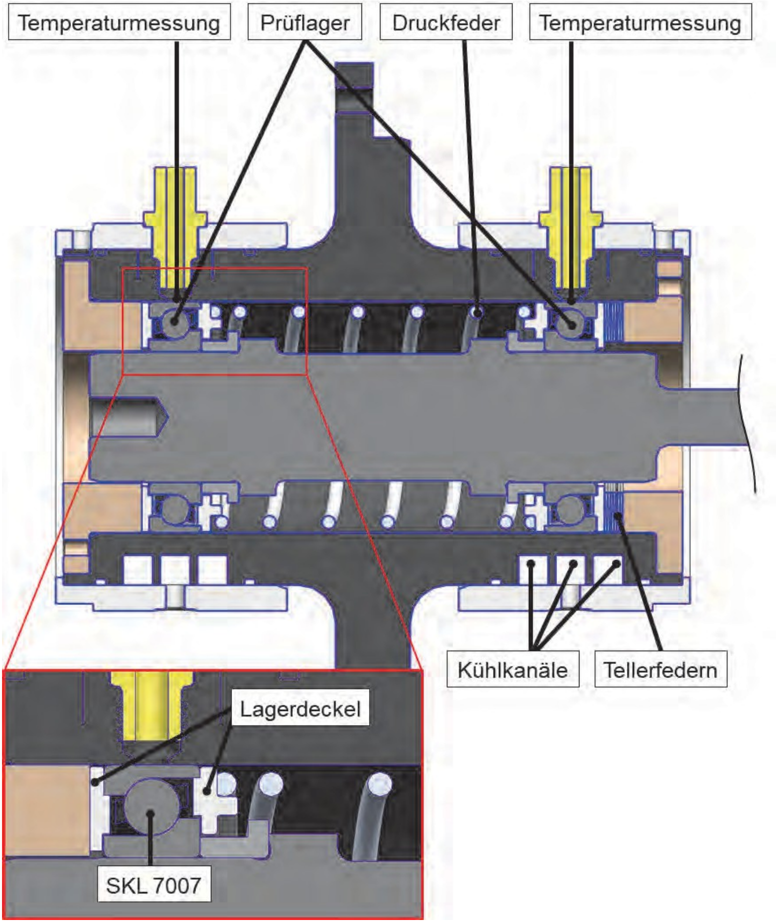


Fig. 9: Cross-sectional views of the high-speed test bench with SKL 7007 and an enlarged view of the bearing covers

The load is applied via a disc spring package. The springs are preloaded using the available installation space in the housing. The test head also has cooling channels. The bearing back temperature of both bearings is recorded and serves as a control variable for temperature control. The measuring shaft (see Fig. 8) is capable of measuring up to 5 Nm of friction torque. It has an accuracy class of 0.1%. This means that the resolution limit is approximately 5 Nmm.

3.1.3 Special tribometer

Glass components were used on the special tribometer to replace these rolling bearing components, which provided an insight into the test bearing. The angular contact ball bearing (SKL 7007) and the axial cylindrical roller bearing (AZRL 81212) were examined.

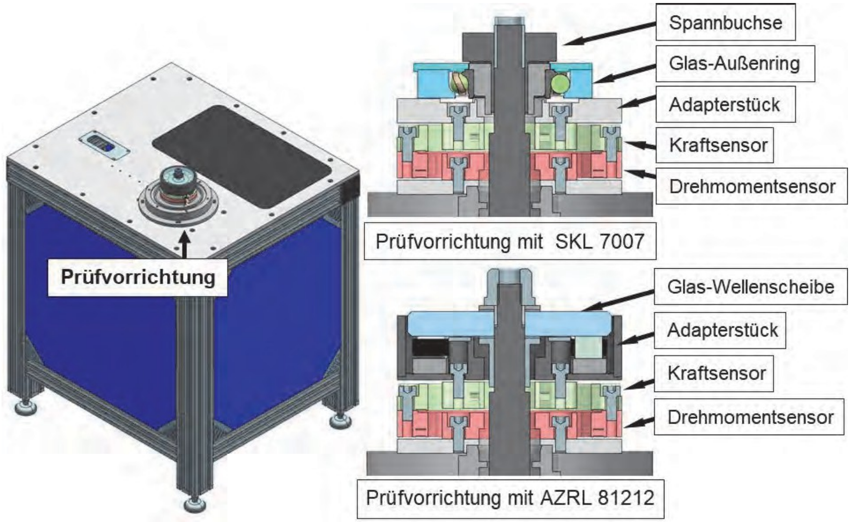


Fig. 10: View of the special tribometer and cross-section of the SKL 7007 and AZRL 81212 test configuration with sensors for force (load) and torque

A sensor package was implemented to measure the test load and friction torque. The sensor package consists of two measuring flanges. One flange measures the normal force (force sensor), whereby the axial test load is measured with this test setup. The other flange measures the torque. Due to the design of the test bench, the torque sensor measures the reaction force from the test bearing, which means that the friction torque from the support bearing is not recorded. This allows the friction torque of the test bearing to be measured directly. The sensor package offers great flexibility, as different measuring ranges can be recorded, providing good sensor resolution for many applications. A torque sensor with a measuring range of up to 1 Nm and an accuracy class of 0.1% was used for the measurements here. This means that the resolution limit is 1 Nmm. Rolling bearing tests up to a maximum speed of 2,000 rpm could be carried out on this test bench.

The load is applied via the drive shaft. Before the test, this moves in the axial direction until the reaction force measured at the force sensor corresponds to the set test load.

test load. Since this is a sensor package, the load is also transmitted through the torque sensor. The load thus generates an offset in the friction torque, which is reset to zero before the test begins. In principle, the load can be changed during operation, as the respective offset has a very high reproducibility.

The aim of the investigations was to use optical analyses to evaluate how the grease distribution develops in the course of such a test and to investigate whether persistence is achieved not only in the friction torque and temperature, but also in the distribution of the lubricating grease. This is followed by explanations of the high-speed recordings with the respective measurement setup that were necessary for these questions.

3.2 High-speed recordings

To gain a detailed understanding of grease distribution in rotating roller bearings, it is necessary to use very high recording frequencies. High-speed cameras were used for this purpose. These were used on the special tribometer. The device specifications and the planned investigations are explained below.

3.2.1 High-speed camera

The "High Speed Star 6" high-speed camera from LaVision with a resolution of 1024x1024 pixels was used for optical recordings. A short exposure time requires a strong aperture, which necessitates an additional light source. This was achieved using a 2,000-watt halogen spotlight. To minimize heat input into the measurement setup, this light source was only switched on for less than three seconds during the high-speed recordings.

3.2.2 Thermal imaging camera

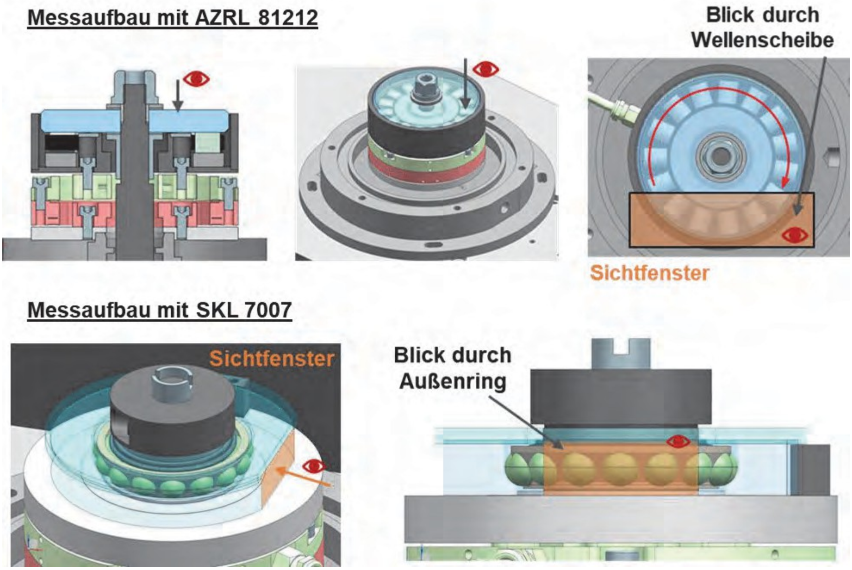
Infrared thermography uses a thermal imaging camera to record a temperature field. The ImageIR®8300 thermal imaging camera from Infratec was used for this purpose. This camera can generate images with a resolution of 640x512 IR pixels in the high-speed range at a speed of up to 1/10,000 s. The thermal resolution of the camera is 0.03 K. IR thermography offers many advantages. For example, after an experiment, the images can be used to evaluate the temperature at any number of points in the temperature field. This allows even small areas of heating to be accurately assessed. The images are evaluated using the manufacturer's IRBIS software.

3.2.3 e measurement setup with high-speed camera

Fig. 11 shows the measurement setup for the AZRL at the top and for the SKL at the bottom on the special tribometer. Various CAD views were selected to illustrate the differences between the tests with the two bearing types. The AZRL is a rotating glass shaft disc, which allows axial viewing of the rolling bearing. This component is solid and made of borosilicate glass. The angular contact ball bearing is a stationary glass outer ring, which allows a radial view into the rolling bearing. To increase the strength of the component, the diameter of the outer ring was made significantly larger than that of the original component. Due to the radii in the raceway, refraction of the

This effect leads to optical distortion of the image and is unavoidable. However, a flat surface has been provided on the outer surface of the glass component. This is intended to prevent further refraction of light by the radius on the outer surface.

Fig. 11: Test setup with AZRL 81212 at the top and SKL 7007 at the bottom



Furthermore, the viewing windows selected for the recordings with the high-speed camera are marked in transparent orange in Fig. 11. At least three rolling elements can be seen in the image section for both types of rolling bearings.

3.2.4 e measurement setup with thermal imaging camera

The outer ring of the angular contact ball bearing is made of sapphire glass. According to the manufacturer, this material is transparent to IR radiation in the spectral range of 0.15-5.5 µm. This spectral range was recorded with an IR camera from IFRATEC, which can capture the spectral range of 2.0-5.7 µm [INFR19].

Fig. 12 shows the measurement setup with the thermal imaging camera and the SKL 7007 on the special tribometer. A wider viewing window was selected for these images than for the images taken with the high-speed camera. This is to capture as large a temperature field as possible across the entire bearing width. The result is that in

There is more optical distortion at the left and right edges than in the area with the flat surface.

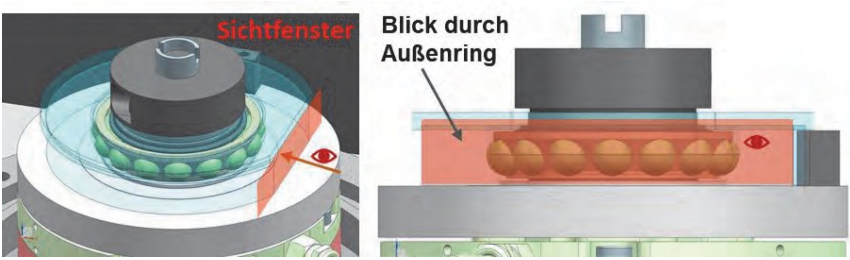


Fig. 12: Left: Temperature measurement using a thermal imaging camera (IR thermography), right: viewing window for recordings

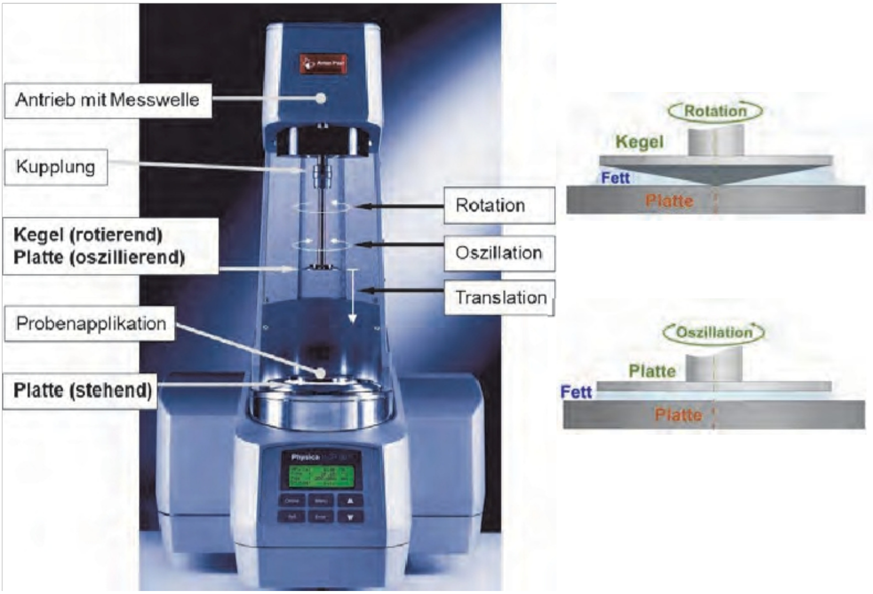
3.3 Rheometer

In this work, numerous rheological properties of the lubricating greases are derived from measurements taken on the rheometer. The rheometers used are explained here.

3.3.1 MCR 301

The Physica Modular Compact Rheometer 301 (MCR 301 for short) from Anton Paar has a drive shaft (Fig. 13). Two different measuring applications can be mounted on it via a coupling. Either a cone or a plate is used. These are operated either rotary (cone) or oscillating (plate). This measuring application can be adjusted translatorily via an additional servomotor. The gap between the rotating test side and the stationary plate can thus be adjusted. For this purpose, the position of the translatorily movable shaft is measured to a tenth of a micrometer and adjusted to a micrometer's precision.

Fig. 13: Design and measurement applications of the MCR 301 rheometer: left: design of the



rheometer [ANTON08], right: sketch of measurement applications used

The stationary plate is equipped with a Peltier element and a temperature measurement system. The temperature is measured in the plate near the lubrication gap. In addition, a media temperature control system is connected to the underside of the plate with the aid of an auxiliary unit. This provides counter-cooling for the system. As a result, temperature control is usually carried out exclusively via the plate. At temperatures above 80 °C

, a heating hood is used. This enables a more even temperature distribution in the stationary and rotating parts of the measuring applications.

Before a measurement, the cone or upper plate is in a locking position at a sufficient distance from the lower plate. The test grease is then applied to the lower plate. The upper counterpart is then moved translationally to the respective measuring position. When measuring with the cone/plate measuring system, the cone is driven in rotation. The resistance caused by the lubricating grease is recorded as torque using a measuring shaft. This measured variable is then evaluated in accordance with the standard [DIN 51810-1]. When measuring with the plate/plate measuring system, the upper plate is operated in an oscillating manner, with the swivel angle being increased successively. Here, too, the torque is measured and the measured variable is evaluated in accordance with the standard [DIN 51810-2]. These evaluations are described in detail in chapter 4.1.

3.3.2 MCR 702e

The Modular Compact Rheometer 702 Evolution MultiDrive (MCR 702e for short) from Anton Paar has two drives (see Fig. 14).

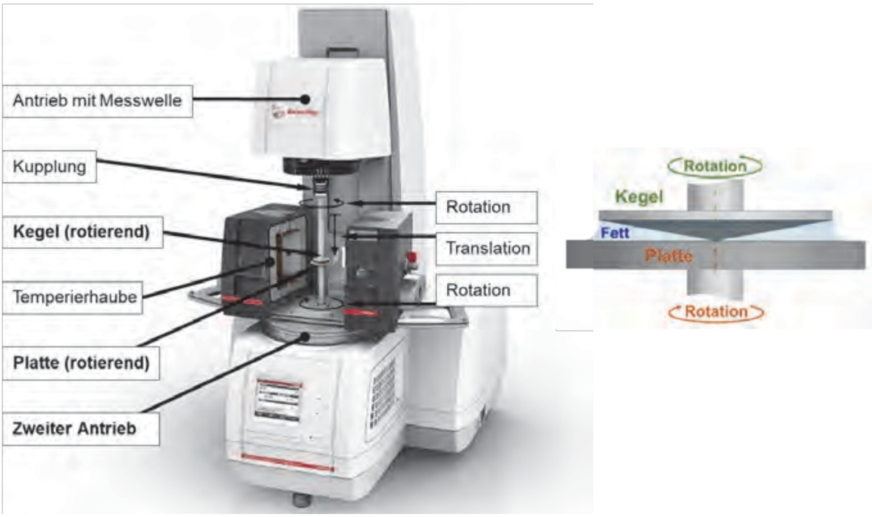


Fig. 14: Design and measurement application of the MCR 702e rheometer: left: design of the rheometer [ANTON19], right: sketch of the measurement application used

In contrast to the MCR 301, the two drives also drive the lower plate. The direction of rotation of the lower plate is always opposite to that of the cone at the same speed. As a result, the total speed prevailing in the gap is twice as high as the respective drive speed.

Temperature control is carried out exclusively via a temperature control hood in the form of a convection heater. Temperature measurement is again integrated into the plate, but is now carried out by means of optical measurement in the measuring gap. Various measuring applications can be used on this rheometer, but in this work only the cone/plate measuring system is used.

3.3.3 Comparison of the rheometers

For a clear comparison of the two rheometers, characteristic values for the MCR 301 and MCR 702e rheometers are shown in Table 4 below. The measurement accuracy is identical. However, the second drive on the MCR 702e achieves a total speed of 6,000 rpm. This doubles the maximum measurable shear rate for the same cone angle.

Table 4: Characteristic values of the MCR 301 and MCR 702e rheometers

	Measurement	Detail	Unit	MCR 301	MCR 702e
Measurement	minimum torque	Rotation	µNm	0.05	0.05
	Minimum torque	Oscillation	µNm	0.01	0.01
	Maximum torque	Rotation/ Oscillation	mNm	200	230
	Torque	Resolution	nNm	0.1	0.1
Speed	Maximum speed	per drive	rpm	3,000	3,000
	Total speed		rpm	3,000	6,000
Shear rate	Cone angle 1.0	Maximum	s^{-1}	17,998	35,999
	Cone angle 0.5°	maximum	s^{-1}	35,996	71,998

4 Experimental investigations

The experimental investigations present both the rheological investigations of the lubricating greases and the test bench trials with roller bearings of different designs. A total of three roller bearing test benches were used: a rotational tribometer, a special tribometer, and a high-speed test bench. Different aspects of grease lubrication were investigated on each of these test benches. The test conditions vary greatly in some cases. Nevertheless, tests with comparable test conditions were also carried out in order to demonstrate the transferability of the results between the test benches.

4.1 Rheological investigations

4.1.1 Apparent viscosity according to DIN 51810- e 1

This standard test is performed on the MCR 301 rheometer using the cone/plate measuring system (CP25-1 measuring cone) [DIN 51810-1]. The maximum shear rate is determined according to the NLGI class. For test greases of NLGI class 1, testing is carried out up to a shear rate of 1,000 s$(^{-1})$(Fig. 15). A torque is measured on the cone shaft. Using the known cone geometry, this torque can be converted into viscosity.

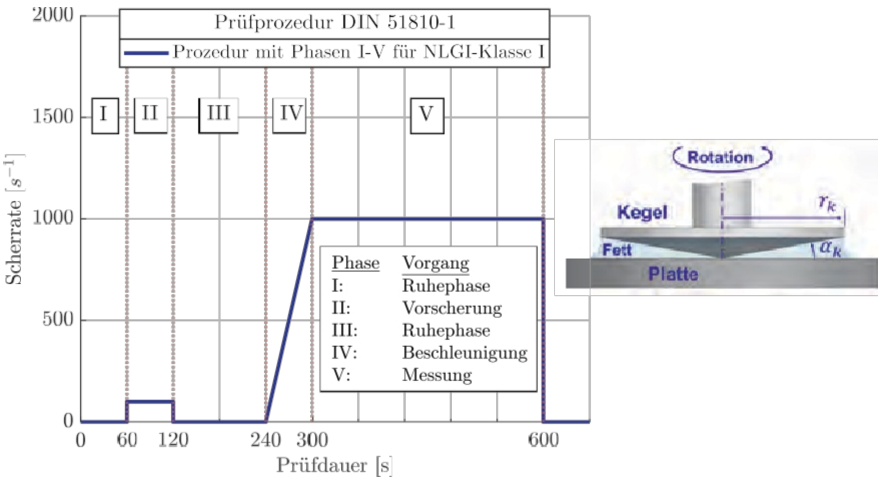


Fig. 15: DIN 51810-1: Procedure in phases I-V, cone-plate test configuration on the right

The cone-plate system offers the advantage that a fixed shear rate prevails throughout the fluid at a specific cone speed. The shear rate $\dot{\gamma}$ is defined as the dependence of the velocity difference du on the height dh. The shear rate is thus calculated according to Eq. *(20)*.

$$\dot{\gamma} = \frac{du}{dh} = \frac{u_k}{h_k} \tag{20}$$

Due to the rotating system (cone), the circular path velocity u_K and the cone height h_K are introduced. The circular path velocity *(21)* and the cone height *(22)* depend on the cone radius r_k. This means that the shear rate in the cone-plate system can be assumed to be constant regardless of the radius. This means that the measured torque can be converted into viscosity using the known angle.

$$u_K = r_k \cdot \omega \tag{21}$$

$$h_K = r_k \cdot \tan(\alpha_k) \tag{22}$$

The measuring cone has an angle of 1° and a diameter of 25 mm. Since both the cone radius r_k and the cone angle α_k can have manufacturing-related deviations, each cone is delivered by the manufacturer with a measurement report.

The measured values are included in the viscosity calculation, as they are crucial for an accurate measurement. DIN EN ISO 3219 describes how the geometric deviations from ideal values are incorporated in the form of correction factors when calculating the shear rate and shear stress [DIN 3219]. Equations *(20)* to *(22)* gives equation *(23)*. For the CP25-1 measuring cone, this results in a speed of approximately 170 rpm for NLGI class 1. In addition, the shear stress is calculated from the torque measured at the cone over the area of the cone *(24)*.

$$\dot{\gamma} = \frac{\omega}{\tan(\alpha_k)} \tag{23}$$

$$\tau = \frac{3 \cdot M}{2\pi r_k^3} \tag{24}$$

Nine tests were performed for each test fat in the range of 20–100 °C. Fig. 16 shows a measurement of fat 1 and fat 2 at 40 °C with phases IV and V (see Fig. 15). In phase V, η_A and η_E are evaluated at 300 and 600 seconds (Fig. 16).

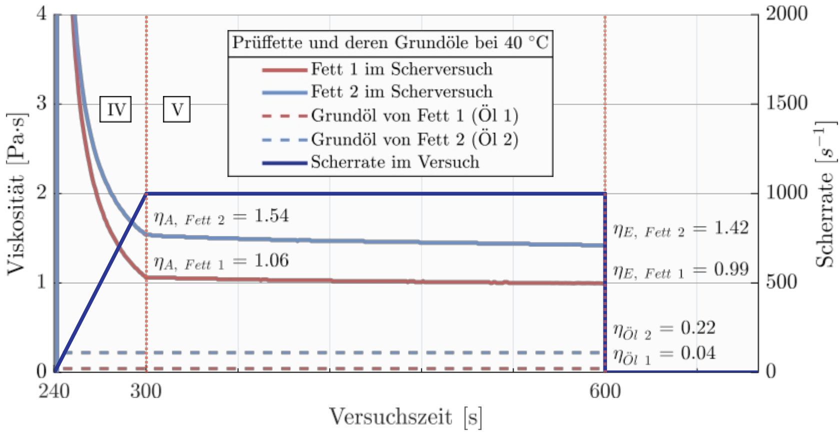


Fig. 16: Fat 1 and Fat 2 in the shear test at 40°C, phase IV up to 300 seconds and phase V up to 600 seconds with the measured variables η_A at 300 and η_E at 600 seconds, as well as the addition of the base oil viscosities Oil 1 and Oil 2.

The kinematic viscosity ꓒ at 40 °C and the density of the base oils are known

ρ are known. Their dynamic viscosity was therefore added to the diagram. At 40 °C, the grease viscosity η_A of grease 1 is 26 times higher and that of grease 2 is seven times higher than the respective base oil viscosity.

A comparison of the viscosity of test greases and base oils clearly shows that the thickener has a significant influence on flow behavior. In greases, viscosity decreases progressively in phase V, as can be seen from the values a t η_A and η_E

. These values are summarized for all tests in the range of 20-100 °C in Table 5

.

Table 5: Apparent viscosity at the beginning (η_A) and end (η_E) of measurement phase IV

	Apparent viscosity [Pa·s]			
Temperature	Grease 1		Grease 2	
[°C]	η_A	η_E	η_A	η_E
20	1,570	1,430	3,170	3,010
30	1,260	1,170	2,030	1,920
40	1,060	0.993	1,530	1,420
50	0.918	0.857	1,200	1,100
60	0.810	0.754	0.898	0.826
70	0.710	0.685	0.749	0.686
80	0.617	0.578	0.642	0.592
90	0.556	0.521	0.533	0.483
100	0.538	0.512	0.489	0.445

In the literature, thixotropy due to shear has been observed in roller bearing greases [KUHN15, REZ16-2, ZHOU18]. KUHN demonstrated this with cone-plate tests, which show the gradual reduction of shear stress during hours of constant shear. The decrease in shear stress is explained by the change in the thickener structure. Qualitative evidence for this was provided using a scanning electron microscope (SEM) or an atomic force microscope (AFM). However, none of the studies provide a calculation model that can predict the structural changes in the thickener. The standard provides the ratio value

η_{Rel}. This relative viscosity change is described in equation *(25)*:

$$\eta_{Rel} = \frac{\eta_A - \eta_E}{\eta_A} \cdot 100 \qquad (25)$$

This value is shown for both test fats in Fig. 17 in the temperature range from 20 to 100 °C. The relative change in viscosity is not linear with temperature and does not follow any trend. This makes it difficult to model this behavior.

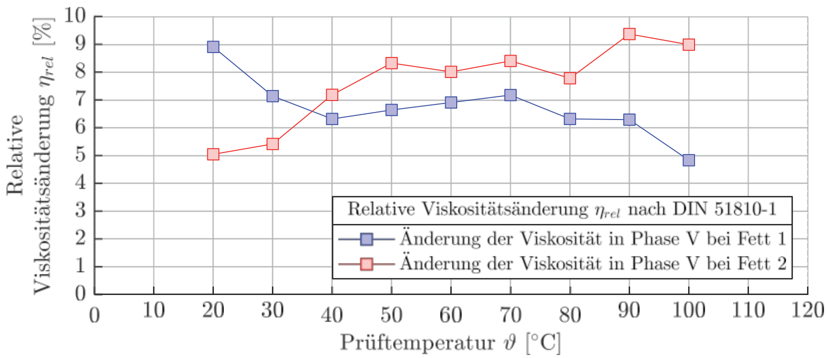


Fig. 17: Relative change in viscosity in phase V of the standard test for the range of 20-100 °C based on the calculation variable η_{Rel} (Eq. *(25)*) according to DIN 51810-1

Furthermore, it is not known how regenerative the test greases are. Furthermore, no model has yet been established that is capable of describing thixotropy (see 2.6). Therefore, the findings from phase V cannot be integrated into the lubricant model. For this reason, only phase IV was used as the data basis for the creation of a lubricating grease model. Here, there is a constant acceleration of the cone, which is why a direct relationship between shear rate and shear stress or viscosity. This is discussed in detail in the description of the grease model development for CFD.

4.1.2 Storage and loss modulus according to DIN 51810- e 2

The plate-plate (PP25/PE) test configuration is used to measure the storage and loss modulus, using the same rheometer as in 4.1.1. During the test, the measuring plate is moved against the base plate in a rotating-oscillating motion. Shear stresses also occur in the lubricant, which are related to the viscoelasticity

of the lubricant. These are the storage modulus G' (elastic
component of the viscoelastic behavior) and the loss modulus G'' (viscous component of the viscoelastic behavior). During the test, the swivel angle is gradually increased. At one point in the test, lubricating greases often fulfill the condition

$G' \equiv G''$ is met. This exact intersection point (crossover point) represents a characteristic lubricant property, which can be important in the context of viscoelasticity in applications with small swivel movements.

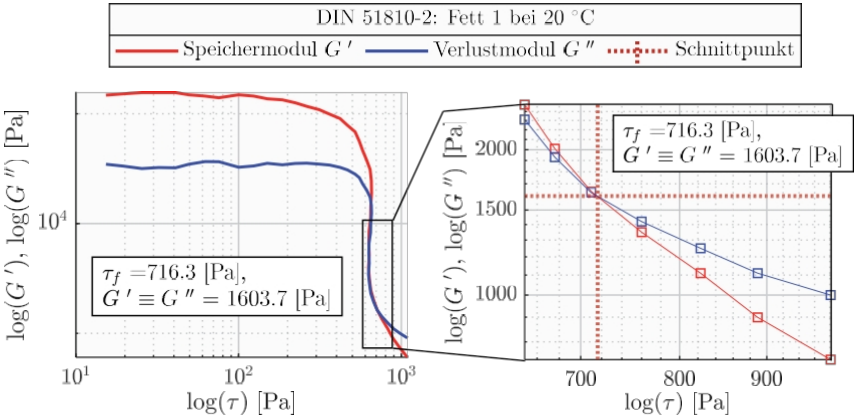


Fig. 18: DIN 51810-2 for determining storage and loss modulus, top: schematic diagram of the plate-plate test configuration (PP-25/PE), bottom: test result for grease 1 at 20 °C (left) with enlarged section of the intersection point determination for deriving the yield point τ_f (right)

The yield point τ_f is derived at the intersection. Fig. 18 shows a test result according to the DIN 51810-2 standard. On the left is the test in a double logarithmic representation and on the right is an enlarged view of the intersection area. The following yield points τ_f were determined for both greases:

Table 6: Yield point τ_f in [Pa] according to DIN 51810-2 in the range of 20-100 °C

Test greases	τ_f 's yield strength in [Pa] at temperature P in [°C]								
	20	30	40	50	60	70	80	90	100
Fat 1	716.3	666.5	482.7	481.1	353.1	299.8	295.4	239.0	207.1
Fat 2	752.0	617.0	553.0	422.3	352.1	292.0	205.4	224.8	178.0

4.1.3 Oil separation according to DIN 51817

Oil separation is performed using a standardized testing device consisting of a pot, a lid with a conical sieve, and a weight. The grease sample is pressed onto a sieve with a defined mesh size under the defined load of the test weight of 100 g. This results in oil separation. The results are presented as the amount of oil separation as a percentage of the original sample weight. Since separation is not only temperature-dependent but also time-dependent, oil separation is recorded at different times. Table 7 shows the results in the temperature range of 40-100 °C and after 18 hours and 7 days for both test greases used.

Table 7: Oil separation according to DIN 51817 and relative oil loss between 40 and 100 °C

	Oil separation in %				Relative oil loss %			
Test temperature	Grease 1		Grease 2		Grease 1		Grease 2	
P in [°C]	18h	7d	18h	7d	18h	7d	18h	7d
40	2.2	4.8	1.5	4.7	1.0	2.2	0.3	0.9
50	2.7	5.6	2.4	6.4	1.2	2.5	0.4	1.2
60	3.4	6.2	3.0	7.9	1.5	2.8	0.5	1.5
70	4.3	7.3	4.1	10.0	1.9	3.4	0.8	2.0
80	4.8	8.0	4.8	11.5	2.2	3.7	0.9	2.3
90	6.4	10.5	7.4	13.2	2.9	5.0	1.4	2.7
100	7.6	12.3	10.1	14.3	3.5	6.0	2.0	2.9

Both fats achieve comparable values in the oil separation test. Therefore, the relative oil loss was also calculated, which takes into account the oil content in the fresh fat. To calculate the relative oil loss from the oil separation, the procedure is explained using a numerical example. A fictitious fat with a thickener content of 20% is used as an example. In an oil separation test, a separation of 50% is achieved. This reduces the total mass of the original sample weight by 50% due to the loss of pure oil. The oil content falls from 80% to 60%, a decrease of 20 percentage points. The relative oil loss is therefore 25%.

In order to better classify the results in Table 7, the values in the following figure are presented in two diagrams (Fig. 19). The upper diagram shows how similar the oil separation is for both fats. However, when looking at the relative oil losses, it becomes clear that fat 1 has a significantly higher oil loss with similar oil separation (Fig. 19, bottom). This proves that despite the high thickener content of grease 1 at 30%, the oil leaves the matrix more easily than with model grease 2. Since both thickeners are chemically identical, this must be related to the lower base oil viscosity of grease 1.

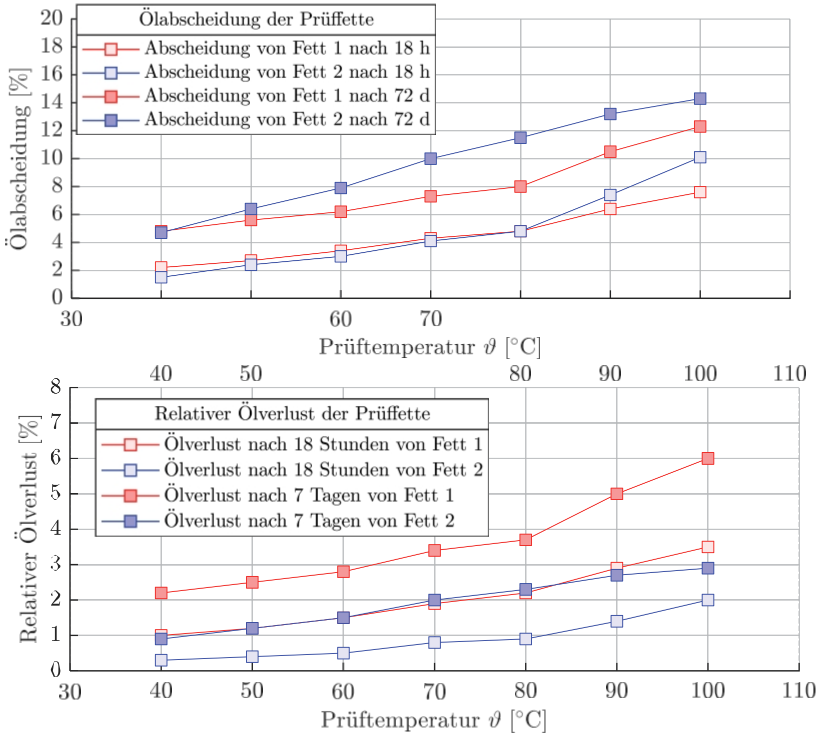


Fig. 19: Oil separation results: top: oil separation of both test fats after 18 hours and seven days; bottom: relative oil loss during oil separation based on the respective thickener content (see Table 2)

4.1.4 Further rheological investigations

In addition to the rheological tests already mentioned, the flow pressure was determined in accordance with DIN 51805-2. According to DIN 51805, the preferred temperatures are - 35, - 20, 0, and+ 25 °C.

Table 8: Flow pressure in [hPa] in the range from -20 to 20 °C according to DIN 51805-2

Flow pressure in [hPa]	Temperature P in [°C]		
	-20	0	20
Fat 1	450	150	100
Fat 2	525	175	125

The dropping point is above 250°C, which is not reached in the tests. When determining penetration in accordance with DIN ISO 2137, the penetration depth is determined [DIN 2137]. A standardized test apparatus (cone) is pressed into a pot filled with grease. The penetration depth of the cone can be read on the device. There are two different types of stress on the test grease. On the one hand, the penetration depth can be determined in an unstressed state, from which the resting penetration is derived. After 60 double strokes with the so-called grease kneader, the penetration depth can be determined again. This penetration depth therefore corresponds to the so-called kneading penetration.

Table 9: Results of the consistency test according to DIN ISO 2137

Test grease	Penetration depth in [0.1 mm]		NLGI class
	Rest penetration	Worked penetration	310 0.1 mm
Grease 1	282	310	I
Fat 2	286	316	I

In accordance with DIN 51818, workability ranges are divided into 9 consistency classes, with class 000 representing a soft, almost liquid grease and class 6 representing a solid grease. For NLGI class I, a workability range of 310-340 · 0.1 mm is defined. Greases of this NLGI class are used particularly in applications with high speeds, which is why the test greases also meet these requirements.

4.2 Basic test methodology for roller bearing tests

When developing a test methodology for measuring grease-related work, several aspects are taken into account that go beyond the actual test. The test methodology includes the reuse of rolling bearings, lubrication before the test, cleaning of the rolling bearings after the test, the design of the bearing covers (seals), the quantification of grease leakage, and the test procedure for measuring grease-related kneading work in the rolling bearing.

When reusing roller bearings, the existing test bearings were used several times during the tests. Certain roller bearings were only used in one test system. In addition, the roller bearings were not replaced during a repeat test on the high-speed test bench. To rule out wear, the roller bearings were weighed before and after use.

Commercially available disposable syringes were used for greasing. The grease was distributed evenly between the rolling elements. A smaller amount was applied between the rolling elements and the inner ring and a slightly larger amount between the rolling elements and the outer ring. This resulted in very good reproducibility.

The type of cleaning and the choice of suitable solvents took into account the polarities of the grease components and the resistance of the cage material. The grease components thickener, oil, and additives have varying degrees of polarity. The base oil can be assumed to have a rather weak polarity. In the case of lithium complex soap, there are polar ends on the fibrils of the matrix, and in the case of additives, there are some highly polar components. However, the exact polarity of the individual components is not known. A mixture of a polar and a non-polar solvent is therefore used to dissolve all grease components. For this purpose, a mixture of four parts cyclohexane (non-polar) to one part 2-propanol (polar) was used. The resistance of the plastic cages of different designs and materials to these solvents was also tested. The resistance was confirmed by the roller bearing manufacturer.

Further conditions are explained using the rotation tribometer and were largely implemented in the same way on the other two test benches.

The test conditions for investigating grease-related flexing require the lowest possible load and the use of non-contact seals. The minimum load of the respective test bearing was calculated according to the catalog method (see 2.4). This minimizes friction from the contact surfaces and ensures virtually slip-free operation. This resulted in the values shown in Table 10. For ball bearings, the minimum load is independent of the speed. For the AZRL, the required minimum load increases with speed, whereby this bearing was tested up to a maximum speed of 500 rpm.

For the SKL, with a static load rating of 17,200 N, the required minimum load is greater than 302 N, which has been rounded up to 400 N. For the RRKL, with a static load rating of 10,300 N and a Y value of 2.0, the required minimum load is greater than 51.5 N, which has been rounded up to 80 N.

Table 10: Minimum loads in [N] of the test bearings according to the catalog method [SCHAE18]

Load in [N]	Speed in rpm				
Rolling bearing type	500	2,000	3,000	10,000	18,000
RRKL 6007	80	80	80	80	80
SKL 7007	400	400	400	400	400

The run-in is important for the formation of a tribological boundary layer in rolling element-raceway contact. In the experimental investigations on rolling work, the minimum load was used at all times. Starting from the minimum load, an increase in load leads to a significant widening of the contact area. In ball bearings, this changes the raceway in the groove radius. Since the rolling bearings are reused, no load increase above the specified load (see 2.4) was carried out. A running-in effect, such as a permanent drastic drop in friction torque at constant test parameters, could not be observed in preliminary tests below the minimum load.

Many roller bearing manufacturers recommend a grease distribution run, especially for spindle bearings [SKF14-2, SCHAE18-2, HQW18]. Each test procedure is characterized by a speed step test, although there are various differences. Some recommend start-stop operation to prevent bearing damage. This is intended to allow the temperature to equalize or the rolling bearing temperature to drop. However, if the bearing back temperature is known and does not exceed a critical value, a shortened procedure can be used (see SKF14-2 p. 111 ff.). In this case, a speed step test can be carried out without interruptions. The end of each speed step is reached when the temperature remains constant. Therefore, Test programs at

Rolling bearing test benches should always be based on grease distribution runs and thus on speed stage tests, as this is more realistic.

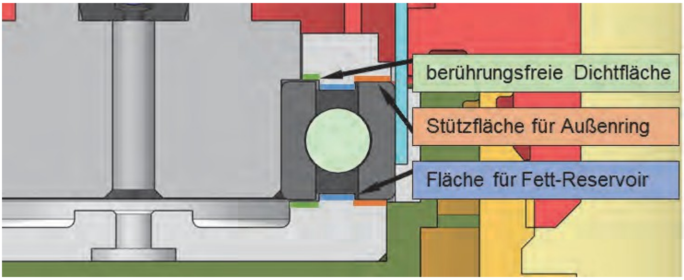
Preliminary tests were conducted to determine the appropriate test duration for investigating the kneading action. In order to clarify the influence of speed on the kneading action, several speed stages were run. The respective holding times for each speed stage were determined based on the findings of preliminary tests. It was important to find the maximum required grease distribution time for both types of roller bearings and both test greases so that the stress duration of the roller bearing greases remained comparable. The holding time is also based on the corresponding test temperature. This results in the following test procedure for both test bearings on the rotation tribometer:

Table 11: Test procedures for the speed stage test on the rotational tribometer

Test temperature	Duration	Rotational speed [rpm]						Σ
		500	1,000	1,500	2,000	2,500	3,000	
Persistence temperature	[h]	5	5	5	5	5	5	**30**
Tempered to 80 °C	[h]	2	2	2	2	2	2	**12**

The geometry of the seals facing the bearing was designed identically for all test benches (see Fig. 20). The side facing the bearing has three flat surfaces (see Fig. 20, red, blue, and green areas). The outer ring rests on the red area at the front. The blue surface is located, like a permanently installed seal, in the geometry of the bearing, near the cage. The green surface is located opposite the front side of the inner ring, with a very narrow gap.

Fig. 20: Sealing concept using the example of the RRKL in the rotation tribometer



When quantifying grease leakage, the grease that is on the blue area after the test is not considered leakage. This amount of grease remained until

Finally, in the test of the internal geometry of the bearing as a reservoir. The quantity was removed during evaluation and the remaining quantity was weighed on the seal. With regard to the different test conditions in the various test benches, there were minor differences in implementation here. For example, it was structurally possible to evaluate the grease leakage in the rotational tribometer exclusively by weighing the seals. On the high-speed test bench, both seals with the greased roller bearing had to be weighed before and after the test, as some of the leakage occurred beyond the seal.

Several options for determining the amount of grease in the roller bearing were explained in Chapter 2.2. When selecting the amount of grease, the geometry of the non-contact seals must now also be taken into account. These are based on the sealed versions of the respective roller bearings. Since the RRKL is an unsealed variant, the free space is more significantly restricted by the use of these seals than in the SKL. In order to establish a basis for comparison in terms of both speed and the two bearing types, a uniform grease quantity of 2.1 g was selected from the available information. Based on this amount, a reduction was made in three stages: 1.8, 1.5, and 1.0

g. In Table 12, the quantities are also given as a percentage by volume, as this percentage is more familiar to users.

Table 12: Lubricant quantities for the RRKL 6007 and SKL 7007 test bearings

Rolling bearing type	Specification	Unit	Quantities			
RRKL 6007	Selected grease quantities	[g	2.1	1.8	1.5	1.0
		[Vol. %]	36	31	26	17
SKL 7007	Selected grease quantities	[g]	2.1	1.8	1.5	1.0
		[Vol. %]	32	27	23	15

Grease leakage is of great importance for the evaluation of rolling bearing tests. Numerous preliminary tests showed that grease leakage could be minimized by improving the sealing gap on the inner ring. Specifically, this means that in the case of the rotation tribometer and the special tribometer, there was no grease leakage in the majority of tests. The amounts of grease that leaked in individual tests were less than 4% of the original amount of grease applied. It can be assumed that this is related to both the seal geometry and the relatively low speed of max.

3,000 rpm. On the high-speed test bench, grease leakage was sometimes greater, which is why this topic is dealt with separately in 4.4.

4.3 Rotational tribometer

The most comprehensive roller bearing tests were carried out on the rotational tribometer. All tests were performed at a maximum speed of 3,000 rpm. This corresponds to an n·dm value of 145,500. The results for both ball bearings at the endurance temperature and at the test temperature of 80°C are explained below. The amounts of grease and oil were varied. Statistically verified values are available for the highest grease quantity level in each case (see Table 12), which were determined by at least five tests.

4.3.1 Tests without temperature control

Fig. 21 shows two example tests with the RRKL 6007 and both test greases at the holding temperature (BT). In accordance with the step test, each new speed level results in a new friction torque and temperature level.

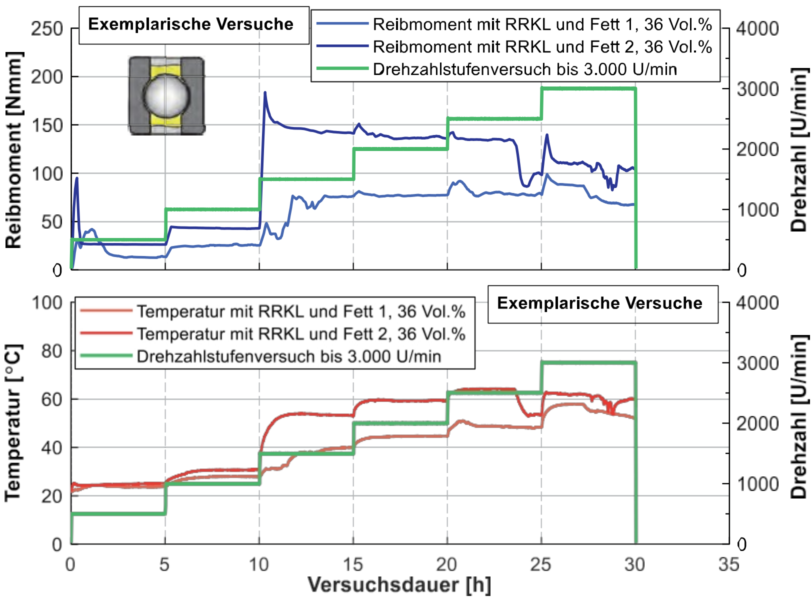


Fig. 21: Roller bearing test with the RRKL 6007 on the rotational tribometer in a speed step test up to 3,000 rpm with grease 1 and grease 2 (top: friction torque curves, bottom: temperature development on the outer ring)

In the first speed stage, the friction torque rises more quickly with grease 2 and is also shorter than with grease 1. This shows a fundamental distribution behavior: the initial grease distribution with grease 2 is significantly faster than with grease 1. Based on the friction torque and temperature, it can be seen that the distribution of grease 1 takes more than three hours. Overall, persistence is achieved relatively quickly in the first stages.

When starting at the speed level of 1,500 rpm, there is a significant increase in the friction torque level and, consequently, also in the temperature. As the test progresses, the friction torque level remains almost constant for grease 1. With grease 2, the friction torque decreases at the last two speed levels. As a result, the friction torque curves of both greases converge slightly.

Fig. 22 shows examples of the results of the speed stage test with SKL 7007 and both greases. With grease 2, there is a comparatively high friction torque at the start of the test. This level decreases more slowly than in the RRKL test described above.

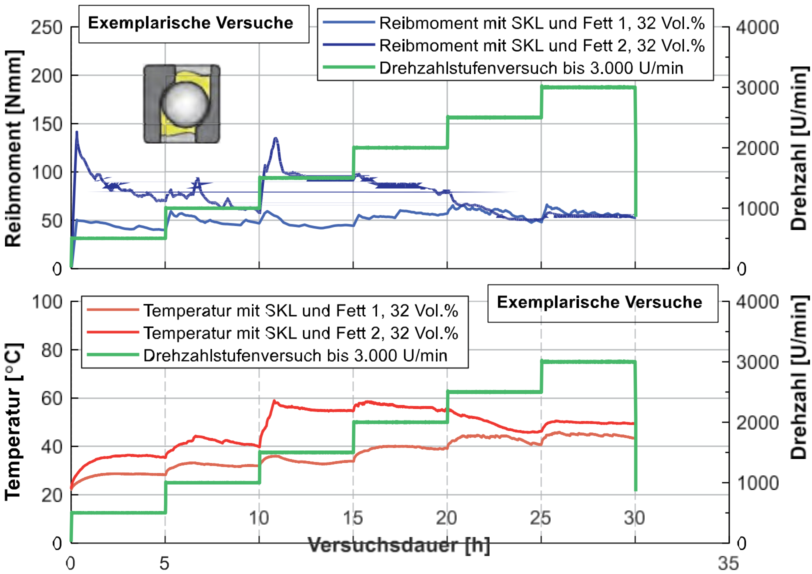


Fig. 22: Roller bearing test with the SKL 7007 on the rotational tribometer in the speed step test up to 3,000 rpm with grease 1 and grease 2 (top: friction torque curves, bottom: temperature development on the outer ring)

The difference between the two greases is also more clearly visible at 500 rpm. This reveals the first differences in friction torque behavior, which are influenced by kinematics and lubricant properties. The significant increase when starting up to the speed stage of 1500 rpm is only present here with grease 2. With grease 1, the level remains almost constant.

The friction torque values derived from this correspond to the value at the end of the respective speed stage. Table 13 lists these friction torque values for both test bearings and both test greases. These values can be used to express the change in friction torque relative to the previous speed stage as a percentage. The friction torque development for the RRKL shows a particularly large increase at one point in the speed stage test. When starting the speed stage of 1,500 rpm, there is an increase of +200% for grease 1 and an increase of +236% for grease 2. This jump is therefore similar for both greases. In comparison, the change for SKL and grease 1 is -2% and for grease 2 +55%. Based on the comparison, this appears to be a rolling bearing-specific effect that comes into play in combination with the rolling bearing grease.

Table 13: Friction torques of both test bearings and test greases at each speed level based on the exemplary tests from Fig. 21 and Fig. 22

Test bearing	Test grease	Base oil viscosity	Unit	Speed level in [rpm]					
				500	1,000	1,500	2,000	2,500	3,000
RRKL 6007	Bold 1	ν_{40}= 49 cSt	[Nmm]	14	25	75	77	76	66
			[%]	-	+79	+200	-3	-1	-13
	Fat 2	ν_{40} = 249 cSt	[Nmm]	26	42	141	136	99	104
			[%]	-	+62	+236	-4	-27	+5
SKL 7007	Fat 1	ν_{40} = 49 cSt	[Nmm]	40	46	45	56	48	53
			[%]	-	+15	-2	+24	-14	+10
	Fat 2	ν_{40} = 249 cSt	[Nmm]	70	60	93	78	51	54
			[%]	-	-14	+55	-16	-35	+6

The exemplary tests presented here without temperature control provide an overview of the different behavior of the test configurations consisting of two roller bearing types and two test greases in the speed step test. The increase in friction torque is significant with the RRKL 6007 and both greases. In order to better understand this effect, various investigations are carried out with this test configuration in the further course of the test. These are carried out on the rotation tribometer, the special tribometer, and the high-speed test bench.

4.3.1.1 Repeat tests

During the repeat tests, seven tests were carried out with both greases for the RRKL and five tests for the SKL. Each individual test lasted 30 hours, with each speed stage being maintained for five hours (Table 11). The results of the repeat tests are shown below based on the mean value and the respective standard deviation.

It is clear that in the case of RRKL 6007 (Fig. 23, above) in conjunction with both test greases, there is a significant increase in the friction torque when starting at the speed level of 1,500 rpm. This level represents the global maximum. After this peak value in the friction torque, the friction torque level tends to drop at higher speeds.

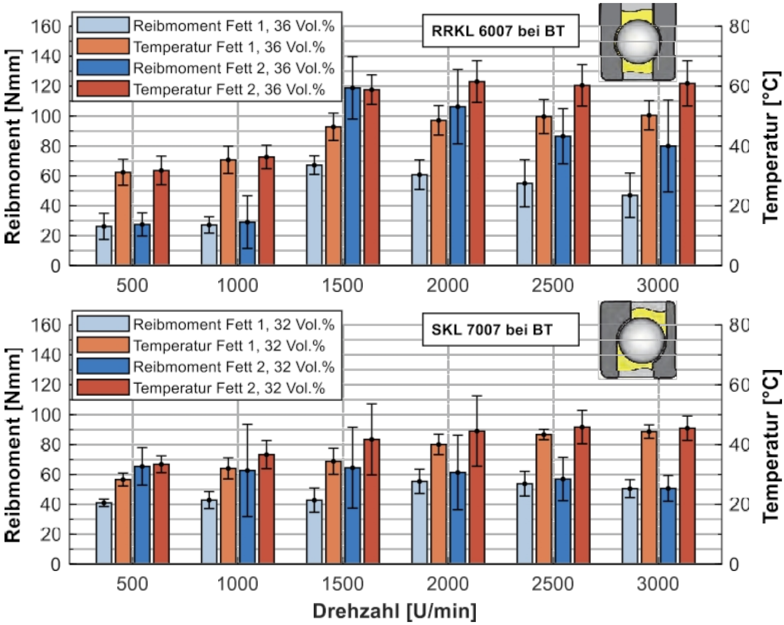


Fig. 23: Repeat tests at persistence temperature on the rotation tribometer with RRKL (top) and SKL (bottom), each with grease 1 and grease 2

In the SKL 7007 (Fig. 23, bottom), the friction torque curve is not characterized by such sharp increases. On the one hand, the friction torque and temperature levels are lower. On the other hand, the changes over the speed step test are rather moderate. However, with SKL and grease 1, there is a very reproducible jump when changing from 1,500 to 2,000 rpm. With SKL and grease 2, the standard deviations are often very high. Until recently, this fact could be attributed to random increases in friction torque at different speed levels. This is a short-term effect that normalizes in terms of friction torque within the five-hour holding period. The temperature increase resulting from the increased friction leads to reduced viscosity. This is why there is a lower friction torque in the subsequent stage. These are probably transient distribution effects.

4.3.1.2 e determination of the kneading work

The work of the walk is determined by differential measurements. Both grease quantities were varied and oil tests were carried out. The friction torque curves obtained were compared by calculating the difference. The assumption is that the difference between the two curves must result from the different grease quantities and thus represent the bearing losses during grease lubrication. These bearing losses are attributed here to the work of the walk.

To explain this procedure, the difference is visualized in Fig. 24 using sample test results from the rotational tribometer. The tests correspond to the grease quantity variations mentioned above. Fig. 24 shows the friction torque curves of two speed stage tests with the RRKL and grease 1 at the top. On the one hand, the selected comparison quantity of 36 vol.% (corresponding to 2.1 g) is shown in orange. On the other hand, the grease quantity of 32 vol.% (corresponding to 1.87 g) is shown in light yellow. The quantity of 1.87 g describes the lower limit of the range specified by the manufacturer (see 2.3). The reduction in the amount of grease results in a lower friction torque in many areas of the speed step test.

The difference between the areas was calculated in the diagram below. This is also given in Nmm and is interpreted here as the friction torque resulting from the walking action. The power can be derived using the speed and the friction torque. This makes it possible to reduce this power loss (bearing losses) by, for example, reducing the amount of lubricant used. The measured temperatures (below) correlate with the measured friction torques (above).

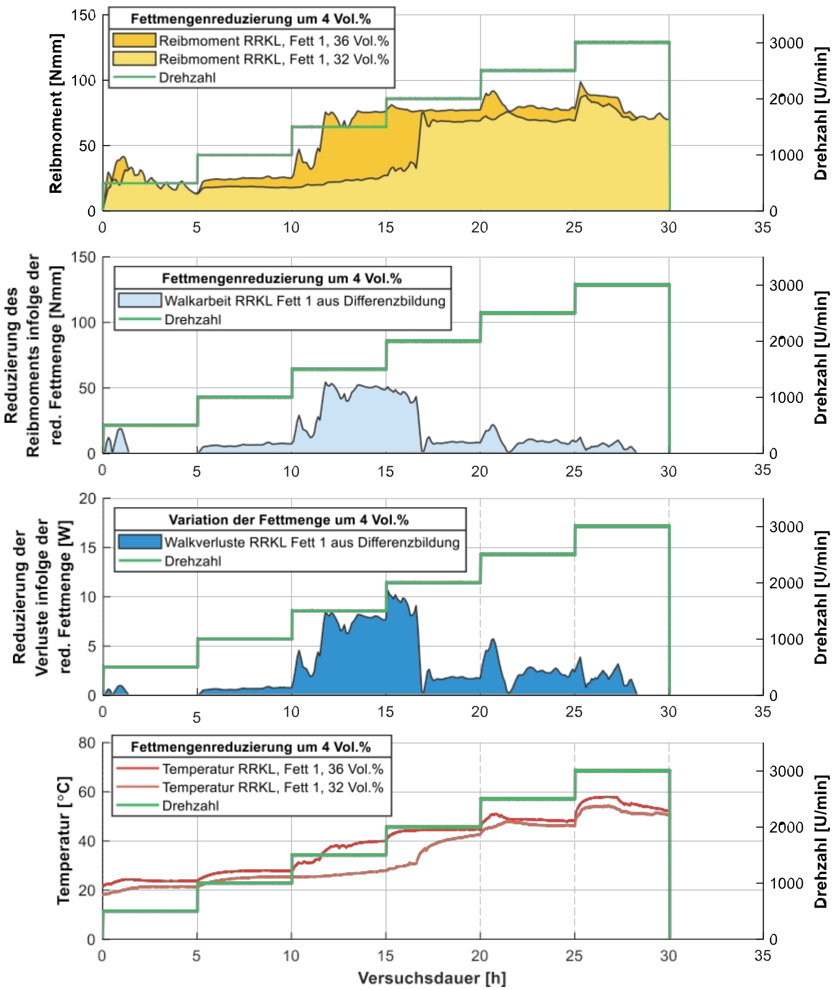


Fig. 24: Determination of the work done by difference measurements: top: Friction torque for RRKL and grease 1 with 36 vol.% and 32 vol.% respectively in the speed step test on the rotation tribometer up to 3,000 rpm, below: friction torque due to the fulling work by difference of both friction torque curves, below: the calculated power, below: temperature curves

Fig. 24 illustrates the basic test procedure. Differences in friction torque were referred to here as friction torque changes due to the reduction in grease quantity. In the further course of the work, this difference is interpreted as fulling work. In addition, this showed how the

Storage losses or power losses from the fulling process can be calculated. Since this is merely a conversion, the focus will be on the fulling process and not on the fulling losses. The example in Fig. 24 illustrates the potential of the following investigations based on this test methodology.

4.3.1.3 Lubricant quantity variation

As has already been shown, a variation in the amount of lubricant can lead to a reduction in the friction torque and, as a result, in power loss (see 4.3.1.2). The variation in the amount of lubricant focused on two areas: the variation in the amount of grease and the variation in the amount of oil. The variation in oil quantity was used to investigate whether losses could occur during oil lubrication (e.g., splash losses). During oil lubrication, minimal quantities were introduced into the roller bearing. A standard disposable syringe was used to introduce the lubricant, whereby each rolling element was wetted with oil. The weight of the lubricant quantity was determined using a precision scale. The comparison point is formed from the statistically evaluated tests at the persistence temperature. The grease quantity corresponds to 36 vol.% for RRKL 6007 and 32 vol.% for SKL 7007 (see Table 14). The respective base oil was used in the oil tests (in short: "Oil 1" for Grease 1 and "Oil 2" for Grease 2).

Table 14: Reference quantity of grease and variation in oil quantity for RRKL and SKL

Bearing type	Unit	Reference quantity of lubricating grease	Oil quantity		
RRKL 6007	[g	2.1	0.3	0.2	0.1
	[Vol. %]	36	5.1	3.4	1.7
SKL7007	[g]	2.1	0.3	0.2	0.1
	[Vol. %]	32	4.6	3.0	1.5

Fig. 25 shows the results of comparing different oil quantities for grease lubrication using the example of the RRKL 6007 with grease 1. The friction torque curves are shown at the top and the temperature curves in the middle. The oil tests were carried out with a shortened test duration of 30 minutes per speed stage. In the cases of minimum quantity lubrication, a steady state was quickly reached. In order to be able to compare the results of the grease tests (with 30 hours) with those of the oil tests (with 6 hours), the oil tests were scaled in terms of time. The ordinate therefore represents the relative test duration of a total of 30 hours.

according to the grease tests. It is to be expected that a reduction in oil quantity will also lead to a reduced friction torque.

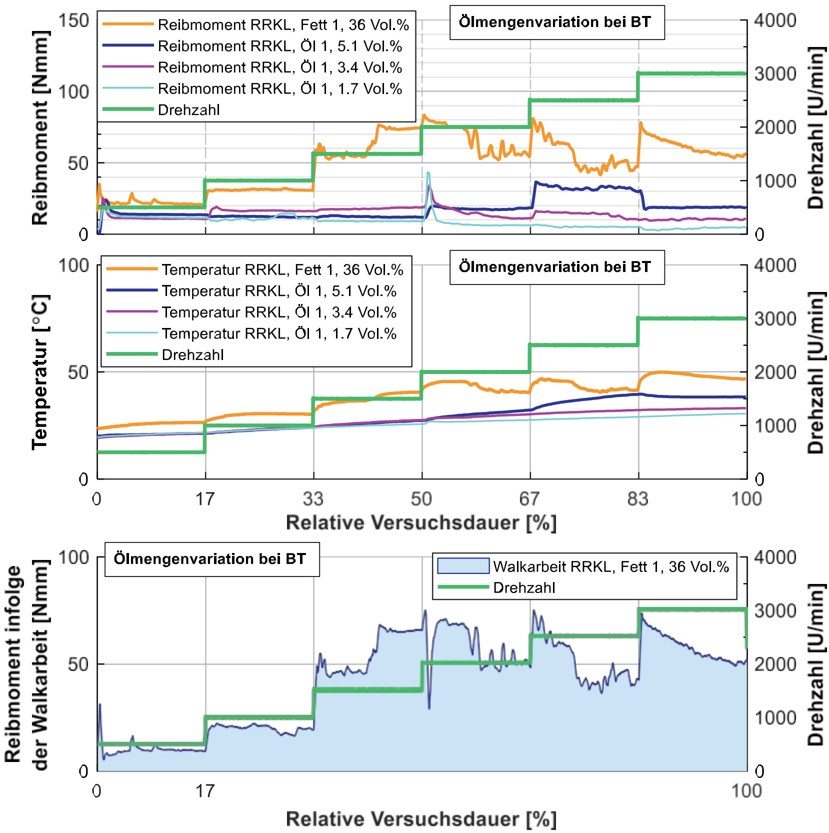


Fig. 25: Oil quantity variation at steady-state temperature, top: friction torque and temperature of grease lubrication (grease test = 30 hours) and different oil quantities (oil test = 6 hours), bottom: difference between grease-related and oil-related (minimum quantity) friction for deriving the work done

Even though the varied oil quantities differ only marginally, there are trends here that are in line with expectations. This is confirmed by the measurement results for friction torque and temperature alike. This shows that, in the RRKL, the quantity of

1.7 vol.% produces the lowest possible friction in many areas of the speed stage test

. With an oil quantity below 1.7 vol.%, higher friction torques were determined at many speed stages.

It is assumed that conditions of insufficient lubrication occur here, leading to more frequent mixed friction. This may also be due to the type of preparation of the test bearings. In order to achieve small quantities, it was not possible to coat every rolling element with oil at less than 1.7 vol.%. Therefore, a further reduction was not considered. As a result of the analysis, the minimum quantity of 1.7 vol.% was set as a comparative measure in order to derive the differences between grease-induced friction and friction during oil lubrication.

The friction determined from the difference is interpreted below as work done by friction and is illustrated in Fig. 25 below as an example for the RRKL with grease 1. The difference shows that the reproducible drastic increases in friction torque can only be observed with grease lubrication and can therefore be attributed to work done by friction. A drastic increase in friction torque when reaching a new speed level usually only occurs very briefly with oils (see Fig. 25 at 2,000 rpm).

By calculating the difference between the determined minimum quantity lubrication with oil and the corresponding grease lubrication, the kneading work can be derived for all bearings and all test greases. The results of all the differences determined are shown in Fig. 26. This reveals various findings. On the one hand, the friction torque appears to depend heavily on the speed as a result of the work of walking. A clear dependence on speed is already evident from the differences determined in Fig. 25.

The work of kneading increases significantly from a speed of 1,500 rpm. There is also a dependency on the lubricating grease. This can be seen from the comparison of grease 1 and grease 2 in the RRKL (Fig. 26, above). Here, grease 2 delivers a higher friction torque at all speeds as a result of the work of kneading. The conditions are similar to those for grease 1. Further insights can be derived from the results of the SKL (friction torque due to work of kneading, see Fig. 26, bottom). It becomes clear that in areas of the speed stage test, the work of kneading is higher for grease 1 than for grease 2. The work of kneading is therefore also dependent on the rolling bearing kinematics. Only above 2,000 rpm does grease 2 deliver higher work of kneading. Rheology is therefore only one of several factors. The extent to which the specific rolling bearing kinematics influence the shearing of the grease and thus the work of kneading remains to be clarified.

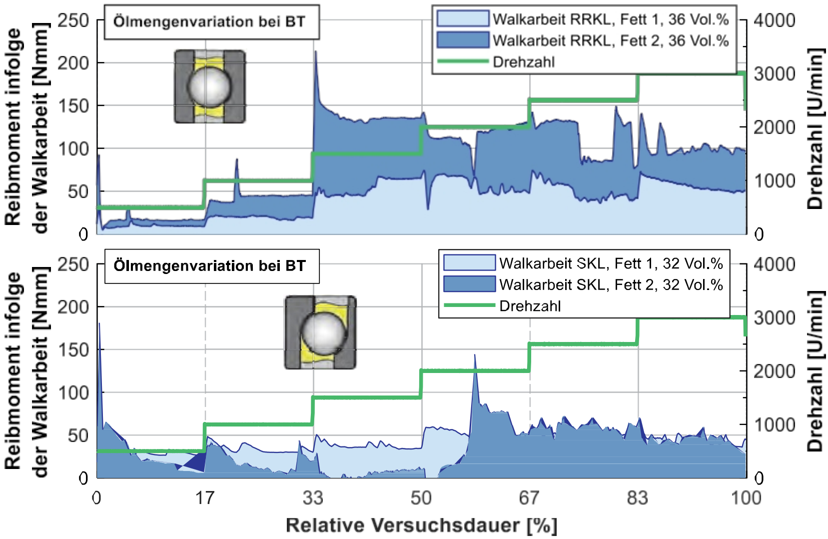


Fig. 26: Friction torque resulting from the walking process, calculated from the difference between the minimum oil quantity and grease lubrication: top: RRKL 6007 with 36 vol.% grease and 1.7 vol.% oil, bottom: SKL 7007 with 32 vol.% grease and 1.5 vol.% oil

In addition to varying the oil quantities, the grease quantities were also changed. Here, too, the quantity was reduced in three stages. The different measurement results were also compared with the usual maximum grease quantity. In contrast to the oil tests, the holding time for each speed stage is five hours. Fig. 27 shows the results of the grease quantity variation. The friction torque measurements are shown above and the temperature measurements from the speed level test using RRKL and grease 2 are shown below. Here, too, the friction torque behavior is as expected. A reduction in the amount of grease results in a reduction in friction torque at many operating points. The lower two stages are very similar in terms of friction torque behavior. The respective lowest amount of grease is used for the comparison. The differences for both roller bearings and both test greases are shown in Fig. 27 below.

With RRKL 6007, grease 2 causes greater friction torque than grease 1 at all speeds. In the first speed stage, no kneading work can be quantified for grease 1 due to a variation in the amount of lubricating grease. Therefore, with regard to practical grease quantity variations, it can be argued that kneading work is unavoidable in such a case. The same applies to SKL 7007 and grease 1 at 500 rpm.

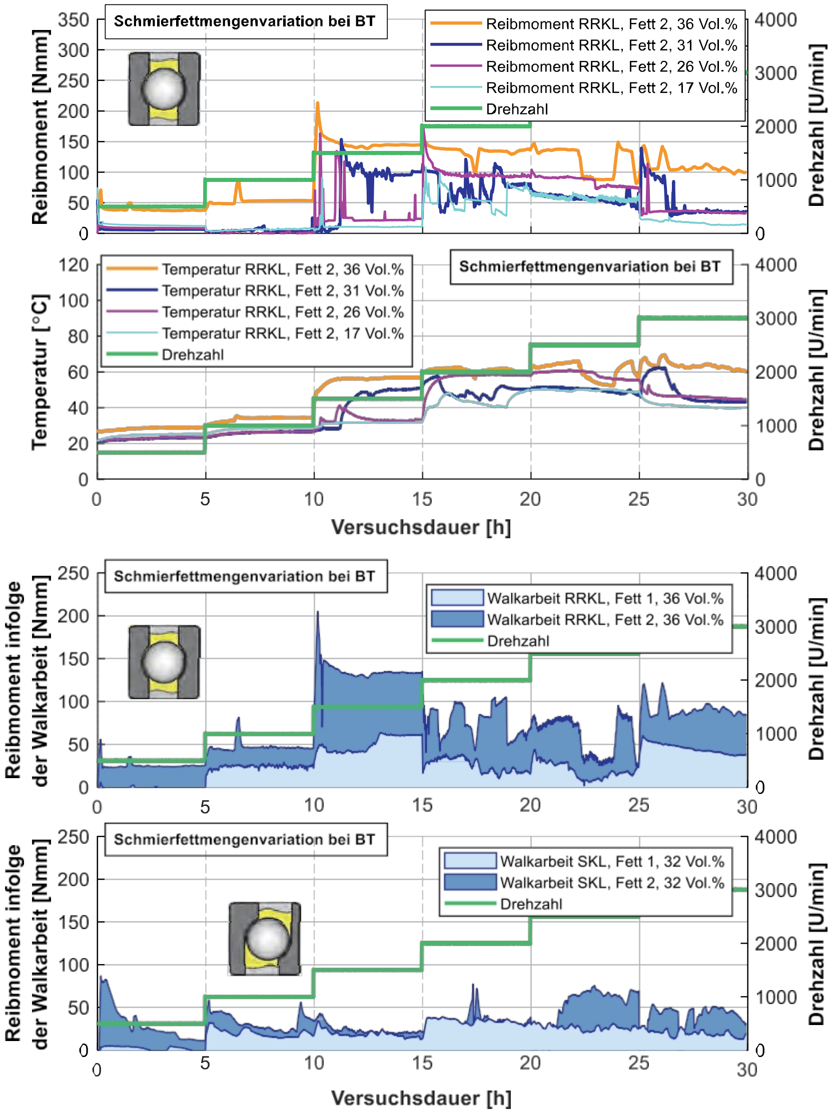


Fig. 27: Lubricating grease quantity variation at persistence temperature (BT) for the RRKL with grease 2 (top) and the derived friction torque due to kneading work from the difference between the maximum and minimum grease quantities for the RRKL and the SKL for both greases (bottom)

In the case of SKL, the kneading work appears to be comparable in many areas due to the different test greases. Grease 2 causes a higher friction torque. In contrast, the derivation of the flexing work from the oil tests painted a different picture. Here, SKL and some speed levels caused higher flexing work with grease 1.

In addition, there was a significant increase in scouring behavior with grease 2 at speeds above 2,000 rpm. The friction torque levels of grease 1 and grease 2 are very similar. Later, there is a comparison of friction torques from maximum and minimum lubrication and the oil lubrication also explained here.

4.3.1.4 Increase in friction torque with RRKL 6007

In order to gain a better understanding of the large jump in friction torque when starting at the speed level of 1,500 rpm, tests were carried out with the RRKL 6007 and a modified speed level test. The modification involves slowly approaching and passing through the critical speed level. The focus was on the test configuration with grease 2, as the effect was most pronounced here. The results of a single test and repeat tests are shown in Fig. 28 below.

In Fig. 28, the selected friction torque curve (top) shows transient states before the actual jump. Apparently, the first effects are already apparent here, but they are not stable. When a critical speed is reached, the effect of the increase is stable and thus stationary. Based on the adapted test procedure and the evaluation of the repeat tests, it was shown that the jump to the steady-state high friction torque level actually occurs between 1,300 rpm and 1,500 rpm. For the repeat tests, different roller bearings of the same type with the same amount of lubricant were used.

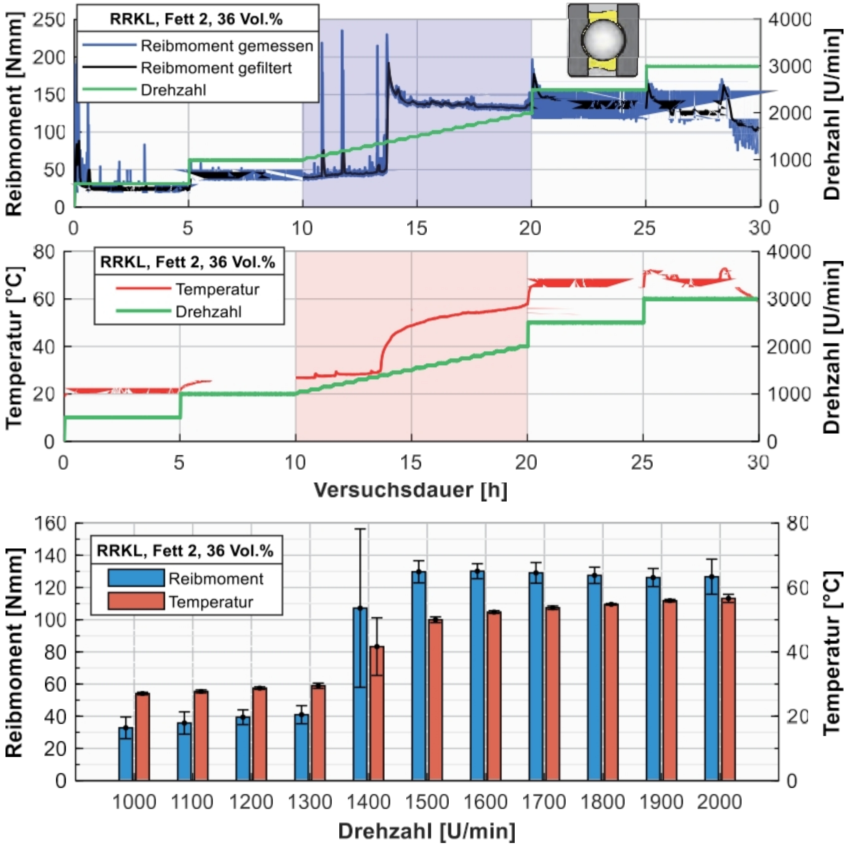


Fig. 28: RRKL 6007 with grease 2: adapted speed step test in the range of 1,000 - 2,000 rpm, top: friction torque and temperature of a single test, bottom: three repeat tests with identical step test

This behavior was later also demonstrated on other test benches. This is therefore clearly an effect resulting from a combination of lubricant rheology, specific rolling bearing kinematics, and speed. This allowed this effect to be clearly limited to a relatively narrow speed range, with initial manifestations already observable beforehand.

4.3.2 Temperature control tests (at 80°C and °C)

The temperature-controlled tests on the rotational tribometer included both repeat tests and tests with varying amounts of lubricant. The repeat tests were carried out with a lubricant quantity of 2.1 g (RRKL 36 vol.%, SKL 32 vol.%).

Furthermore, preliminary tests showed that the steady state is reached after two hours at the latest for each speed level. In the tests without temperature control, this state was reached after 5 hours for each speed level.

4.3.2.1 Repeat tests

Fig. 29 shows repeat tests with temperature control using the RRKL (top) and the SKL (bottom) with both grease 1 (pale colors) and grease 2 (strong colors). With the RRKL, there is no significant jump in the friction torque level at a certain speed for either grease 1 or grease 2 (Fig. 29, top). As expected, the overall level of friction torque is significantly lower. The increase in friction torque over the step test is relatively uniform. In both cases, there is a slight drop in the friction torque level at 3,000 rpm. This reduction is particularly noticeable with grease 2. The deviation in temperature clearly shows that, despite the control inertia associated with air temperature control, slight deviations in the outer ring temperature occur.

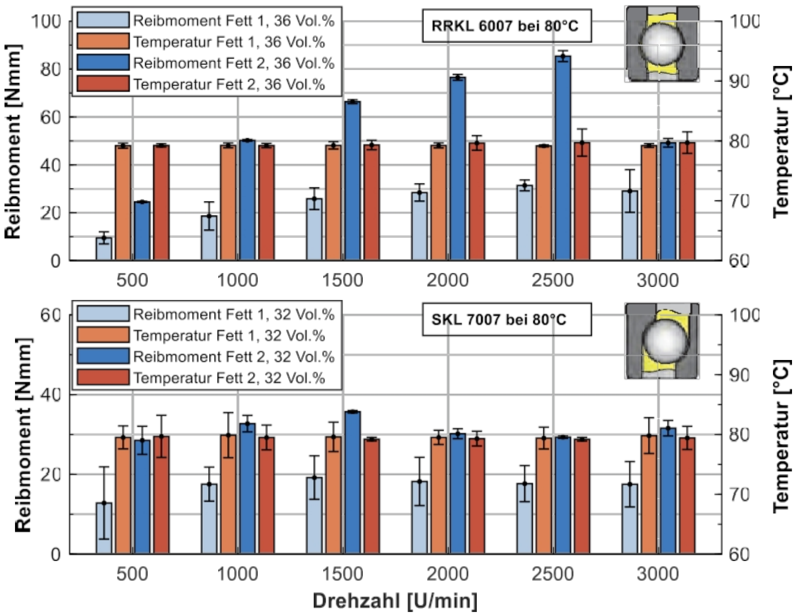


Fig. 29: Repeat tests at a test temperature of 80 °C with the RRKL 6007 (top) and the SKL 7007 (bottom), each with grease 1 and grease 2

In the SKL (Fig. 29, bottom), the friction torque for grease 1 and grease 2 is low throughout the entire test. The reproducibility is relatively good. Due to the overall very low friction torque levels, the differences compared to oil or smaller amounts of grease are difficult to quantify. For example, with both roller bearings and grease 1 at 500 rpm, a friction torque of less than 10 Nmm occurs in some cases. Overall, grease 2 causes a higher friction torque for both roller bearings and all six speed levels. The tests once again provide the basis for the subsequent comparisons of grease quantity reductions and oil tests.

4.3.2.2 Lubricant quantity variation

No variation in oil quantity was used in the temperature-controlled case, as the minimum oil quantity had already been determined at the persistence temperature for both bearing types. Consequently, an oil quantity of 1.7 g was used again. The oil tests were also carried out again with a shortened step duration of 30 minutes per speed, which was scaled in time for the graphical representation. Fig. 30 shows an example grease test with SKL and grease 1 compared with an oil test with oil 1. The grease tests are statistically validated.

The comparison reveals clear differences. With RRKL 6007 (Fig. 30, center), grease 2 again dominates the friction torque in large parts of the speed step test. Only at 3,000 rpm does the friction torque drop as a result of the flexing action. This friction torque behavior was observed in the grease tests at 80 °C and was reproducible here (Fig. 29). At 500 rpm, a friction torque due to the flexing action cannot be quantified for grease 1 based on oil tests.

In contrast, with SKL 7007 (Fig. 30, below), changes in friction torque can still be observed up to 1,000 rpm. In many areas of the speed step test, the work done by grease 1 is hardly quantifiable. This also applies in part to grease 2. With SKL 7007 and a test temperature of 80 °C, there are therefore areas of the step test in which oil lubrication and grease lubrication produce the same friction torque.

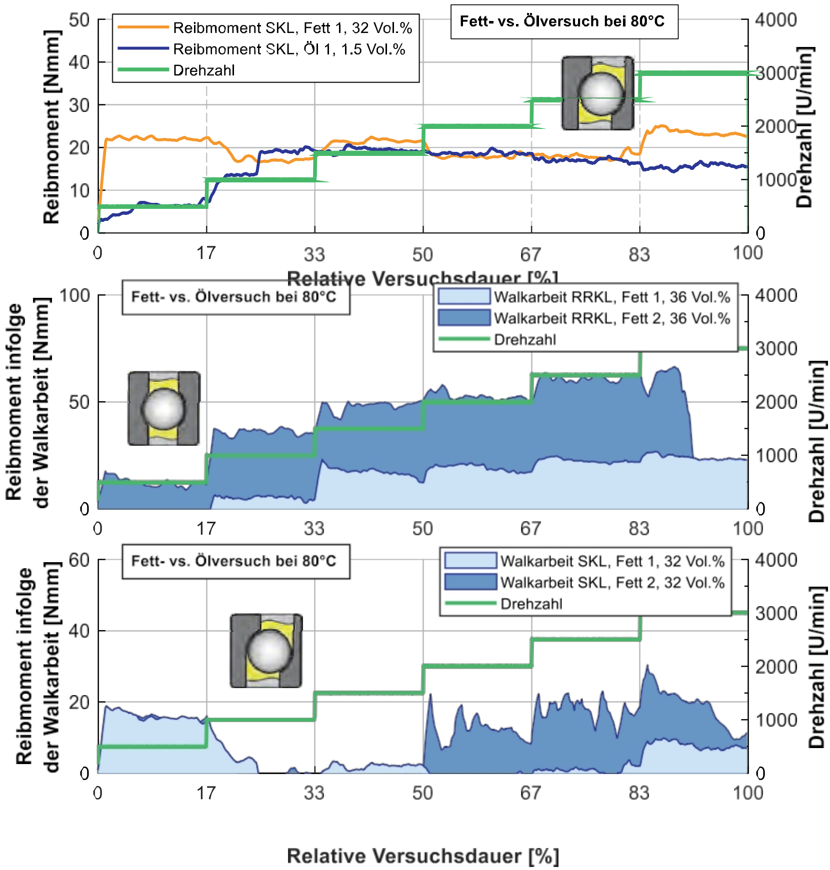


Fig. 30: Grease versus oil test at 80 °C, top: test with SKL 7007 and grease 1 (grease test = 12 hours) and oil 1 (oil test = 6 hours), bottom: formation of the kneading work based on the difference between the grease and oil tests

Although the level of friction torque is very low due to the test temperature, a gradual reduction in the amount of grease has a measurable effect on the friction torque. These differences in friction torque are clearly visible in the example at the end of the test (Fig. 31, top). The work done as a result of the reduction in grease quantity indicates a behavior that could already be observed in the tests at the persistence temperature.

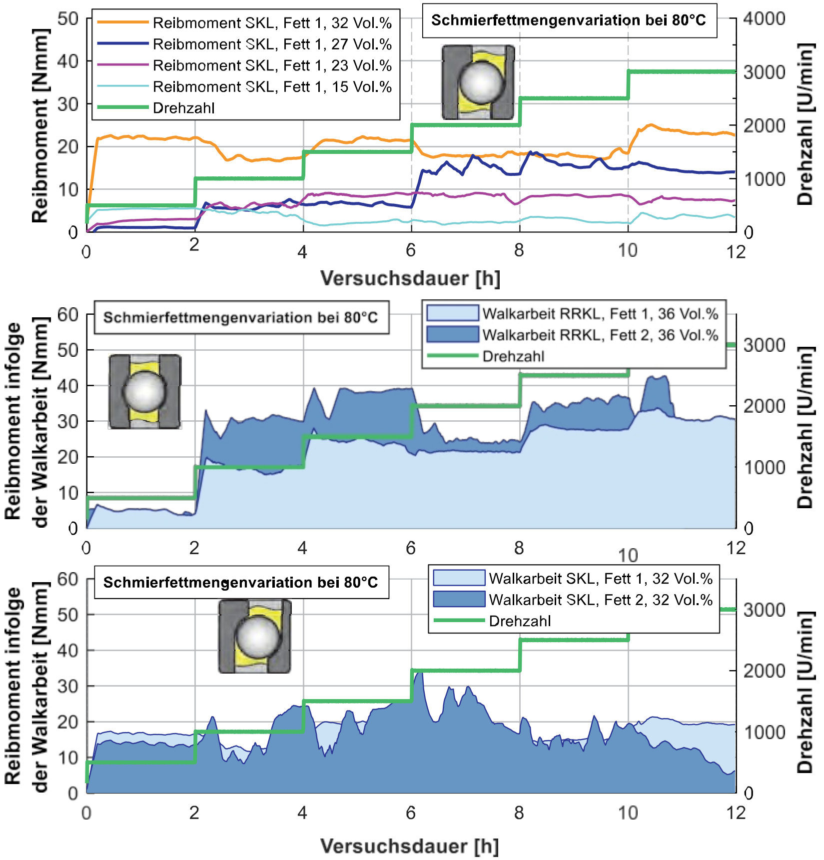


Fig. 31: Grease quantity variation at 80 °C, top: example test with SKL and grease 2, bottom: formation of work of kneading based on the difference between maximum and minimum grease quantity

In RRKL 6007, grease 2 delivers the greater friction torque in large parts of the speed stage test, whereas in SKL 7007, grease 1 causes the higher friction torque in parts of the test. Both the friction torque levels and the differences between the two greases are very small.

4.3.3 's work from all tests

Section 4.3.1.2 graphically illustrates how the friction torque resulting from the work of kneading is determined based on difference measurements. Two methods were used: firstly, the reduction in grease quantity and, secondly, the comparison with base oil tests using a minimum quantity of oil.

In order to bring together all the findings on work of rolling to date, these are presented in a compact form as bar charts in Fig. 32. Each bar chart shows the friction torques determined in the tests for a rolling bearing type and a test grease on the y-axis on the left. The speed levels are plotted on the x-axis. Each diagram contains friction torques from tests with the largest amount of grease (green), the smallest amount of grease (light blue), and the smallest amount of oil (dark blue). This comparison is followed by the calculation of the respective difference to derive the friction torque resulting from the kneading process. These bars have a dashed edge and are read on the y-axis on the right. Either the difference to the oil test (red) or to the grease test with a reduced amount (orange) is calculated. In contrast to previous representations, only the inertia torque at the end of a speed level is compared.

Fig. 32 shows the results for RRKL and both test greases in the two upper diagrams. Statistical analysis of the grease tests (MG 36 vol.%) again clearly shows a significant increase when the speed level of

1,500 rpm. This increase does not appear to occur with oil lubrication (oil 1.7 vol.%) or with a reduced amount of grease (grease 17 vol.%). However, with a reduced amount of grease, an increase in the friction torque is recorded when the speed jumps to 2,000 rpm, although this is now significantly lower than with the usual amount of grease (grease 36 vol.%). In all tests with grease lubrication, the friction torque level then decreases gradually up to the maximum speed. This trend is evident with both test greases and both grease quantities shown. In the oil-lubricated system, the friction torque remains constant up to

3,000 rpm consistently low. Thus, the sharp increase in friction torque observed here can clearly be attributed to the kneading action. These results show that there are also operating points with regard to speed that react differently to the amount of lubricating grease present. For example, the test with 17% grease by volume and the test with 1.7% oil by volume have a similar friction torque level up to a speed of 1,500 rpm.

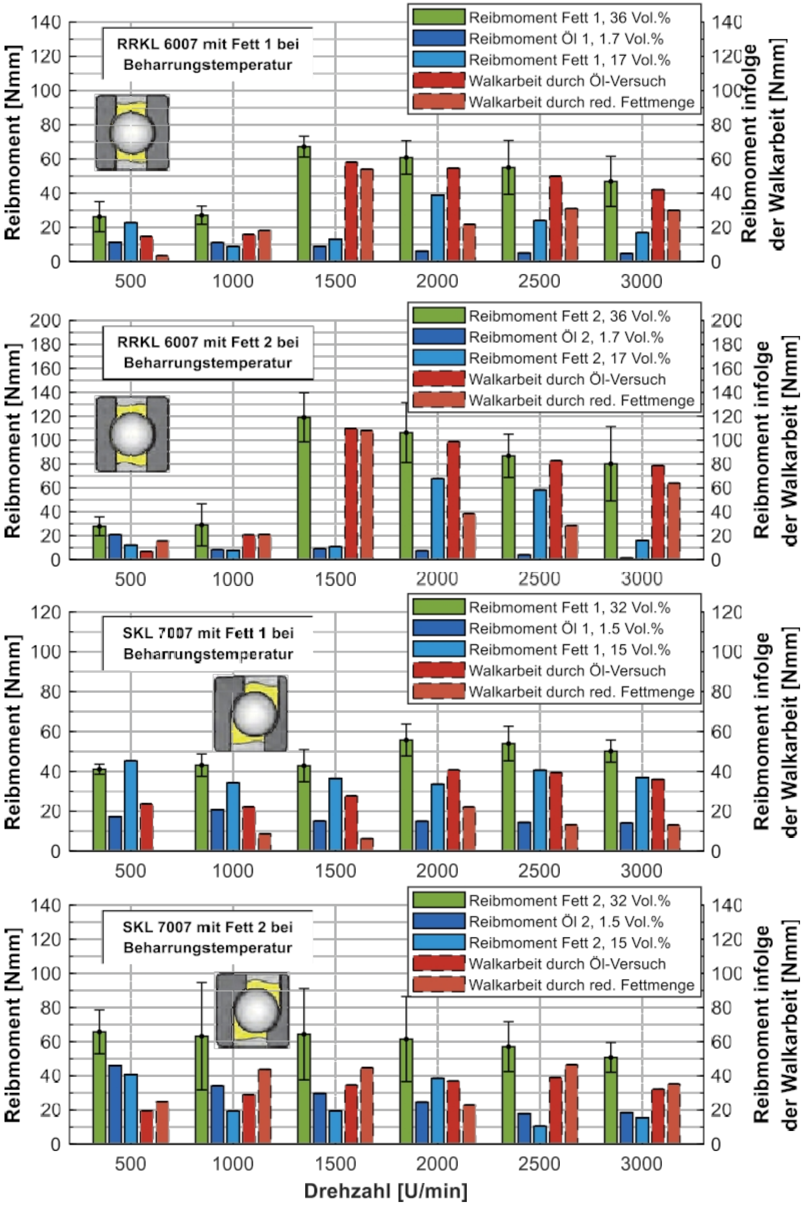


Fig. 32: Friction torque due to kneading at steady-state temperature in the speed stage test up to 3,000 rpm with RRKL and SKL, each with grease 1 and grease 2 and their base oils (oil 1 and oil 2)

Thus, the sharp increase can only be avoided by reducing the amount of grease at 1,500 rpm. However, the increase is already clearly visible at 2,000 rpm, even with the lower amount of grease. The cause of this can be assumed to be a conveying mechanism resulting from the roller bearing design. As the speed increases, so do the centrifugal forces, which cause the grease to flow at a certain point. The design of the RRKL may promote centrifugal force-induced delivery into the outer ring groove. At 2,000 rpm, an increase in friction torque due to the kneading action is apparently unavoidable. Only the amount of kneading can be influenced here by the amount of lubricating grease.

In the SKL (Fig. 32, bottom), oil lubrication also results in a lower friction torque throughout the entire step test compared to the standard grease quantity of 32% by volume. With grease 1, reducing the amount of grease does not have as significant an effect as with grease 2. Here, the lowest friction torque is often achieved with the reduced amount of grease (grease 2, 15 vol.%). This means that the kneading work with grease 2, which can be saved by reducing the amount of grease, is apparently greater than with grease 1.

Fig. 33 summarizes the results of the tests at 80 °C in the same way. Due to the test temperature, there are clear differences between the RRKL and the results at the holding temperature (Fig. 32). The 80 °C tests (light blue) clearly show that the enormous increase when starting the third speed stage is no longer present. Instead, the increases from stage to stage are uniform. With grease 1, the use of the base oil (oil 1, 1.7 vol.%) tends to produce rather small differences compared to grease lubrication (grease 1, 36 vol.%). The reduced amount of grease, on the other hand, causes the greatest difference in the long term. The situation is different again with grease 2. Here, minimum quantity lubrication with oil produces the lowest friction torque and thus illustrates the greatest amount of kneading work.

In the SKL, the 80 °C tests (light blue) clearly show that the friction torque changes very little as a result of the speed change up to 3,000 rpm compared to the RRKL. The friction torque level is very low. In addition, with grease 1, the work done by friction is almost impossible to quantify due to differences compared to oil lubrication. In contrast, the reduction in grease quantity (15 vol.%) offers a way of evaluating the work done in this test configuration. However, the derived work done is very low at a maximum of 40 Nmm. The situation is similar with grease 2. Here, too, there are only a few operating points where oil lubrication causes a significantly lower friction torque.

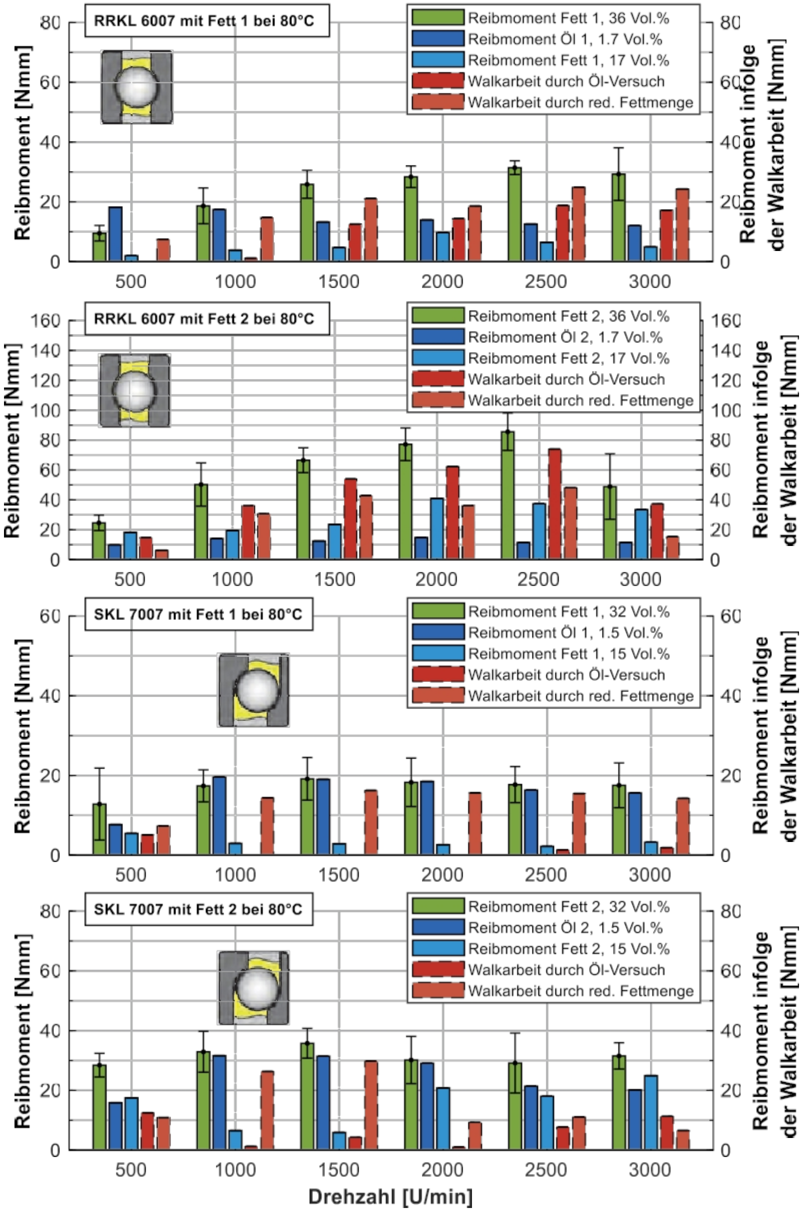


Fig. 33: Friction torque due to work done at the practical temperature of 80 °C in a speed stage test up to 3,000 rpm with RRKL and SKL, each with grease 1 and grease 2 and their base oils (oil 1 and oil 2)

Each method is capable of determining a minimum of friction in the rolling bearing under certain test configurations. From this, the maximum possible degree of bearing losses with grease lubrication can be derived.

On the one hand, lubricating greases can cause unfavorable flexing. On the other hand, grease lubrication can also cause starvation effects, which are favorable for friction [FVA 388]. In the best case, this results in ideal minimum quantity lubrication, which is comparable to oil mist lubrication in terms of friction torque level. In tests on the rotational tribometer, both methods deliver the lowest friction torque at various operating points. Ultimately, it is not possible to assess why a small amount of oil in some cases and a smaller amount of grease in others results in lower friction.

In summary, both methods offer advantages for deriving the friction torque as a result of the kneading work. It is practical to use as little base oil as possible in the rolling bearing. A more practical and realistic target value is the comparison with grease quantity reductions. Ultimately, there is currently no standardized test method for quantifying kneading work in grease-lubricated rolling bearings. The measurements presented here illustrate two ways of determining kneading losses.

4.4 High-speed rolling bearing test bench

This chapter presents a test methodology for grease-lubricated rolling bearings at high speeds. Using this methodology, tests were carried out with a variation in grease quantity and a variation in lubrication at 80 °C. In these tests, a speed of 18,000 rpm was achieved and the shaft position was varied.

4.4.1 Test methodology for high-speed rolling bearings

In preliminary tests, a test methodology for high speeds was developed. A classic grease distribution run served as a model for this. The preliminary tests showed that at high speeds and a temperature of 80°C, the necessary persistence was not achieved even after a holding period of five hours. With the aim of conducting the tests within a reasonable total duration, a grease distribution run was preceded by the test run (see Fig. 34, green area "Run-in"). This involves reaching the maximum speed before the actual measurement is taken.

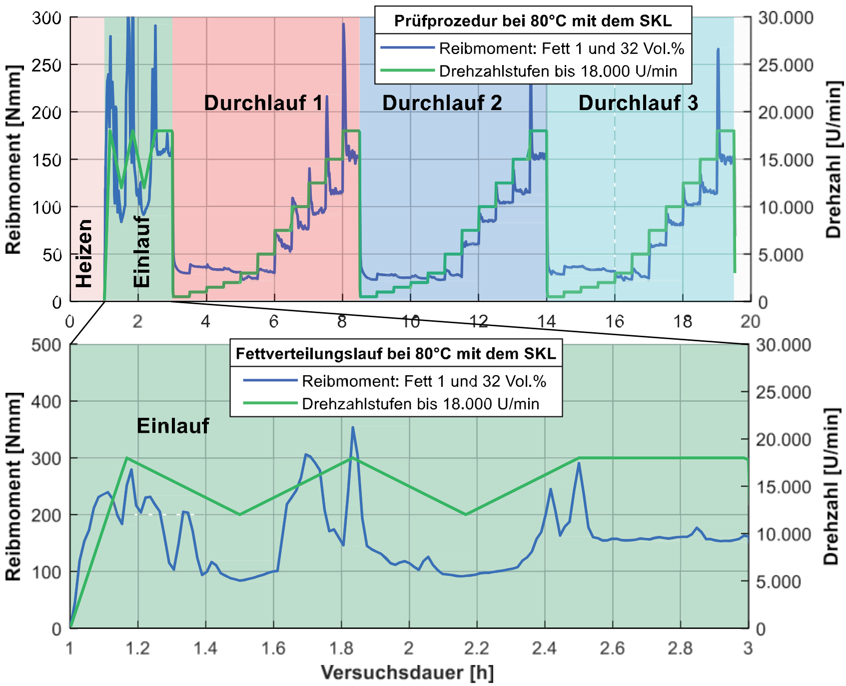


Fig. 34: Test procedure for grease-lubricated rolling bearings up to maximum speed 18,000 rpm with upstream grease distribution run (shown as "run-in") using the example of SKL 7007 with grease 1 at the highest fill level of 32 vol.% in horizontal shaft position with friction torque (blue) and speed (green)

The grease distribution run comprises six steps (see Table 15). It should be noted that the bearing back temperature is kept constant at 80 °C throughout the entire test, including the grease distribution run, by means of the cooling channels integrated in the test head. In the first step, the maximum speed of 18,000 rpm is reached in ten minutes. This is followed by four consecutive steps in which the speed is changed between 18,000 rpm and 12,000 rpm in a period of 20 minutes each. Preliminary tests showed that the friction torque increases significantly from around 7,000 rpm. Therefore, the lowest speed during the alternating run in steps 2-5 should be above 7,000 rpm. The aim is to provoke rapid grease distribution. At the same time, continuous operation at maximum speed should be avoided, as this could cause excessive damage to the lubricant.

Table 15: Speed stages in the grease distribution run (short run-in) for high speeds

Step	**Speed**	**Step duration**
[No.]	[rpm]	[min]
1	0-18,000	10
2	18,000-12,000	20
3	12,000-18,000	20
4	18,000-12,000	20
5	12,000-18,000	20
6	18,000	20
-	-	**Σ 120**

After the distribution run, the speed stage test is repeated three times to evaluate reproducibility. The displacement of the lubricant from the bearing plays a role, especially at high speeds. Non-contact seals were also used in the high-speed test bench. In addition, preliminary tests showed that grease leakage can be very high and occurs mainly radially on the outside. To improve the sealing effect, elastic sealing rings were printed on a 3D printer and inserted at the front of the outer ring (support point) between the metal sealing disc and the bearing. The grease leakage should be less than the difference to the next stage of grease quantity variation (Table 12). This was ensured by the modification, which can be verified by the grease leakage at the end of the test results.

Fig. 35 shows the test methodology (green line) based on a test with SKL and grease 1 at a quantity of 32 vol.% (blue line). The test procedure is divided into colored areas and shows the heating phase (heating), the grease distribution run (run-in), and the three repetitions of the speed stage test (runs 1-3).

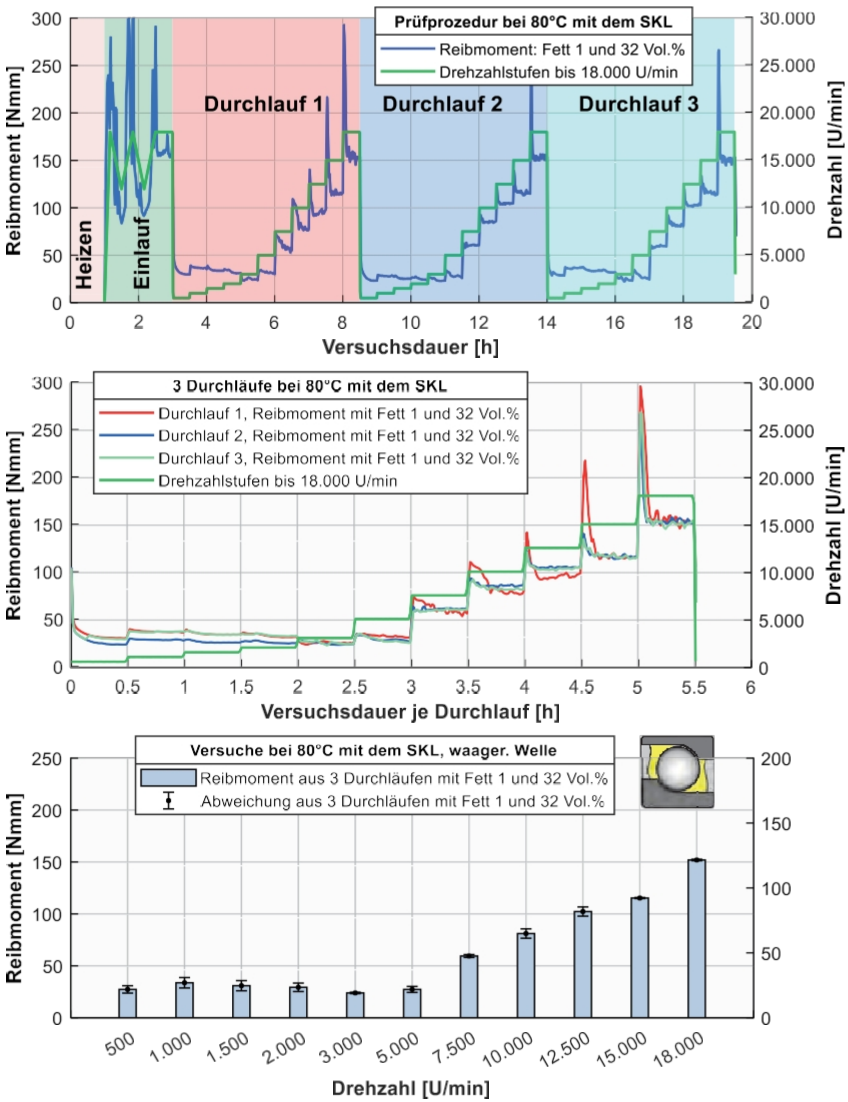


Fig. 35: Test procedure and evaluation based on an example friction torque curve: grease distribution followed by three repetitions up to the maximum speed of 18,000 rpm, example: SKL 7007 with grease 1 at the highest filling level of 32 vol.% in a horizontal shaft position

In the grease distribution run, the maximum speed of 18,000 rpm is reached in ten minutes in the first step. This corresponds to an acceleration of 30 rpm^2. Then is then switched to a lower acceleration of 5 U/s^2 between 18,000 rpm and 12,000 rpm. In the subsequent repetitions the acceleration corresponds to 25 rpm^2 and is thus similar to the acceleration of the other test benches. All these modifications, together with the grease distribution run, lead to good reproducibility of the subsequent repetitions. The actual test run is repeated three times. Table 16 shows all further steps with the respective step duration and the total duration, including the heating and cooling phases.

Table 16: Complete test procedure for the high-speed roller bearing test bench up to 18,000 rpm

Step	Speed	Step duration	Test time
[Designation]	[rpm]	[h]	[h]
Heating	0	1	1
Inlet	0-18,000	2	3
Trial	500	0.5	
	1,000	0.5	
	1,500	0.5	
	2,000	0.5	
Total 3 Repetitions	3,000	0.5	
	5,000	0.5	
	7,500	0.5	
	10,000	0.5	
	12,500	0.5	
	15,000	0.5	
	18,000	0.5	19.5
Cooling	0	0.5	**20**

The reproducibility can be illustrated using the measurement results in Fig. 35 in the middle. In Fig. 35 below, the repetitions have been converted into a bar chart. Here, the deviation is shown by error bars. For this purpose, the steady-state friction torques achieved at the respective speed levels were statistically evaluated with those of all runs. It can be seen that the largest deviations occurred in the low speed range and can be classified as rather small.

Using this test methodology, the angular contact ball bearing was tested with both greases, different grease quantities, and using a minimum amount of base oil in two shaft positions. These results are presented below.

4.4.2 Lubricant quantity variation at 80 °C

Fig. 36 shows the results with the SKL 7007 and grease 1. The amount of grease was varied in stages according to Table 12. The results with vertical shaft position are shown at the top and those with horizontal shaft position at the bottom. The image of the stylized angular contact ball bearing symbolizes the corresponding shaft position. With regard to the variation in grease quantity, grease 1 showed rather small differences in friction torque. The friction torque behavior with respect to speed is characteristic, as it increases up to 2,000 rpm and reaches another minimum at 5,000 rpm. From 10,000 rpm, the friction torque then increases significantly. At 18,000 rpm, the maximum is reached for every grease or oil quantity.

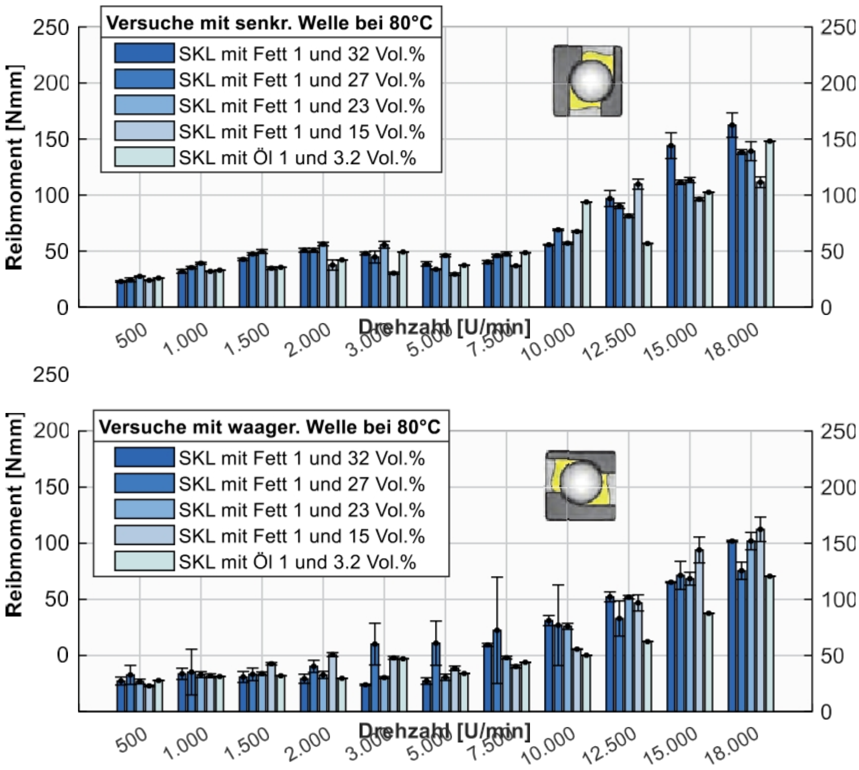


Fig. 36: Lubricant quantity variation with SKL 7007, grease 1, and oil 1 (base oil) in both shaft positions

Fig. 37 shows the results with SKL 7007 and grease 2 for both shaft positions. Here, too, the friction torque level of 500 rpm is again reached at 5,000 rpm.

achieved. A variation in grease quantity appears to have a more pronounced effect with grease 2. However, this mainly applies to the horizontal shaft position. With the vertical shaft position, the friction torque levels resulting from the variation in lubricant quantity are very often similar. With the horizontal shaft position, the reductions in quantity are clearly noticeable from 1,500 rpm. Here, oil lubrication also delivers lower friction torque levels. The levels equalize up to 10,000 rpm. It can be assumed that grease 2 flows back into the area of the rolling elements or the raceway more frequently due to the shaft position, gravity, temperature, and flow behavior.

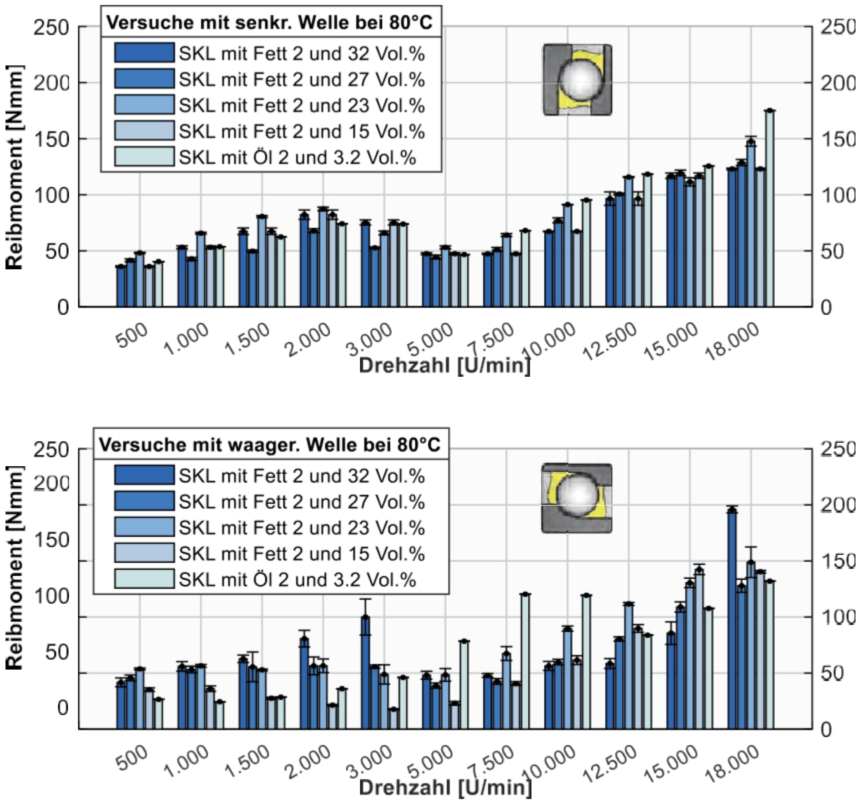


Fig. 37: Lubricant quantity variation with SKL 7007, grease 2, and oil 2 (base oil) in both shaft positions

It is difficult to draw clear conclusions about the kneading action at high speeds from the test results obtained here on the high-speed test bench. Various aspects of the test are responsible for the limited interpretability:

Firstly, the rotation tribometer has already shown that at 80 °C, differences in the friction torque are significantly lower when the amount of grease varies than in tests without temperature control (persistence temperature at each speed level). This means that changes in the lubricant properties, such as viscosity at the test temperature of 80 °C, play a role here. However, since the rheology of greases is complex, viscosity alone cannot be the sole cause.

Another reason is the test procedure itself with the upstream grease distribution run. This did not take place during the tests on the rotational tribometer. Since the maximum speed was already reached several times before the actual measurement, which was evaluated in Fig. 36 and Fig. 37, the differences in grease quantities appear to even out to a certain extent.

Furthermore, in preliminary tests, grease leakage proved to be a challenge during the tests. Design measures (see 4.4.1) reduced grease leakage during the tests. The leakage was always less than the difference between the grease quantities tested. Measured against the respective grease quantity before the start of the test, the grease leakage was between 2.9 and 18.7%, as summarized in Table 17:

Table 17**:** Grease leakage with varying grease quantities on the high-speed test bench with SKL 7007 and both greases and shaft positions for each bearing (see Fig. 9)

Shaft position	**Rolling bearing**	**Amount of fat**	**Fat leakage in %** relative to the amount of grease applied			
		Greasing	Grease 1		Fat 2	
		Vol.	Bearing 1	Bearing 2	Storage 1	Storage 2
Horizontal	SKL 7007	32	4.8	3.8	7.1	4.8
		27	13.9	16.7	11.7	15.0
		23	18.7	14.0	14.0	8.0
		15	12.0	14.0	15	10.0
Vertical	SKL 7007	32	2.9	2.9	10.0	7.1
		27	8.9	4.4	10.6	6.1
		23	4.0	7.3	6.7	7.3
		15	6.0	8.0	16.0	14.0

The grease leaks in Table 17 show that in all tests, less than 20% of the amount of grease applied leaked out during the test. They also show that grease leakage varies greatly. There are no discernible trends that correlate with the amount of grease applied, the shaft position, or the test grease. This suggests that small differences in the lubrication of the roller bearings have an influence on the results. Especially at high speeds, even the smallest differences in quantity around the bearing circumference resulting from the initial lubrication could be the cause of the variation in grease leakage on the high-speed test bench. Overall, however, the leakage can be classified as low.

In summary, grease quantities were varied on the high-speed test bench and a test was presented using pure oil lubrication. The differences in friction torque between the different grease quantities or the different types of lubrication allow hardly any conclusions to be drawn about the kneading work. The friction torque curves for the different grease quantities are very similar in some cases. Fluctuations from identical tests with the same grease quantity are sometimes greater than the resulting difference from the grease quantity variation. As a result, only limited recommendations can be derived on the basis of the available results:

For grease 1, all results in Fig. 36 show very similar friction torque curves. The friction torque level is also comparable. These friction torque curves with grease 1 were achieved despite differences in shaft position, grease quantity before the start of the test, and grease leakage. It can therefore be concluded that neither a varying grease quantity nor a shaft position has a major influence on the distribution in the rolling bearing and thus on the rolling work. This offers enormous advantages for applications in this speed range with this typical ambient temperature.

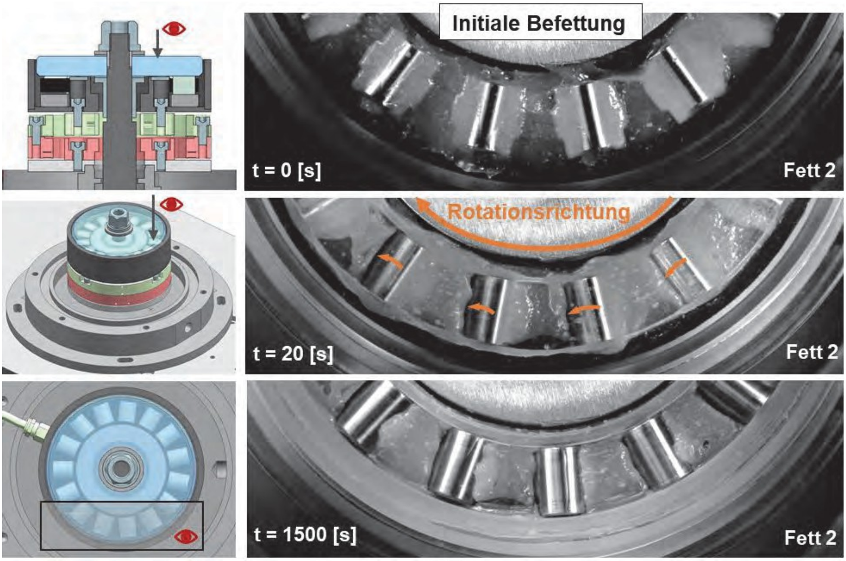
4.5 Special tribometer

Rolling bearing tests with transparent rolling bearing components were carried out on the special tribometer. Speeds of up to 2,000 rpm were achieved without temperature control in a vertical shaft position.

4.5.1 Tests with the AZRL 81212

The axial cylindrical roller bearing offers a particularly good view of the grease distribution thanks to the glass shaft disc. This provides a visual validation option for 3D CFD simulations. In Fig. 38, three CAD views of the test bench on the left illustrate the perspective chosen for the recordings on the rolling bearing (right). The images (right) show different test times: the initial greasing of the bearing (top), the distribution status after 20 s (middle), and the distribution status after 1,500 s (bottom).

Fig. 38: Images for evaluating the initial grease distribution with AZRL 81212 and grease 2



when starting up at a speed of 500 rpm with a standard camera, view through glass shaft disc

During initial lubrication, each cage pocket was filled completely with grease, as was the area between the cage and the glass shaft disc. This initial lubrication is relevant to the CFD simulation and was taken into account there in exactly the same way. In addition, during initial lubrication in the CFD, this value was selected taking density into account. It follows that the grease quantity of 5.45 g corresponds to just under

30% by volume of the free space in the rolling bearing. In the simulation, the initial greasing is carried out in the same way. Important distribution properties can be derived from the time curve.

After just 20 seconds of test running (Fig. 38 center), a grease collar forms on the front faces of the rolling elements. This adheres to the shaft disc and moves with it in the direction of rotation. As a result, large parts of the grease collar no longer flow, which is why the shear rate in these areas is sometimes zero. At this point, it is also possible to see how strongly the lubricant is sheared on the surface of the cage between the rolling elements. Depending on the direction of rotation of the rolling element, this lubricant is conveyed to the next rolling element. Lubricant accumulates on the rolling element.

After 25 minutes (t= 1,500 s), this redistribution process appears to be complete (Fig. 38 below). The accumulated quantity in front of the rolling element is now distributed. Only a small amount of lubricant is still visible on the shaft disc in the raceway area. Large quantities of the lubricant that was between the cage and the shaft disc were transported radially inwards and outwards. As a result, the two grease collars have become significantly wider. On the surface of the cage between the rolling elements, it can be seen that the lubricant has distributed itself relatively evenly. These quantities of grease are now no longer in contact with the rotating shaft disc. They rest on the cage surface and are transported on it. Only the grease collars described at the beginning now adhere to the shaft disc. Visually, the lubricant adhering to the rotating shaft disc can be distinguished from that resting on the cage surface by the reflective parts of the lubricant on the cage surface. The distribution properties described here are similar for grease 1 and grease 2. The first images show that the distribution on the individual segments differs slightly.

Measurements with grease 1 and grease 2 are shown in Fig. 39 below. The friction torque for a test with grease 1 and grease 2 is shown in blue at the top, and the corresponding temperature curves are shown at the bottom. It is clear that the static friction torque is already reached after about one hour with grease 2. With grease 1, this takes approximately two hours. The magnification shows the respective starting friction torque, which is significantly greater and lasts longer with grease 2. The temperature (Fig. 39 below) only reaches a steady state after three hours with both greases.

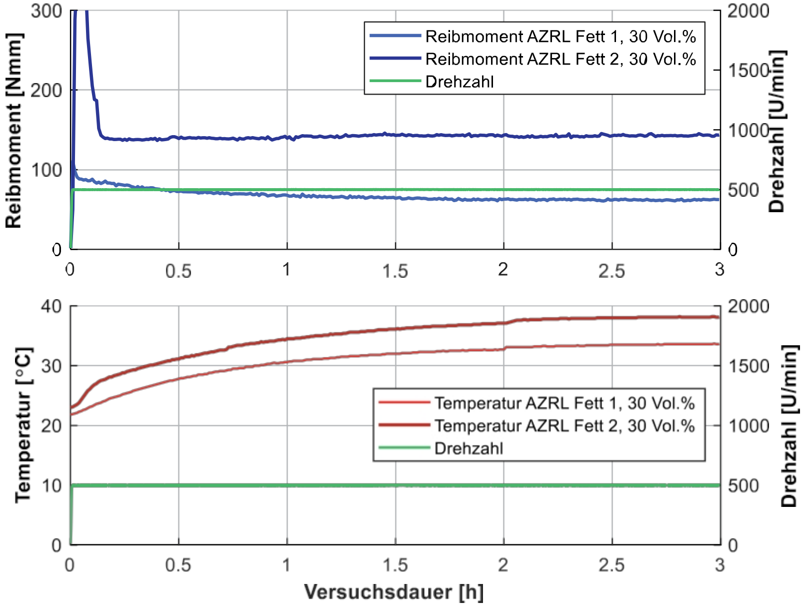


Fig. 39: Tests on the special tribometer with the AZRL and both greases

Fig. 40 now shows a comparison of the fat distribution of fat 1 (left) and fat 2 (right). The images were taken with a high-speed camera at different times during the experiment. The same angle segment was always selected for illustration based on the derived videos. The attainment of a steady state after a test duration of three hours is also shown. Here, the aspects of fat distribution explained in Fig. 38 were also evident, with additional fat-specific differences now becoming apparent.

After three hours, clear differences became apparent between grease 1 and grease 2. This point in time is based on the persistence of the friction torque and the temperature (Fig. 39). In the tests, the grease distribution was recorded at regular intervals using high-speed recordings. This allows the path to the steady state to be visualized below.

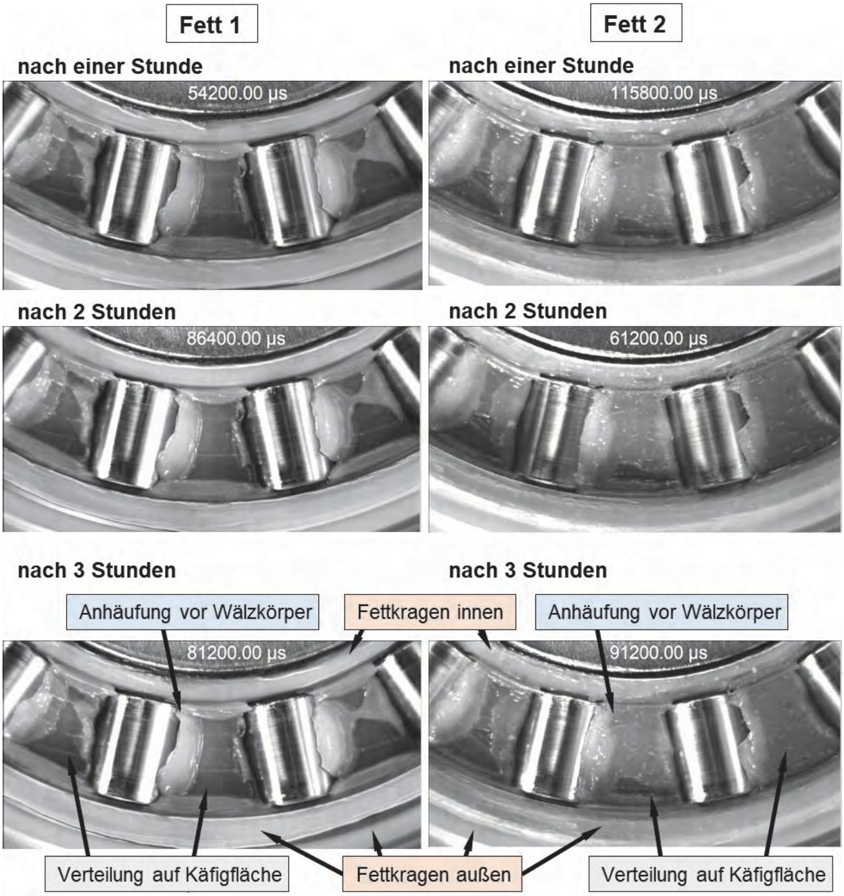


Fig. 40: High-speed recordings on the special tribometer with the AZRL 81212: Differences in grease distribution with grease 1 and grease 2 after a test duration of three hours.

The optical distribution on the cage surface differs significantly. With grease 2, a uniform layer thickness forms, whereas with grease 1, local accumulations can be seen. The surface of the cage is partially visible through a very thin layer of grease. With grease 2, this is only the case in a few segments. The grease collar shows another reproducible difference on the outer radius. With grease 1, it is interrupted in many places. With grease 2, on the other hand, it is continuous, but adheres to the shaft disc with a different layer thickness. The lubricant that is conveyed into contact with the rolling element offers further differentiation. With grease 2, a less thick bead appears to form and the lubricant extends further into the contact than with grease 1.

4.5.2 Tests with SKL 7007

The tests to evaluate the grease distribution in SKL 7007 were carried out in a speed stage test up to a maximum speed of 2,000 rpm. Fig. 41 shows the perspective of the high-speed images as a CAD view (top) and an example image with grease 1 at 100 rpm (bottom). The initial basic distribution properties can be seen here.

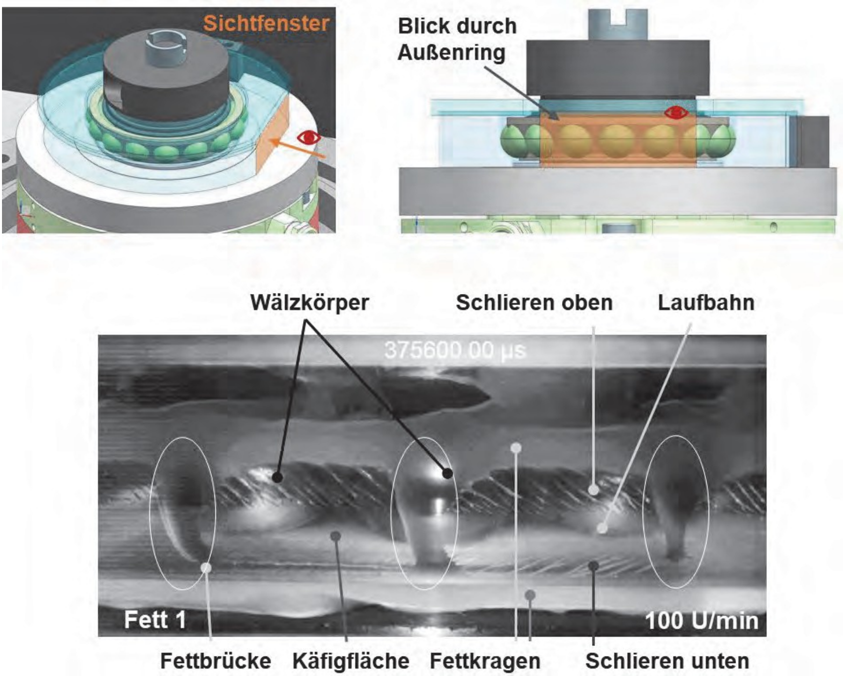


Fig. 41: Grease distribution in the SKL 7007, top: illustration of the perspective of the images through the sapphire crystal outer ring, bottom: example image with grease 1 at 100 rpm

Two grease collars can be seen in the video clip. The grease collars rest on the outer ring and are not sheared. In the area between the grease collars, a grease bridge is clearly visible on each of the three rolling elements. This creates a connection between the two grease collars. The grease collars typically serve as a lubricant reservoir in a rolling bearing as Reservoir. The Presence of a grease bridge is for the lubricant supply to the rolling element-raceway contact. This grease bridge travels around with the rolling element over the stationary outer ring. Fine streaks can be seen between the grease bridges of the individual rolling elements, extending over the entire area between the grease collars. The orientation of the streaks changes across the width of the outer ring. These streaks (top and bottom) indicate the position of the raceway and thus the narrowest gap. With each rollover, the streaks are picked up in front of the contact area and reformed behind it.

Fig. 42 shows a total of eight images taken with each lubricating grease at the end of each speed stage from 500 to 2,000 rpm (two greases, four stages). A holding time of 30 minutes per stage was selected for the tests. This was to protect the glass component, as the fatigue strength of the ground raceway is unknown.

At a speed of 500 rpm (Fig. 42, top left), the two grease collars have become wider. The upper collar in particular has shifted further upwards and widened. The situation is similar with grease 2 (Fig. 42, top right), whereby the grease collar has been displaced further upwards overall. The formation of grease collars and their significant displacement from the contact area has already been observed with AZRL. At this speed, there was hardly any lubricant left on the raceway. In the SKL, the area between the two grease collars on the outer glass ring is almost completely covered with a thin layer of lubricant. This difference to the AZRL is due to the narrowing gap at point contacts compared to line contact. The layer thickness decreases towards the contact area.

At 500 rpm, the first differences between the two lubricants become apparent. Grease 1 forms a wider collar at the top than grease 2. The grease collar appears to be smaller in width and layer thickness with grease 2. Differences in transparency could also be a factor here. Furthermore, the rolling element appears more blurred with grease 2 and at all speed levels than with grease 1. Since the grease quantities are the same at the start of the test and the grease collar is obviously thicker with grease 1, it can be assumed that there are different quantities between the raceway and the respective grease collar for the greases. With grease 2, it should therefore be larger.

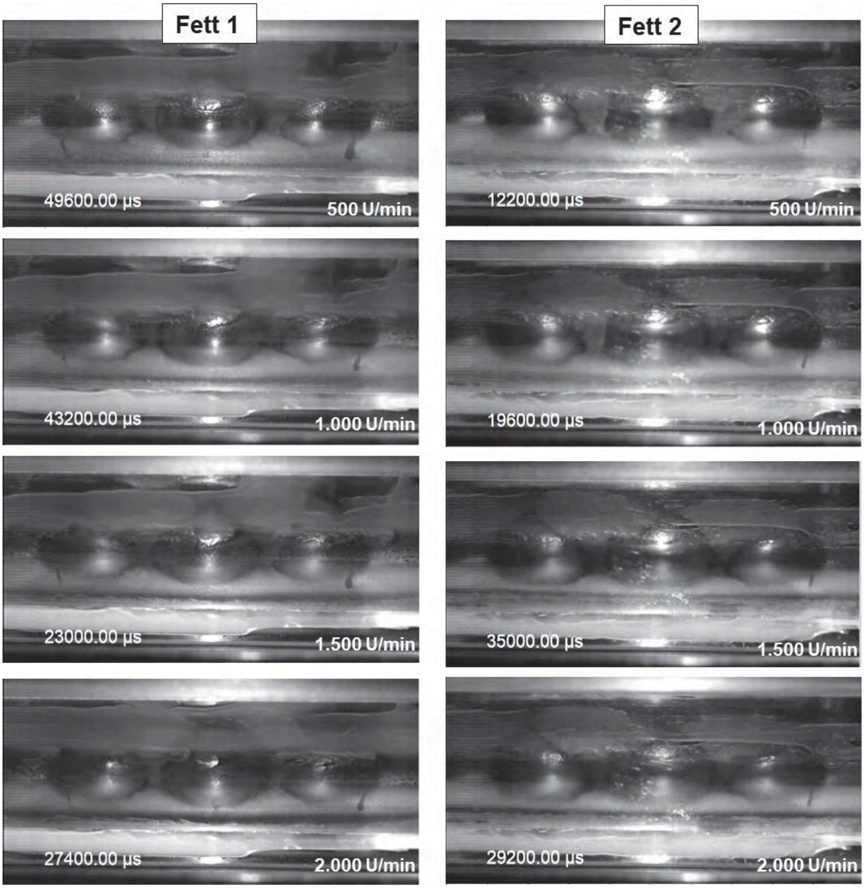


Fig. 42: High-speed images of the grease distribution in the SKL 7007 in the speed stage test from 500 to 2,000 rpm, left: grease 1, right: grease 2

A quantity of lubricating grease is also visible on the cage surface. Here, too, there are parallels to the results from the AZRL. Larger quantities of lubricant are transported on the cage. This picture changes only slightly over the course of the speed stages. In the case of grease 1, slightly larger quantities of lubricant appear to be transported on the cage. This would be consistent with the distribution properties that were also observed at the AZRL. There, grease 1 had piled up higher in some places, whereas grease 2 had formed an almost uniform layer thickness on the cage surface. The most significant change in the speed stage test occurs with grease 1 and a speed of 2,000 rpm. Here, there was obviously an axial shift to the outside. The grease collar has become significantly wider at the top.

Fig. 42 shows all the results of the speed stage tests up to 2,000 rpm. In order to better compare the views, all video excerpts were selected so that there is always a rolling element in the center of the image. Two others are still clearly visible next to the center rolling element.

The high-speed recordings enabled the distribution in the SKL to be evaluated visually. The speed change was used to locate the raceway and describe basic displacement properties. A comparison of grease 1 and 2 also enabled assumptions to be made about differences in the layer thickness on the outer glass ring. However, this impression may be misleading, as it can be assumed that the significant difference in the thickener content of the two greases also influences transparency. The thickener content is 30% for grease 1 and 18% for grease 2. When starting up at a speed of 2,000 rpm, the rotation tribometer showed a reproducible increase in the friction torque level in conjunction with grease 1 (Fig. 23). The use of the sapphire glass outer ring apparently does indeed result in a redistribution. However, this was not noticeable in the friction torque.

4.5.3 e measurements using a thermal imaging camera

Infrared thermography allows temperature fields to be recorded across the entire bearing width. For this purpose, a thermal imaging camera was used to take images of the bearing at regular intervals. Following a test, the manufacturer's software can be used to evaluate the temperature at any number of points across the entire temperature field (3.2.2).

The measurement using a thermal imaging camera was performed for the configuration with SKL and grease 1 at the persistence temperature. This configuration achieved the best reproducibility on the rotational tribometer. In addition, there was a measurable increase in the friction torque when starting up to the speed level of 2,000 rpm (see Fig. 23). Therefore, a temperature field of the rolling bearing with grease 1 was recorded using the glass outer ring and a thermal imaging camera. The aim was to check, if possible, whether shear heating could also be quantified in the test in this speed range.

During the 2-hour test, an image was captured every ten seconds. Fig. 43 shows a measurement with grease 1 at the top. This is a speed stage test up to a speed of 2,000 rpm, which was carried out with a shortened holding time of 30 minutes per stage.

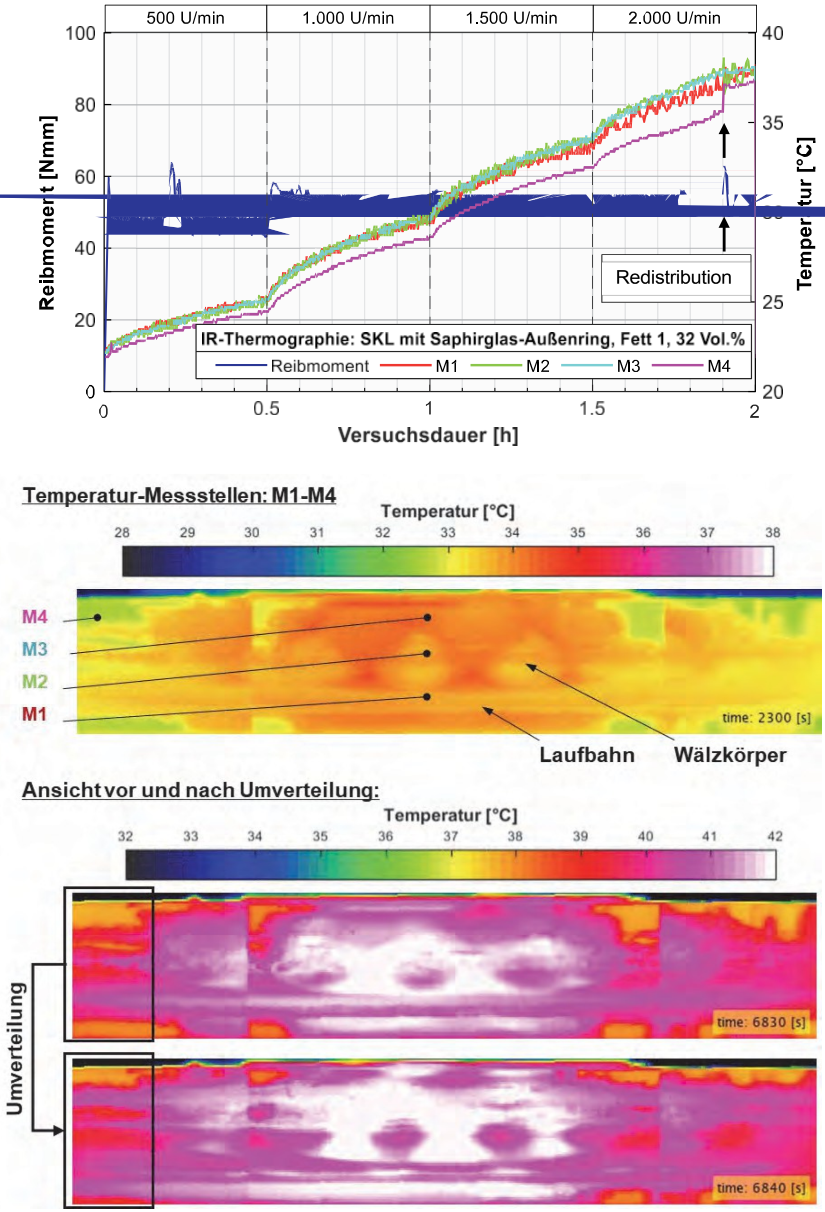


Fig. 43: Temperature measurement using a thermal imaging camera, top: diagram with measured values, bottom: images taken with the thermal imaging camera at 2,300, 6,830, and 6,840 seconds into the test

The friction torque was measured and the temperature was evaluated on the basis of images taken with a thermal imaging camera. However, the glass component did not show a comparable increase in friction torque to that observed on the rotational tribometer.

The temperature was evaluated for the diagram at four points, which are entered in the recording below. In a rolling bearing test, the temperature is typically measured on the outer ring, near the raceway. Points 1-3 were selected to check whether there are significant differences across the width of a bearing.

The diagram shows that, on average, there are no major deviations across the bearing width. The reasons for this may lie in the measuring point. The recordings are based on the principle of infrared thermography (see 3.2.2). Steel rolling elements reflect IR radiation and have a low emissivity. This makes it difficult to differentiate between the slightest temperature differences across the bearing width. Ceramic rolling elements would be recommended for further investigations. Immediately after the speed stage test, the grease temperature between the rolling elements was measured with a thermocouple. A temperature of 38 °C was determined there. Therefore, there is no doubt about the measured temperature, but only about the differentiability of the temperature near the steel rolling elements. Based on measuring point 4, an increase in temperature can be recorded during the speed stage of 2,000 rpm, which is not present at the first three measuring points. There is a brief increase in friction torque there. The point in time is enlarged in the diagram (Fig. 43, top right).

In addition, IR images were taken before (6830 [s]) and after the event (6840 [s]) (Fig. 43, bottom). The images show a redistribution. This means that the lubricant in the rolling bearing is redistributed within ten seconds. The event is not very significant in terms of friction torque. Nevertheless, only IR thermography was able to prove that a redistribution had actually taken place. This redistribution process also generated shear heating in this case.

4.6 cal comparison of the roller bearing test benches

In order to verify that comparable results were achieved on all test benches, the results from all test benches are now being compared with each other. Since the most extensive tests were carried out on the rotational tribometer, these tests serve as the reference for the comparison. For this comparison, reference tests were also carried out on the special tribometer and the high-speed test bench in accordance with the test methodology used for the rotational tribometer. To ensure a uniform basis for comparison, no temperature control was used in the tests, as this is not possible on the special tribometer. The maximum speed on the special tribometer is limited to 2,000 rpm. Tests of up to 3,000 rpm were possible on the high-speed test bench. All subsequent tests were carried out with a vertical shaft.

Fig. 44 shows the comparison between the rotational tribometer and the special tribometer in two diagrams. The test configuration with the RRKL and grease 2 was selected as the reference test. The upper diagram shows two individual tests up to 2,000 rpm. The friction torque curve showed a significant increase at 1,500 rpm, which was also evident on the special tribometer (Fig. 44, top).

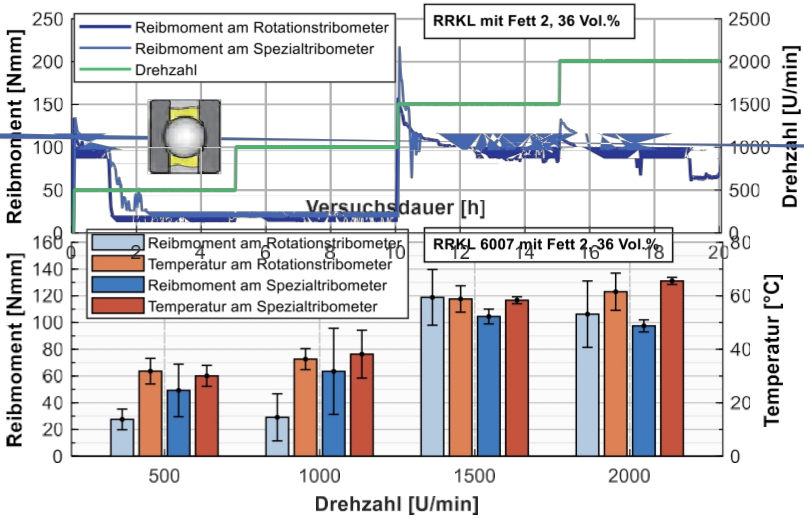


Fig. 44: Comparison of the rotational tribometer and the special tribometer, top: friction torque curves of exemplary individual tests with RRKL and grease 2, bottom: repeat tests on the rotational tribometer (seven tests, light colors) and on the special tribometer (five tests, dark colors)

Fig. 44 shows the statistically evaluated results of both test benches up to a speed of 2,000 rpm. It is clear that the jump in friction torque is repeated on both test benches. The friction torque level is also comparable from a speed of 1,500 rpm. The special tribometer in particular provides very good reproducibility here. In the first two speed stages, however, the friction torque differs more significantly, although the average temperatures of both test benches are similar. In terms of friction torque, the standard deviation is also greater for the special tribometer. Since the confidence intervals overlap, the first two speed stages are also comparable with each other.

In order to transfer the tests from the high-speed test bench to the rotational tribometer, reference tests had to be carried out on the high-speed test bench under the same test conditions. Since the high-speed test bench has a double bearing arrangement, the friction torque measured there was divided by two (reference: single bearing).

Fig. 45 shows four roller bearing tests. At the top, you can see the friction torque curves from tests with the RRKL and both test greases at steady-state temperature. Since the test head on the high-speed test bench has a significantly greater mass than the other test benches and media temperature control has been integrated there, the conditions for heat dissipation are different. Nevertheless, significant increases in friction torque are recorded with the RRKL and both lubricants when starting up at the speed level of 1,500 rpm. The results from the high-speed test bench with the RRKL are therefore consistent with those from the other two test benches.

Fig. 45 below shows tests with the SKL and both test greases. For grease 1, the first speed stages correspond almost completely to the curve on the rotational tribometer. The grease leakage on both test benches is less than 0.1 g (less than 4%) up to this speed. On the rotation tribometer, random jumps in the friction torque repeatedly occurred with the SKL and grease 2 configuration. These were not noticeable on the high-speed test bench. Overall, all curves are very similar.

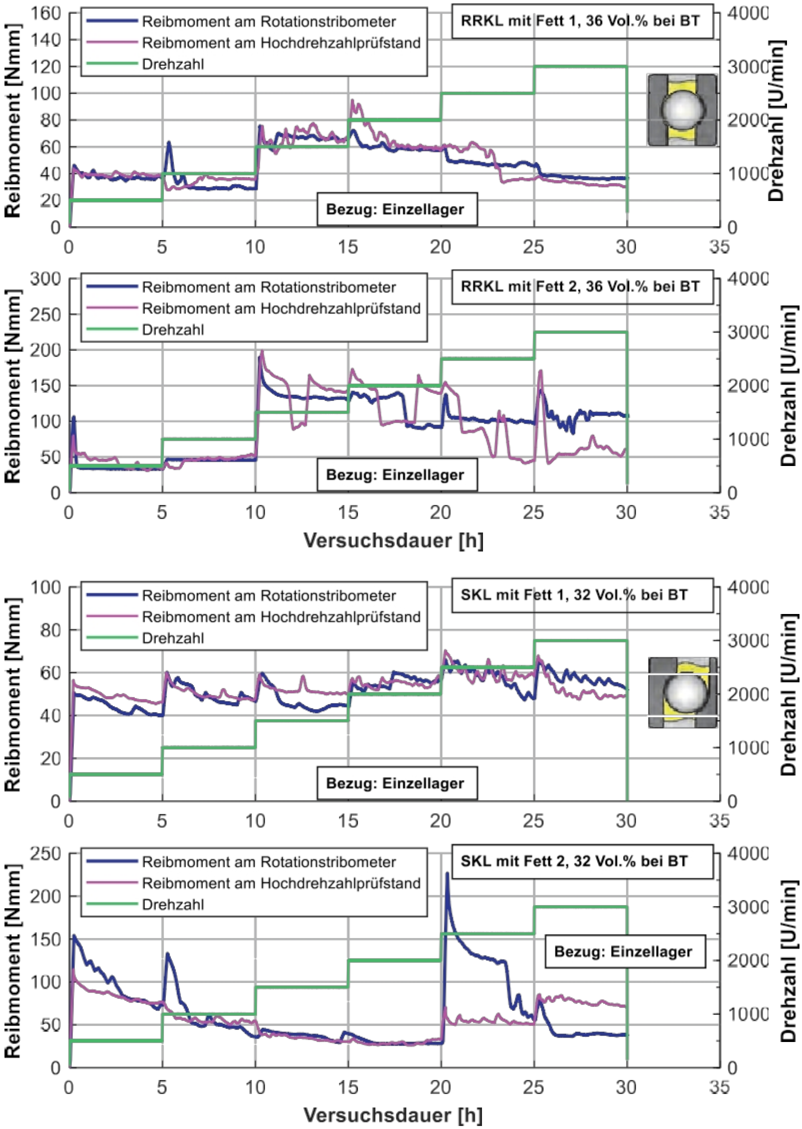


Fig. 45: Comparison of the rotational tribometer and high-speed test rig: top: RRKL 6007 with both test greases (36 vol.%), rotational tribometer (blue), high-speed test bench (magenta), bottom: SKL 7007 with both test greases (32 vol.%), all tests at steady-state temperature and based on a single roller bearing

The high-speed test bench also clearly shows the dramatic increase with the RRKL for both greases. This noticeable increase in friction torque was therefore observed on all three roller bearing test benches under the same test conditions. It can therefore be concluded at this point that this is not an effect of a test bench, but rather a superimposed effect of grease rheology, grease quantity, rolling bearing kinematics, speed, and temperature. The friction torque recorded on the high-speed test bench for SKL is also comparable to that of the rotation tribometer. This shows that a characteristic friction torque curve for a given test configuration can be verified independently of the test bench. Overall, the transferability of the results from the different test benches is guaranteed.

4.7 e summary of the rolling bearing tests

In the rolling bearing tests, a test procedure for analyzing grease-lubricated rolling bearings in terms of their work of kneading was presented. This is essentially based on a single speed step test with a fixed amount of lubricant. With the help of the speed step test, the degree of kneading work could be visualized by calculating the difference between two rolling bearing tests with different lubrication. On the one hand, the difference in friction torque could be illustrated by varying the amount of grease. On the other hand, this difference in friction torque between a grease-lubricated and an oil-lubricated rolling bearing test could be quantified. The respective differences in friction torque are interpreted as work of kneading. Consequently, the findings from this test procedure for work of kneading are based on difference measurements, whereby the amount of lubricant or the type of lubrication was influenced in the individual tests. Building on this basic test procedure, it was also presented how the work of kneading can be converted into kneading losses from the known speed.

This basic test procedure was developed on the rotational tribometer, where the most extensive tests were also carried out. These tests reveal numerous dependencies of work of kneading. These include grease rheology, grease quantity, temperature, rotational speed, and rolling bearing kinematics.

The influence of roller bearing kinematics on the flexing behavior is particularly noteworthy here. For example, grease 2 in the RRKL exhibited stronger flexing behavior than grease 1 over large ranges of the speed stage test. In contrast, grease 1 in the SKL caused stronger flexing work in certain areas of the speed stage test. Furthermore, with the RRKL without temperature control, both greases showed a significant increase (jump) in the friction torque when starting at the speed stage of 1,500 rpm, which can be attributed to both the rolling bearing kinematics and the exact speed.

Furthermore, the flexing behavior in the tests at 80 °C is significantly lower than in the tests at the holding temperature (max. 65 °C). This temperature influence on the degree of flexing was also confirmed on the high-speed test bench, where only tests at 80 °C were carried out. At higher test temperatures, the differences in friction torque due to the variation in grease quantity decrease and are sometimes in the single-digit Nmm range. In the 80 °C tests on the rotational tribometer, the differences in friction torque in the speed step test were often below 10 Nmm. These differences are only slightly above the confidence range of the sensor technology (+ 3Nmm, see section 3.1). This means that the percentage of measurement error is greater. Apart from the sensor accuracy, the tests with temperature control show a different friction torque behavior than without temperature control. In all 80 °C tests, the friction torque for grease 2 was higher than that for grease 1. In addition, RRKL did not show a drastic increase in friction torque when starting at the speed level of 1,500 rpm, which could be reliably observed in the tests without temperature control on all three test benches. This clear effect of the rolling action is therefore not only dependent on the rolling bearing kinematics and the speed, but also on the temperature.

A time-lapse test procedure was developed on the high-speed test bench, which is valid for grease-lubricated rolling bearings up to 18,000 rpm. At the start of the test procedure, a grease distribution run is performed up to the maximum speed. This is followed by three repetitions of the speed step test. The three repetitions showed good reproducibility. The test procedure included variations in grease quantity and an oil test. Despite a variation in the amount of grease before the start of the test, the same friction torque was observed in some cases, regardless of the amount of grease, especially with grease 1. With grease 1, it was found that neither the amount of grease nor the shaft position had a significant influence on the friction torque curve. It is assumed that the grease distribution run equalizes some of the differences with regard to the variation in grease quantity. With grease 2, relatively high friction torques occurred sporadically in a few tests, which ultimately cannot be explained experimentally. It is also possible that these were short-term problems with the presence of grease at a contact point in the roller bearing. These fluctuations make it difficult to draw clear conclusions about the flexing behavior at high speeds and 80 °C. This is partly due to the test temperature, as the rotation tribometer has already shown that the amount of flexing work is highly temperature-dependent.

Using high-speed recordings, information on grease distribution in rotating roller bearings was collected for the AZRL and SKL on the special tribometer. Both roller bearing types differ significantly in terms of design, size, and kinematics. Nevertheless, there are many parallels between the two types of roller bearings in terms of grease distribution: hardly any lubricant is visible in the area where the rolling elements come into contact with the raceway. Outside this area, grease collars form in the shape of tracks. Furthermore, a considerable amount of grease is deposited on the cage. The aspects of grease distribution differ between the two greases and are therefore grease-specific.

The findings from the special tribometer provide an important reference for the grease distributions calculated in the CFD simulation of rolling bearings.

In summary, the rolling bearing tests made it possible to derive numerous factors influencing the kneading work. These include the rheology of the lubricating grease, which can change as a result of temperature. Furthermore, the speed and the rolling bearing kinematics, which can be attributed primarily to the type of rolling bearing, also influence the kneading work. Finally, it was demonstrated that even the smallest changes in grease quantity can lead to significant differences in kneading work. All influencing factors are closely interrelated and could be verified by extensively varying the test parameters on the roller bearing test benches.

5 Numerical simulation of the " "

The aim of the numerical simulation was to simulate the distribution of lubricating grease in roller bearings of different designs. This raised a number of questions, which were summarized in three key areas:

An analytical lubricating grease model was developed to simulate the flow behavior of the test greases (see 5.1). This model combined various dependencies and lubricant properties. These include the dependence on the shear rate, the temperature, and the viscosity of the base oil. The majority of the data used to develop the lubricating grease model was obtained from the results of the standardized shear test (see 4.1.1 and DIN 51810-1).

Numerical flow simulations (computational fluid dynamics, or CFD for short) were also carried out. ANSYS Fluent software was used for this purpose. This software is used to solve the complete Navier-Stokes equations approximately. Compared to Reynolds solvers, this is a complex and time-consuming process. The flow area in a rolling bearing includes both very small gaps and larger free spaces. This results in large differences in the shear rate. In addition, a rolling bearing contains sliding and rolling contacts. Both shear and pressure flow are present. Locally, these flows can overlap, leading to complex flow conditions. For these reasons, this solution method was considered necessary and appropriate. In addition, the state of the art has shown that CFD simulation was often chosen for simulating lubricant distribution in rolling bearings (see 1.2). In the 3D CFD simulation of rolling bearings, the bearing types angular contact ball bearings, radial deep groove ball bearings, and axial cylindrical roller bearings were considered (5.3).

In an intermediate step, the analytical grease model was verified. For this purpose, a CFD model was constructed based on the shear test mentioned above (5.2). Shear flow is dominant in this test. This enabled subsequent validation of the grease model (see 6.1.1).

5.1 Analytical grease model

To describe the flow behavior, the shear rate dependence of viscosity must be taken into account to a particular degree. To date, the Herschel-Bulkley model is most commonly used in CFD simulations of grease-lubricated systems [YU15, EHRE16, SARK17, WEST18, AUGU19, RAJ21, MAST21]. Further flow models were explained in Chapter 2.6. In the following, the formation of the lubricating grease model will be based on the Herschel-Bulkley model. For this purpose, numerous

Regressions were performed. The database covers a temperature range between 20 and 100 °C in 10 °C increments. The Herschel-Bulkley coefficients derived in this way form the basis for describing the temperature dependence for the CFD grease model. Furthermore, the Vogel model is used to take into account the base oil viscosity (see 2.6), which is intended to supplement the plastic-structural viscous flow behavior of the Herschel-Bulkley model in the high shear rate range.

5.1.1 Determination of the Herschel-Bulkley coefficients ( )

The coefficients for the Herschel-Bulkley model can be determined using regression. For this purpose, Eq. *(10)* was rearranged according to the shear stress τ :

$$\tau=\tau(\ 0_{)} + \kappa \cdot \dot{\gamma}^{n} \tag{26}$$

The yield strength τ_0, the consistency coefficient κ , and the flow index n
are determined using regressions. To do this, the relationship between shear stress and shear rate must be known. This was determined using the standard test with the cone-plate test configuration. The standard is usually used to derive η_A and η_E from
Phase V (4.1.1). Instead, phase IV of the standard test forms the data basis for the
Herschel-Bulkley model. This phase involves constant acceleration up to a shear rate of 1,000 s^{-1}. Regression tools can be used to describe the dependency with a good degree of accuracy, as can be seen in Fig. 46, above.

Two criteria were used to evaluate the quality of a regression. The most commonly used quality criterion is the coefficient of determination. The model provides a coefficient of determination of at least 98% (Fig. 46, bottom left). The diagram shows the coefficient of determination between 98 and 100%.

The lowest value of the coefficient of determination is 98.83% for fat 1 at 80°C. The highest value is 99.99% for fat 2 at 20°C. To illustrate this range of result qualities, these two examples are shown side by side in Fig. 46 above. In addition, a reference line based on the test data was introduced as an evaluation criterion (Fig. 46, center). This is an ideal reference line. This is contrasted with the data points of the regression (red), whereby the difference is evaluated as a percentage. In the negative example of fat 1 at 80 °C (Fig. 46, center left), the deviations reach values of just under 3%. In most cases, the deviation is less than 1%.

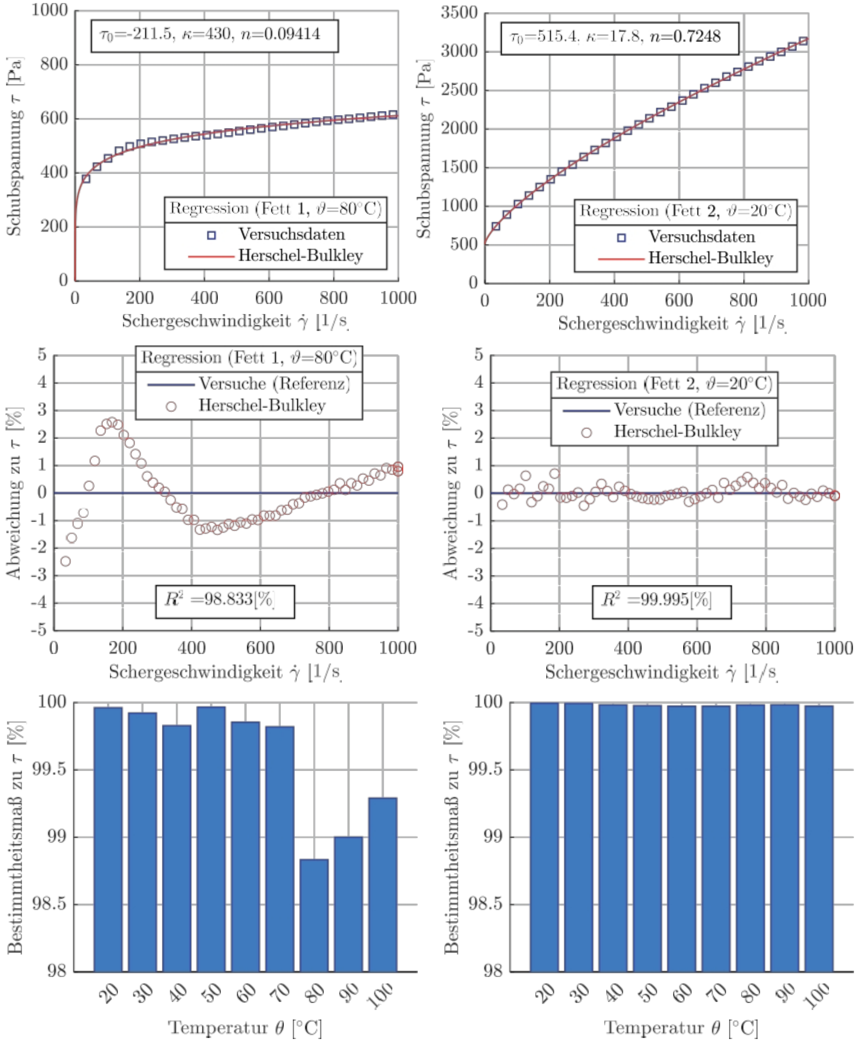


Fig. 46: Regression with the Herschel-Bulkley model for fat 1 at 80 °C (left) and fat 2 at 20 °C (right), top: shear stress versus shear rate, middle: difference between regression and experiment, bottom: coefficient of determination for all regressions

Table 18 shows the results of the regression with the Herschel-Bulkley model. It shows the three coefficients for each fat at different temperatures in 10 °C increments. It can be seen that the yield point τ_0 decreases with

decreases with increasing temperature. The consistency coefficient κ increases with increasing

Temperature up to 50 or 60 °C and then significantly up to 100 °C. The flow index n ranges between 0 and 1 and tends to decrease as the temperature rises. In some cases, the yield point takes on negative values.

However, a liquid medium can only transmit very low negative shear stress or tensile stress, which is why these values are not correct from a physical point of view. The reasons why these values were nevertheless used for the lubricating grease model and the additional problems that can arise during regression are explained below.

Table 18: Herschel-Bulkley coefficients according to Eq. *(10)* in the range of 20-100 °C

HB regression		Grease 1			Grease 2		
$\tau(\dot{\gamma}) = \tau_0 + \kappa \cdot \dot{\gamma}^n$ [Pa]		τ_0 [Pa]	κ [Pa·s]	n [-]	τ_0 [Pa]	κ [Pa·s]	n [-]
Temperature [°C]	20	555.06	48.304	0.441	515.40	17,787	0.7248
	30	438.76	59,219	0.382	494.74	9.8978	0.7308
	40	394.92	57.411	0.356	494.03	8.2077	0.7011
	50	487.55	19,999	0.445	441.70	8.9869	0.6437
	60	441.61	15,661	0.459	318.71	18,010	0.5027
	70	425.29	8.7257	0.506	236.53	37,390	0.3795
	80	-211.53	429.82	0.094	115.68	87.114	0.2610
	90	-867.22	1059.9	0.042	-30.403	177.60	0.1679
	100	-977.61	1198.5	0.034	-184.10	307.90	0.1139

5.1.2 Special features of regression and Model modification

Various problems can arise when determining the coefficients. On the one hand, the selected value range has a considerable influence on the regression. In particular, the flow behavior at very low shear rates poses a problem. On the other hand, the model modification can also cause problems.

For others, it is the determination of the yield point τ_0 . This had assumed negative values in some areas of high
temperatures (Table 18). Both points are explained in detail below, along with the reasons why the regression can produce negative values for the shear stress.

Measurement data is available from acceleration phase IV according to DIN 51810-1. The phase lasts one minute and values are stored every second as standard. The data set of 60 points is within a shear rate range of [0-1,000] s^{-1}. The regression of the flow curve is performed according to Eq. *(26)* and describes the dependence of the

shear stress τ on the shear rate $\dot{\gamma}$. In the range of low shear rates, the gradient of the
Large flow curve. The representation of large gradients can be problematic with Eq. *(26)*.

With the aim of achieving the best possible representation of the flow behavior, a shear rate range of [16-1,000] was selected. The data point 0 s($^{-1)}$was neglected. The sensitivity of the regression to the value range used is shown in Fig. 47. For clarification, 5 additional data points were omitted as examples, and the regressions were thus performed for the following shear rate ranges (Fig. 47 above):

- W_I = [16-1,000] s^{-1} and
- W_{II}= [120-1,000] s^{-1} .

The Herschel-Bulkley coefficients differ significantly (Fig. 47 above). The yield point$\tau_{(0)}$ shows how the flow behavior at low shear rates can be influenced by the choice of value range. The "deviation from shear stressτ " also reveals noticeable differences resulting from the illustrated value ranges. By reducing the data set,
medium and high shear rates. The coefficient of determination even increases for the selected example. However, greater mapping difficulties arise in the range of very low shear rates (Fig. 47 below, left). Here, the deviation rises to over 4 percent.

However, with regard to the application of grease-lubricated rolling bearings, it is important to be able to accurately represent even small shear rates. From a physical point of view, a negative yield strength is not correct, but from a mathematical point of view, it can be useful to make an assumption here. For this reason, a model modification of the Herschel-Bulkley model used was carried out at this point. The modification includes a non-negativity condition for the entire value range (in Fig. 47 W_I).

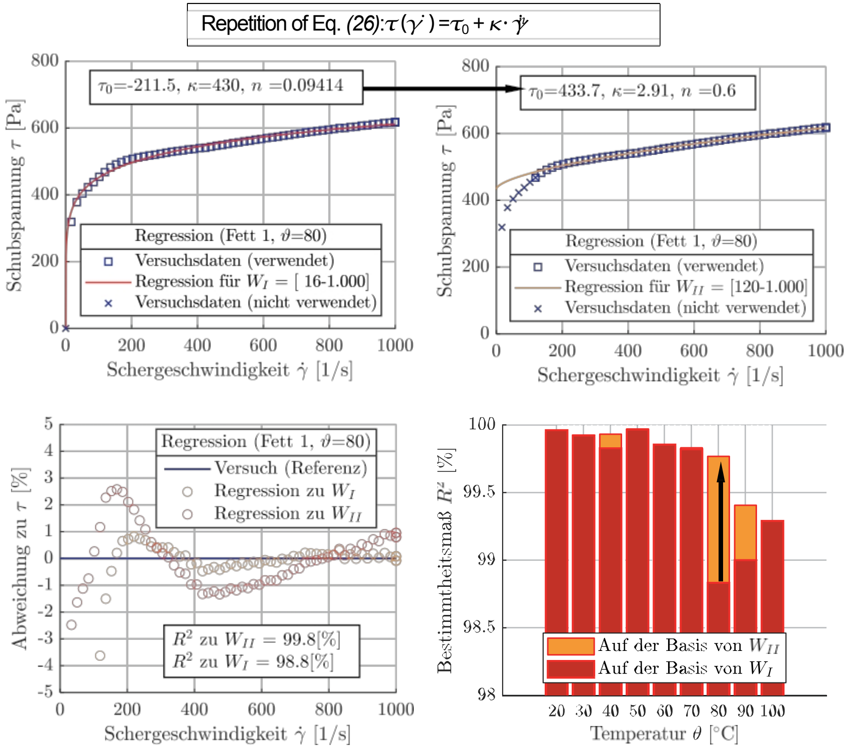


Fig. 47: Influence of the value range on the regression of the flow behavior, top: Fat 1 at 80°C with shear rate value range in [1/s] on the left [16-1,000] and on the right [120-1,000], Bottom: Deviation from the shear stress of both regressions on the left and coefficients of determination R(2)on the right

Fig. 48 illustrates this model modification using the example of fat 2 at 20 °C. The figure shows the shear stress on the left and the viscosity on the right, each over the shear rate. For fat 2, the Herschel-Bulkley model (red), the experimental data from the rheometer (blue dashed line), and the model modification (red dotted line) are shown.

D u e to the non-negativity condition, until the critical shear rate $\dot{\gamma}_c$ = 16 s^{-1} Newtonian flow behavior is assumed. A comparison of the modified HB model and the experimental data regarding flow behavior is clearly shown in Fig. 48 up to the plotted value $\dot{\gamma}_c$.

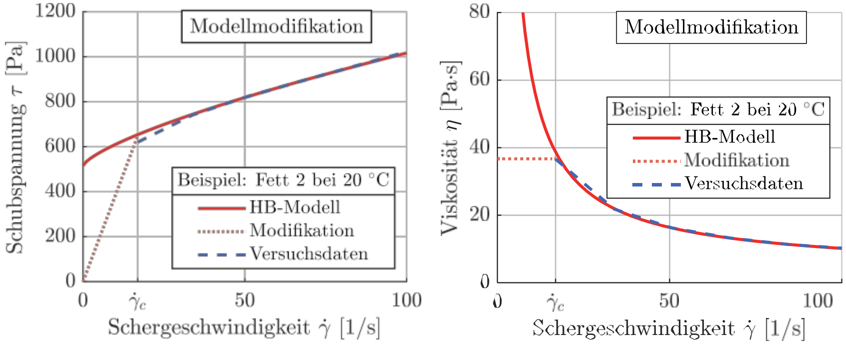


Fig. 48: Flow behavior according to the Herschel-Bulkley model (HB model for short) with model modification and experimental data using the example of fat 2 at 20 °C, left: Herschel-Bulkley model according to Eq. *(26)* and right: according to Eq. *(10)*

There are basically three ways to determine the yield point. One method is to determine the crossover point of the storage modulus and loss modulus in accordance with DIN 51810-2. These results are available and were explained in section 4.1.2. Another method is the regression of phase IV according to DIN 51810-1, which was explained in section 5.1.1. The last option is to determine the inflection point from the deformation behavior according to DIN 51810-2, also known as the log-log method. Table 19 compares the first two methods, as these are the most commonly used. To distinguish between the two determination methods

The value determined by regression was denoted byτ_0 and the value determined by the crossover point

$\tau_f$were labeledτ_f and rounded in Table 19. To illustrate the difference between the two methods, the respective

deviation (abbreviation: Abw.) in percent has been added. The difference between the two methods is considerable in some cases.

Table 19: Methods for determining the yield point between 20-100 °C

Fat	Value	Standard	Test temperatures in P in [°C]								
		DIN	20	30	40	50	60	70	80	90	100
Fat 1	τ_0 [Pa]	51810-1	555	439	395	488	442	425	-212	-867	-978
	τ_f [Pa]	51810-2	716	667	483	481	353	300	295	239	207
	Deviation [%]	-	23	34	18	-1	-25	-42	172	463	572
Fat 2	τ_0 [Pa]	51810-1	515	495	494	442	318	237	116	-30.4	-184
	τ_f [Pa]	51810-2	752	617	553	422	352	292	205	225	178
	Deviation [%]	-	31	20	11	-5	9	19	44	114	203

Since the determination by means of the plate-plate test involves oscillatory stress, the yield strength determined here can be attributed more to viscoelasticity. In summary, based on a comparison of the two determination methods, the explained mapping accuracy of the flow behavior at low shear rates, and the specific application case, the integration of the crossover point into the lubricant model has been omitted.

5.1.3 Extension of the model to include the temperature dependence of the

After the coefficients for describing the shear rate dependence according to the Herschel-Bulkley model have been formed, the coefficients are considered over the temperature rangeϑ (Eq. *(27)*). From each isothermal experiment, as shown in Fig. 46 above

, three coefficients were determined. This made it possible to create a diagram for each coefficient coefficient, a diagram can be created that shows its temperature dependence (Fig. 49). This was used to check whether the dependence can be described by a functional approach. For fat 2, a clear trend can be seen for each coefficient. This observation therefore led to a second-order regression.

$$\tau(\dot{\gamma},\vartheta)=\tau_0(\vartheta)+\kappa(\vartheta)\cdot\dot{\gamma}^{n(\mathrm{P})} \qquad (27)$$

To interpolate the trend, a separate polynomial approximation function was first created for each HB coefficient (Fig. 49, red trend line). Since the polynomial approximation function for fat 1 has large deviations, particularly at some support points, the polynomial interpolation approach was replaced by a piecewise defined function (Fig. 49, blue trend line). This offers absolute accuracy for each coefficient that has already been determined. Fig. 49 shows all the results of the second-order regression for both fats.

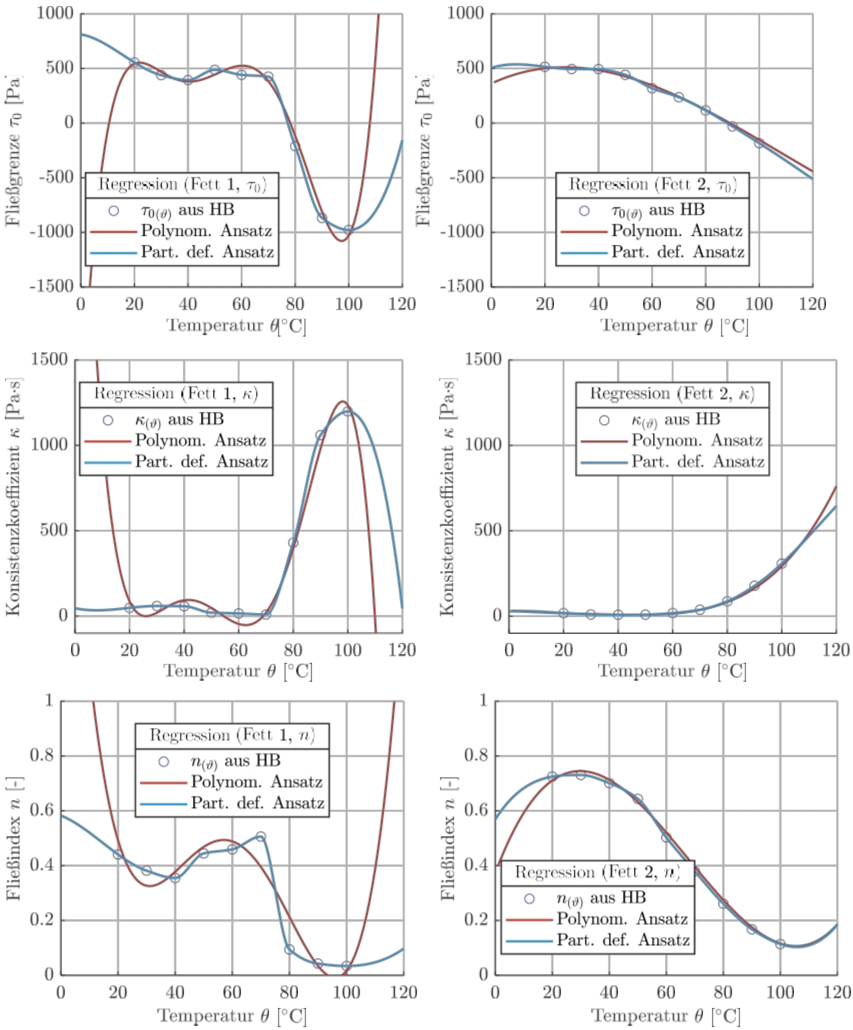


Fig. 49: Regression of temperature-dependent Herschel-Bulkley coefficients according to Eq. *(27)*, fat 1 on the left, fat 2 on the right with (from top to bottom) τ_0 , κ and n

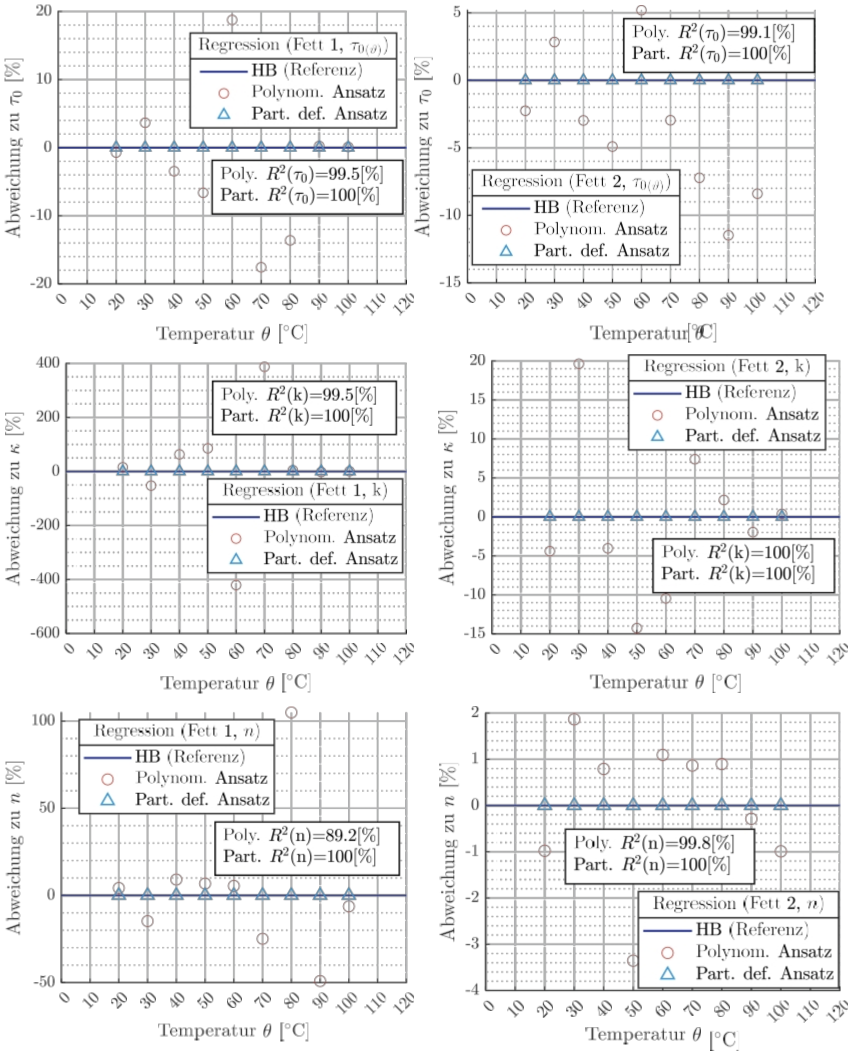


Fig. 50: Evaluation of the interpolation methods for Eq. *(27)* with reference line (dark blue) formed by Herschel-Bulkley regression and, in comparison, two mathematical approaches to interpolation

The yield strength τ_0 lies in a negative range for temperatures above 70 °C or 80 °C. This is due to the structure of the Herschel-Bulkley function according to Eq. *(10)*

. Since large changes in shear stress must be mapped, especially in the range of small shear rates, the y-intercept of the regression lies in the negative range. A fluid can only transmit very low tensile stresses. A non-negativity condition therefore applies to the CFD (see 5.1.2).

The example of fat 1 and 80°C (Fig. 46, above) shows that even at very low shear rates, the HB regression (red line) yields positive values for shear stress. For this reason, with regard to modeling for CFD, the HB model is only activated at a shear rate that yields physically correct shear stresses. This topic is explained in more detail in Modeling in CFD. In addition

Fig. 49 also shows that the flow index n is always in the range between
0 and 1. The consistency coefficient increases significantly above 60°C for both greases.

To evaluate the interpolation method, the results of the interpolation methods were compared to the reference points determined in 5.1.1. This makes it possible to explain the quality based on percentage deviations and to compare the explained approach functions. The two approaches were implemented in the CFD using a C-based script. The Ansys Fluent software provides an interface for this purpose. To check the regression, a model of the cone-plate test was developed, which is described in detail in Chapter 5.2.

5.1.4 3D representation of the developed grease model

Fig. 51 shows a 3D representation of the generated lubricant model for both greases (grease 1 above, grease 2 below). It describes the dependence of the shear stress on the shear rate and temperature, with the model being represented as a surface. Each grease has its own color scale for the shear stress.

The figure shows the shear rate dependence up to a shear rate of 1,000 s^{-1} and the temperature dependence in the range from 20 to 100 °C. In this range, all temperature conditions can be clearly described in terms of flow behavior. Sectionally defined functions were used to interpolate the temperature dependence. These often contain polynomials, which makes them unsuitable for extrapolation. The presented lubricating grease model is not suitable for use above 100°C and below 20°C, as there are no adequate reference points in these ranges.

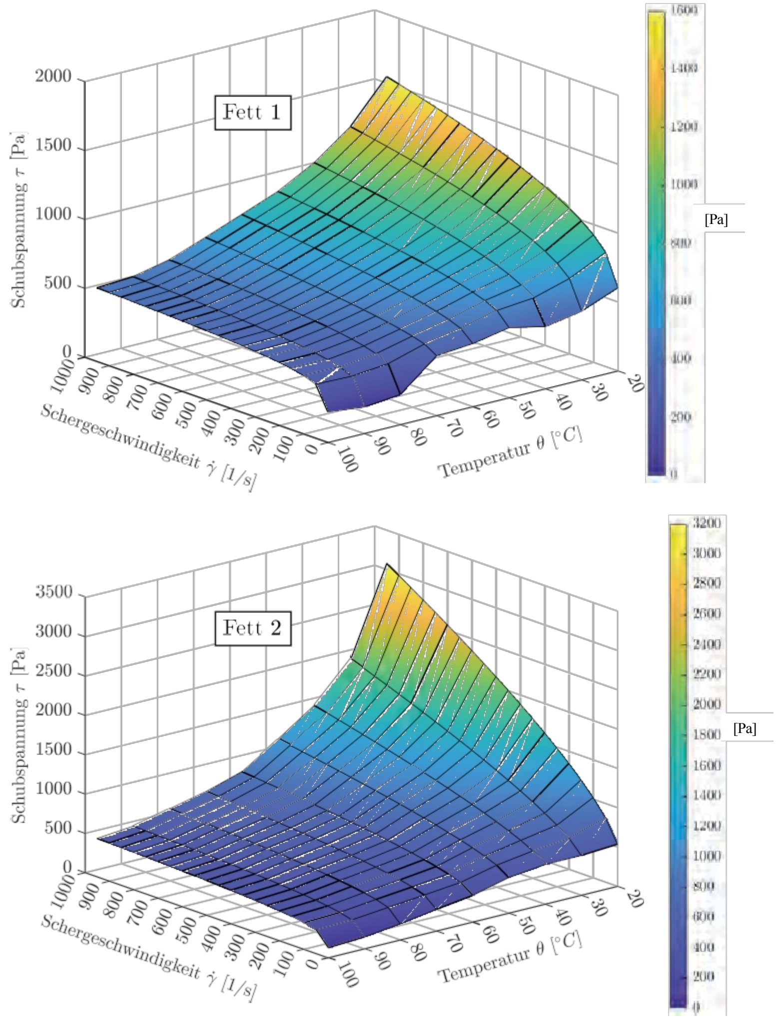


Fig. 51: Shear stress versus shear rate and temperature according to Eq. *(27)* for fat 1 (top) and fat 2 (bottom), whereby the dependencies of shear stress, shear rate, and temperature were generated by the piecewise-defined function shown in Fig. 49

The Herschel-Bulkley model, on which the overall model is based, also offers the possibility of extrapolation with regard to shear rate. In the case of very high shear rates, an extension was made to include the base oil viscosity, which is explained below.

5.1.5 Extension of the model to include base oil viscosity

The Vogel model is used to take into account the temperature-dependent base oil viscosity. The link to the lubricating grease model explained above is made via a condition at high shear rates. This means that the base oil viscosity cannot be lower than the base oil viscosity according to Vogel in the lubricating grease model. Mathematically, this leads to a discontinuity with a very small change in slope.

Table 20 shows the results of the Vogel equation calculations for the base oils of the test greases. Both the conversion values for dynamic viscosity and the coefficients of the Vogel equationC_1 andC_2 $_{\text{are}}$given. Thus

, viscosities at other operating temperatures can also be derived from Eq. *(17)*. For use in Eq. *(1)* and Eq. *(2)*, the base oil viscosities are required, for example. The viscosity at a later point in the CFD simulation is also relevant.

Table 20: Temperature dependence of the base oils for both test greases (see Table 2) calculated according to Vogel (Eq. *(17)*)

Lubricant data	Symbol	Unit	Grease 1	Grease 2
Density Oil	ρ	kg/m³	900	900
Kinematic viscosity	ƌ40	mm²/s	46	249
	ƌ100	m²/s	8	33
Dynamic viscosity	η_{40}	Pa·s	$41.400^{\cdot 10(-)\ (3)}$	$224,100^{\cdot 10(-)\ (3)}$
	η_{100}	Pa·s	$7,200^{\cdot 10(-)\ (3)}$	$29.700^{\cdot 10(-)\ (3)}$
Coefficients of the Vogel equation	C_1	-	333.304	385.085
	C_2	-	-3,851	-3.502

5.1.6 Extrapolation with regard to the shear rate

Fig. 52 shows the extrapolation of the model based on viscosity. On the left, a shear rate limit of 10,000 s^{-1} has been selected in each case to illustrate that the model provides plausible behavior beyond the data base of the standard test. In Fig. 52, the scaling on the right has been chosen so that the approximation of the flow model to the base oil viscosity is clear. In general, a grease approaches the viscosity of its base oil at high shear rates [PALA84].

At 40 °C, the viscosity of the grease intersects with that of the base oil at very high shear rates. For grease 1, this occurs above a shear rate of 50,000 $s^{(-1)}$ and for grease 2 above 100,000 $s^{(-1)}$ ·A modification was therefore made when applying the flow model in this work. As a result

, the model does not assume a value lower than that of the base oil viscosity above the intersection point described.

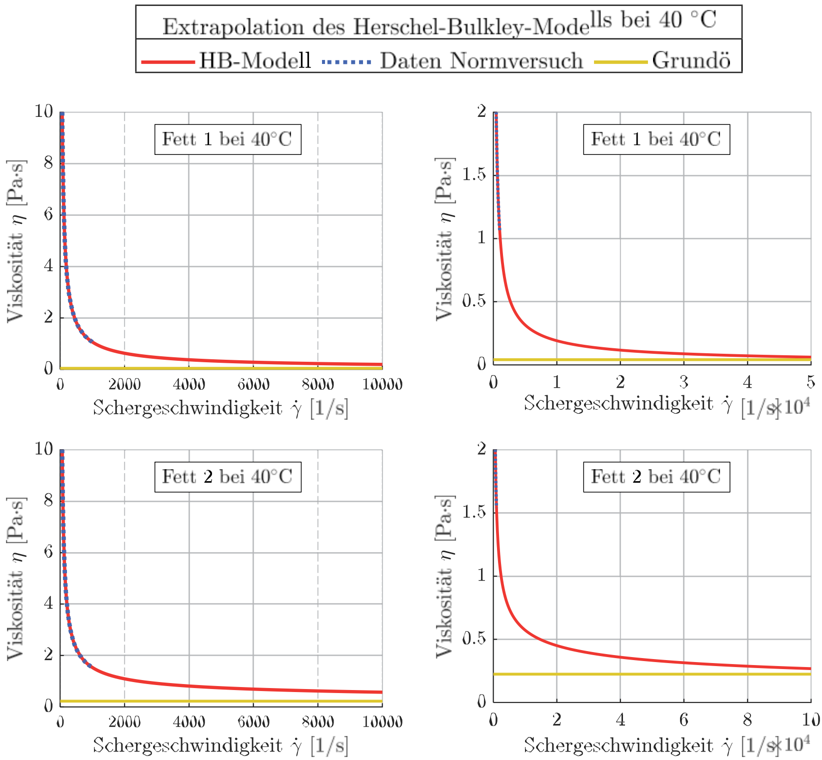


Fig. 52: Possibilities and limitations of the model by extrapolating the shear viscosity at 40 °C and approximating the grease viscosity (red) to the respective base oil viscosity (turquoise)

5.2 CFD model for rheometer

In a rolling bearing, there are many factors that influence the distribution of the lubricant. It is essentially a multiphase flow consisting of grease and air. In addition, the geometry and kinematics are complex, which is why, as already explained at the beginning of Chapter 5, flow effects occur in a superimposed manner. In order to verify the analytical grease model developed (see 5.1), a CFD model was therefore required in which the flow processes are easier to describe than in a rolling bearing. For this purpose, a CFD model was created for a standardized shear test from the rheometer with the cone-plate test configuration. This is a single-phase flow, with the focus of the test being on shear. This allowed various development stages and properties of the lubricant model to be checked step by step. The existing results from the standard tests serve as reference values (see 4.1.1) and thus represent a direct validation option for the analytical grease model.

5.2.1 Boundary conditions

Fig. 53 shows the geometry of the cone-plate model. The cone and plate have their original dimensions in the top views.

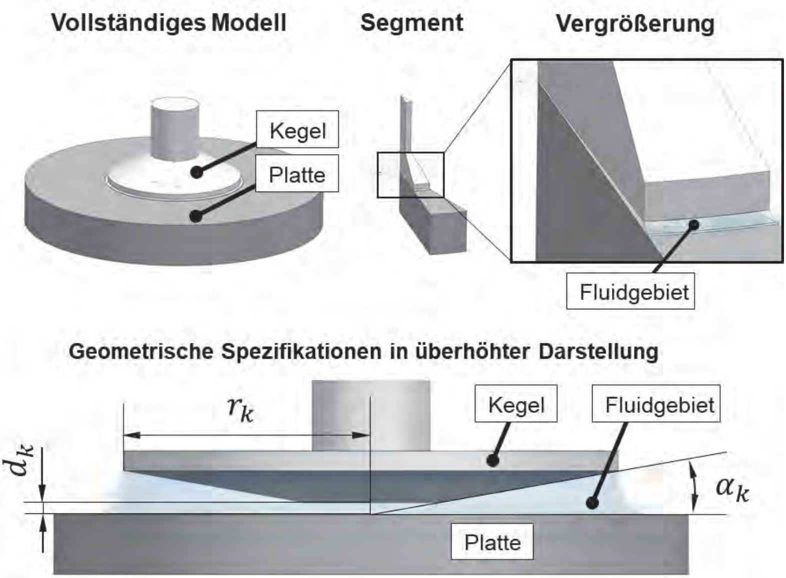


Fig. 53: CFD model for the cone-plate experiment, top: segment of a complete model, bottom: geometric specifications exaggerated

The material properties of the solids were taken into account in the CFD simulation using stainless steel available in FLUENT. Chapter 4.1.1 describes in detail how the temperature control and viscosity determination are carried out. In this way, the temperature of the surface of the cone and the underside of the plate was fixed. In addition, the grease has the exact test temperature at the start of the simulation. After that, heating can take place through shearing, whereby the temperature boundary condition applied to the solids simulates the temperature control of the system. The manufacturing-related deviations described above were taken into account when creating the geometries.

Furthermore, during production, the tip of the cone is reduced by the amount d_k, resulting in a flat surface with a diameter of approximately 6 mm. The cone tip reduction is taken into account both in the cone geometry and in the distance between the cone and the plate. In the experiment, the theoretical cone tip touches the plate. Therefore, the distance between the plate and the flat surface of the cone center always corresponds to the exact cone tip reduction of a respective cone. In many Anton Paar devices, precise adjustment of the distance is ensured by an inductive measuring system, which is integrated into both the cone and the plate and was explained by LÄUGER [LÄUG14]. The specifications on the cone tip decrease d_k, the cone radius r_k and the cone angle α_k are always included in the cone data sheet. The following applies here: Cone angle α_k = 1.02°, Cone radius r_k = 12,487 mm and cone tip reduction d_k = 53 µm.

The third geometry is that of the fluid area, which also takes into account the geometric adjustments to the cone described above. At the outermost radius of the fluid area, the instructions from the standard for correct filling are taken into account [DIN 51810-1]. Since this is a rotationally symmetric model, only a 2-degree segment of the overall model was used for the simulation. Fig. 53 shows an angle segment at the top, which has a larger angle for illustrative purposes. A mesh study was also carried out for this segment (see 5.2.3).

5.2.2 e CFD calculation of the friction torque

Two methods were developed for calculating the friction torque on the CFD model of the rheometer:

1. Method: Friction torque from shear stress.
2. Method: Friction torque from power losses.

The results of the calculation methods were always compared with those of the measurement. For the first method, the shear stress on the cone surface was evaluated. The shear stress was converted into a torque, as the torque corresponds to the actual measured variable of the shear test.

Fig. 54 shows two views of the cone-plate model. In both views, an exemplary volume element E within the fluid area is shown in turquoise . The top view also shows the shear stress components τ_x and τ_z representative of the volume element E entered at its element surface A_E . The element surface A_E of the volume element E should be located at the cone surface here.

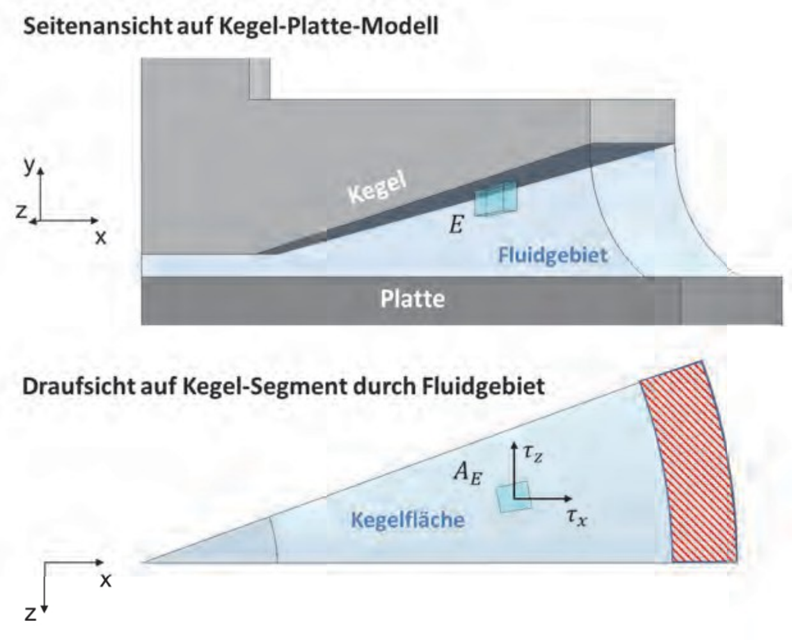


Fig. 54: Views of the CFD model for the rheometer in exaggerated representation: top: side view with exemplary volume element E , bottom: top view of cone surface (blue surface) with surface element A_E on the cone surface and the components of the wall shear stress τ_x and τ_z

The shear stress components are converted into an integral quantity M_{Y-Int} according to Eq. *(28)* :

$$M_{Y-Int} = \frac{1}{n_E} \cdot \sum_{i=1}^{n_E} (\tau_x \cdot A_E \cdot z + A_E \cdot \tau_z \cdot x) \quad (28)$$

First, the respective shear stress component is calculated using the element area at the cone.

A_E converted into a force component. Each force component is converted into an area moment using the corresponding lever arm.

force component is converted into an area moment. The sum of all calculated area moments is divided by the number of area elements on the cone, which is why the resulting moment M_{Y-Int} represents an integral value.

The second calculation method is based on calculating the friction moment from the power losses due to shear heating. This power loss is formed from a term in the energy equation. The energy equation is the first law of thermodynamics for an open system. It includes the part of internal heating that is also referred to as shear heating. In the ANSYS documentation, shear heating is described by the product of the total shear stress tensor

τ''_{eff} and the resulting total velocity vector $\vec{v}$ [ANSYS15,

p. 130]. The components refer to a representative volume element in the calculation area. However, a different formulation can also be used to describe the energy equation (cf. Eq. *(29)* according to OERT02). The thermal energy equation generally applies to incompressible single-phase flow and consists of four terms: convection, heat conduction, friction, and heat radiation [BART10]. It describes volume-related quantities, all of which have the unit watts per cubic meter.

$$\rho \cdot c_v \cdot \overbrace{\left(\frac{\partial\vartheta}{\partial t} + u\cdot\frac{\partial\vartheta}{\partial x} + v\cdot\frac{\partial\vartheta}{\partial y} + w\cdot\frac{\partial\vartheta}{\partial z}\right)}^{Konvektion} - \overbrace{\left(\lambda_x\cdot\frac{\partial^2\vartheta}{\partial x^2} + \lambda_y\cdot\frac{\partial^2\vartheta}{\partial y^2} + \lambda_z\cdot\frac{\partial^2\vartheta}{\partial z^2}\right)}^{Wärmeleitung} = \underbrace{\eta\,\Phi}_{Reibung} + \underbrace{\rho\,\dot{q}_s}_{\text{Wärmestrahlung und/oder chemische Prozese}} \quad (29)$$

The equation contains various spatial and temporal derivatives of temperature, the specific heat capacity at constant volume c_v , and the mass-related

Energy source or sink $\dot{q}_s$. In addition, the term for shear heating (friction) is included

is included. This is formed from the dynamic viscosity and the dissipation function Φ, for which the following applies:

$$\Phi = 2\cdot\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2\right] + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial w}{\partial x}\right)^2 - \frac{2}{3}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)^2 \quad (30)$$

The dissipation function Φ is directly related to the shear rate $\dot{\gamma}$. Using the volume of the element V_E, this relationship can be used to determine the power loss P according to Eq. *(31)*.

$$P/V_E = \sum_{i=1}^{n_E} \eta \cdot \Phi \tag{31}$$

The angular velocity ω is known at each time step, which is why the friction torque M can be determined according to Eq. *(32)*.

$$P = M \cdot \omega \tag{32}$$

Both calculation methods were implemented using user-defined functions (UDF) in ANSYS via a C script.

5.2.3 Mesh study

The mesh study aims to assess the mesh independence of the calculation results by means of an error analysis. To this end, a structured mesh of the fluid domain was first created. The fluid domain is described by a 2-degree segment (see 5.2.1) between the cone and the plate. The number of elements for each edge of this structured mesh was increased step by step. In this way, five meshes with increasing numbers of elements were created. These meshes were used to perform stationary simulations at a shear rate of 1,000 s($^{-1)}$ with grease 2 at 20 °C. The moments derived from the cone surface are shown graphically in Fig. 55. The number of elements (on the x-axis) and the moment at the cone surface (on the y-axis) are shown. The meshes are represented as circles and the selected mesh is shown in red.

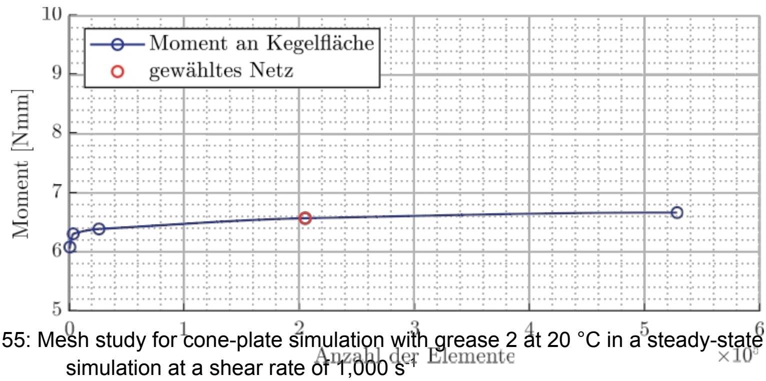


Fig. 55: Mesh study for cone-plate simulation with grease 2 at 20 °C in a steady-state simulation at a shear rate of 1,000 s^{-1}

It becomes clear that the result value (the moment) asymptotically approaches a limit value as the number of elements increases. The deviations between the selected grid with 204,800 elements and the finest grid with 528,000 elements are less than 2%. At the same time, the computing time for the finest grid is significantly higher than for the selected grid. Since time-dependent (transient) solutions are planned with the mesh, a compromise had to be made here in terms of computing time.

5.3 CFD model for the rolling bearing

For the simulation of grease-lubricated rolling bearings, 3D CFD models were created for the SKL 7007, the RRKL 6007, and the AZRL 81212. The model properties and calculation settings are explained below, and the calculation results are presented. The analytical grease model developed in Chapter 5.1 is used to describe the flow behavior of greases.

5.3.1 Boundary conditions

In principle, every rolling bearing can be divided into rotationally symmetrical segments according to the number of rolling elements. To reduce the calculation time, rotational symmetry was exploited and an angular segment of the overall bearing was considered according to the number of rolling elements (Fig. 56, center). The fluid area is derived from the internal geometry of the rolling bearing (Fig. 56, right).

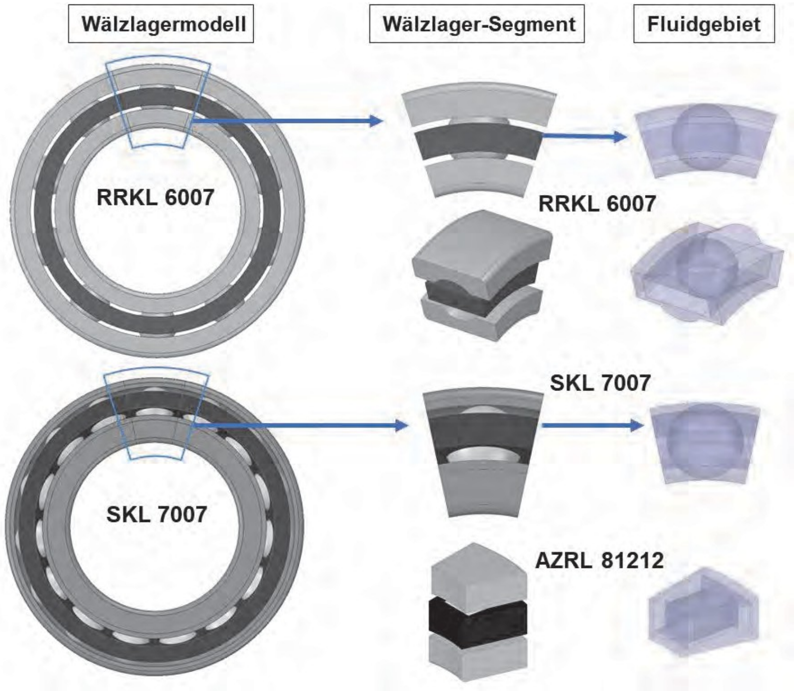


Fig. 56: Rolling bearings of different designs, left: full model, center: angle segment, right: fluid area of the test bearings (see Table 1)

No solids were considered. The system is considered adiabatic. In addition, the bearing back temperature measured in the rolling bearing tests was set as a uniform isothermal temperature throughout the entire fluid area.

Due to the reduction to a rolling bearing segment, special kinematic conditions apply, as illustrated in Fig. 55. In the rolling bearing simulation, a rolling bearing segment is a system of the moving cage. The centrifugal forces act on the fluid due to the rotational speed of the entire system.

This is measured by the cage speed ω_K . The speeds of the inner ring ω_{IR} and outer ring ω_{AR} are expressed by the difference to the cage speed. The rotational speed of the rolling element ω_{WK} is measured by the prevailing operating pressure angle in the roller bearing. The periodic boundary condition is applied at the cut surface. In the experiment, the seals are attached to the side surfaces and are supported by the outer ring. Therefore, they have the outer ring speed ω_{AR}.

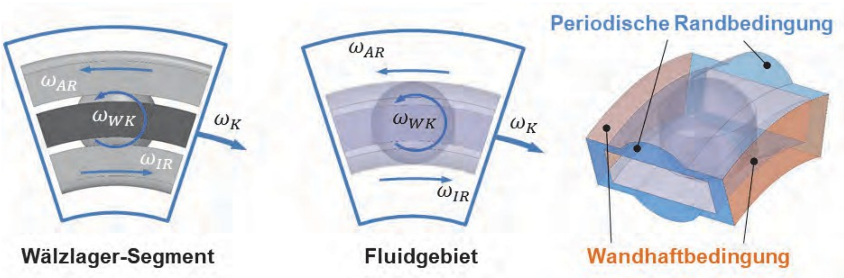


Fig. 57: Boundary conditions in the CFD model of a rolling bearing segment (Fig. 56) using the example of the RRKL 6007

The values listed in Table 21 for the contact kinematics were determined using the BEARINX calculation software. These include the contact angle α_p , the contact normal force F_{KN} and Hertzian pressure p_{max} in the contact area. The results clearly show that for both roller bearings, the angle at the inner and outer rings changes only marginally up to 3,000 rpm.

Since meshing is complex due to the complex geometry, solid-state movement is not taken into account here. Mesh movement was also omitted due to the high cost involved. Instead, the contact angle was taken into account in each case by a mean value in the CFD model. Furthermore, rigid, non-elastic bodies and smooth surfaces were assumed as conditions.

Table 21: Values determined with BEARINX for contacts in ball bearings 6007 and 7007

Rolling bearings	Contact	Value	Unit	Speed in rpm					
				50	100	1500	2000	2500	3000
RRKL	IR	α_p	[°]	9.1	9.1	9.1	9.2	9.2	9.3
		F_{KN}	[N]	46	46	46	46	45	45
		p_{max}	[MPa]	819	819	818	817	816	814
	AR	α_p	[°]	9.1	9.1	9.1	9.1	9.0	9.0
		F_{KN}	[N]	46	46	46	46	46	46
		p_{max}	[MPa]	887	887	887	887	887	887
SKL	IR	α_p	[°]	40.4	40.5	40.7	41.0	41.3	41.7
		F_{KN}	[N]	41	41	41	41	40	40
		p_{max}	[MPa]	741	740	739	738	736	735
	AR	α_p	[°]	40.4	40.3	4.2	40.1	39.9	39.6
		F_{KN}	[N]	41	41	41	41	41	41
		p_{max}	[MPa]	719	719	719	720	720	720

The simulations were initialized with the respective temperatures determined in rolling bearing tests as the temperature boundary conditions. This means that the simulation is not (yet) capable of being performed independently of tests.

5.3.2 Networking

The complete meshing of a ball bearing poses a particular challenge in simulation. This is partly due to the large differences in clearance, which results in a wide range of shear rates. In addition, pressure and shear flow overlap at the concentrated contact. Several million elements were required in project FVV 1277 to correctly map a concentrated contact. Since the distribution simulation in the entire rolling bearing requires just as many elements in the vicinity of the rolling element-raceway contact, a simplification is made here. The meshing of the concentrated contact is based on the assumption that the contact between the rolling element and the raceway plays a rather minor role in the grease distribution throughout the rolling bearing. This applies in particular to the minimum load test condition applicable here. Examples from the literature are given below: In the CFD model of rolling bearings, the narrowest gap was either enlarged [CONCLI20] or completely closed [LIEBR15, FELD17]. In this work, the contact area between the rolling element and the raceway was set to 10 µm.

Fig. 58 shows the meshes used in the simulation. It should be noted that the AZRL 81212 mesh is fully structured, whereas the ball bearing meshes were created using a free mesher. Each of the meshes has a refinement in the area of the rolling element-raceway contact, which was achieved in the AZRL by a structured split height of ten elements. All meshes were created using ANSYS ICEM software.

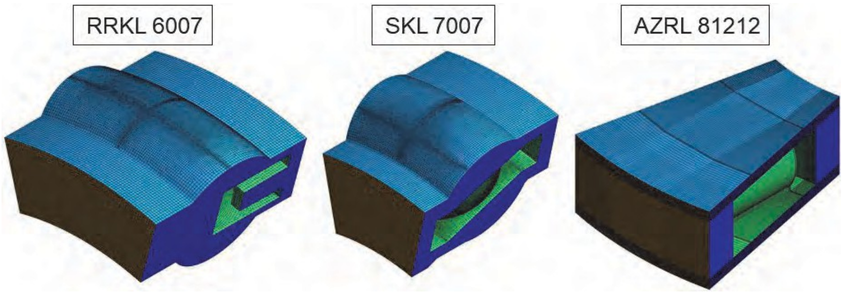


Fig. 58: Meshing of the fluid areas of different types of rolling bearings

In principle, structured meshes are advantageous for CFD simulation because the geometry of the elements can be adapted to the flow in the fluid domain. However, there are very large differences in shear rate in rolling bearings due to their complex geometry. For example, neighboring areas in the rolling bearing have very low shear rates on the one hand and very high shear rates on the other. These areas result in very different objectives for the meshing. Areas with high shear rates require a significantly finer resolution (higher number of elements) than areas with lower shear rates. In some cases, however, it is not possible to reconcile the different numbers of elements in such adjacent areas using clever meshing strategies. This leads to compromises, whereby areas with low shear rates, for example, must have a (too) fine mesh in order to cope with the adjacent areas with high shear rates. This is a major disadvantage of structured meshes for roller bearings. The biggest disadvantage of free meshers is the mesh quality and the very high number of elements. This is due to the way free meshers work. First, a good surface mesh is formed. Then, areas with large gap dimensions are filled with hexahedral elements. This is also a good feature, as low shear rates prevail here due to the large gap dimensions. However, the gaps between the hexahedrons and the surface mesh are filled with a large number of pyramid elements, as can be seen from the widely varying number of elements in Table 22. In addition, these pyramid elements have a significantly poorer mesh quality than hexahedrons (specifically: cuboids).

Table 22: Networking type and number of elements in the networks presented in Fig. 58

Type of network	Rolling bearing	Number of elements
Freely networked	RRKL	2,039,715
	SKL	1,757,007
Structured networked	AZRL	789,296

In addition to free networking and fully structured networking, there is also the option of implementing a hybrid form. This allows different areas to be networked as desired. Conformity of the elements at the transitions between areas (edges) is not necessary here. However, these networks are very time-consuming to simulate, as the transitions between different networks require additional computing power and thus computing time. For this reason, a hybrid form of networking was not used in this work.

A network study, as presented in 5.2.3, was not carried out in full here. Instead, isolated tests with networks were carried out during the first simulations to determine whether further refinement of the network would be worthwhile. In the case of the AZRL, this was not carried out in full due to the computational effort involved, as a grease distribution simulation, for example, took several weeks to months. For the RRKL and SKL, the meshes were created with a free mesher. During creation, there were often meshes that contained arbitrarily faulty elements, e.g., with negative volume, or elements that protruded beyond the geometric boundaries of the rolling bearing models. These errors in the meshes could not be corrected using repair tools. As a result, it was only possible to generate different gradations of mesh fineness to a very limited extent.

5.3.3 Multiphase flow

Rolling bearings are often lubricated with a quantity of grease equivalent to 30 percent of the free bearing volume (see 2.3). The remaining bearing volume contains ambient air. A multiphase model was used for this purpose, which takes into account the two phases of grease and air. No distinction was made between thickener and oil, making this a two-phase model. The FLUENT software offers three models for mapping multiple phases:

- Mixture Model
- Volume of Fluid
- Eulerian Model

In the order listed, the models depict the boundary between fluids with increasing accuracy. In the Mixture Model, a homogeneous mixture of both fluids is considered, whereby a local mixture density and mixture viscosity are calculated. This allows for very large time steps in a transient simulation, for example. However, in the Mixture Model, the phase boundary is depicted only very imprecisely and blurred. In contrast, the Volume of Fluid Model determines the volume fraction in each cell for each phase, which ranges between 0 and 1. In relation to the fat, a volume fraction of 1 would correspond to a full filling of the volume element. In the Eulerian model, pressure, temperature, and velocity fields are solved for each phase throughout the entire fluid domain. This means a considerable amount of additional work for the simulation and significantly longer computing time. In order to generate a compromise between a sufficiently good phase boundary and saving computing time, the Volume of Fluid (VOF) model was chosen to represent the two phases of fat and air.

5.3.4 Initial e greasing

Before a real rolling bearing test, a rolling bearing is greased using a standard syringe. The exact amount of grease is distributed evenly in the spaces between the rolling elements, the cage, and the respective bearing rings. In relation to the cross-section of a rolling bearing, the grease is therefore applied "centrally." This procedure was taken into account in the CFD simulation.

Fig. 59 illustrates various aspects of the initial lubrication in the CFD simulation and the kinematic boundary conditions using the RRKL 6007 as an example. On the left, you can see the initial lubrication of the rolling bearing in the CFD, which is based on the actual lubrication of the rolling bearing. This type of rolling bearing has a snap cage. Before the roller bearing tests, the lubricating grease is therefore introduced on one side, away from the cage. To visualize the grease phase, the phase boundary was displayed in orange-transparent. The phase boundary is a result of the phase model (VOF), which takes the phase volume fraction into account. All values between 0 and 1 are located between the two phases. A value above 0.4 was specified for all representations of the phase boundary. The phase boundary is also shown in orange-transparent to provide a view behind the boundary.

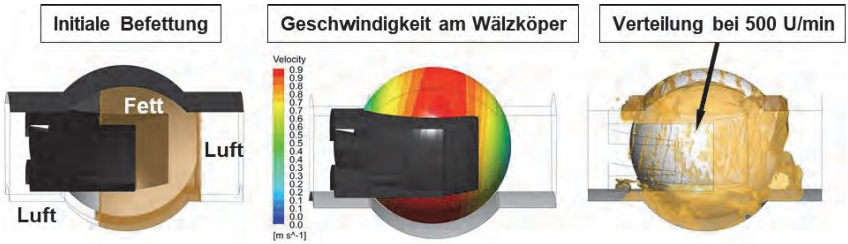


Fig. 59: Stationary simulation with RRKL 6007 and grease 1 at 500 rpm, showing the phase boundary between air and grease, left: initial greasing, center: speed on the surface of the rolling element, and right: example simulation result

Fig. 59 shows the speed on the surface of the rolling element in the center. This illustrates the operating pressure angle of 9.1° in the RRKL shown in Table 21. On the right, you can see the result of the simulation for grease 1 at 500 rpm. The distribution in the cage pocket can be seen due to the hidden cage. Here, the movement of the lubricant is dominated by the movement of the rolling element, which is evident from the speed (Fig. 59, center).

5.3.5 Convergence

In computational fluid dynamics, the complete Navier-Stokes equations are solved approximately, which is time-consuming compared to Reynolds solvers. As shown in Section 3.3, steady states are only reached after minutes or hours of test running time. Therefore, two strategies were pursued in the simulation of rolling bearings. On the one hand, a steady-state friction torque is established in the test at constant test parameters (speed, temperature, load). On this basis, the assumption of a quasi-stationary grease distribution was made. On the other hand, a simulation can also be performed in a non-stationary manner. However, since it is necessary to reduce the time step size as the speed increases, a direct consequence is an increase in computing time. With a view to the long-term simulation goal (high speeds), a non-stationary simulation approach is therefore fraught with the problem of long computing times. In order to be able to map grease distribution at higher speeds with a large number of elements, as is the case with ball bearings (RRKL and SKL), a steady-state simulation was used. In the case of multiphase flow, the convergence of a steady-state simulation is not easy to assess, as the end of grease distribution in the rolling bearing is theoretically never complete.

In principle, it is advisable to consider result variables in addition to classic residuals when assessing convergence in CFD simulation. The friction torque caused by the grease was also evaluated in order to assess the quality of the solution. In the case of RRKL and SKL, this was done exclusively on the surfaces of the roller bearing. This result variable is based on the measured variable: the friction torque measured across the outer ring. With each new speed level, good convergence should be achieved with regard to the torque on the outer ring. Convergence problems frequently occurred during the stationary simulation, particularly with RRKL, which is why these calculations became increasingly implausible at speeds above 500 rpm. Fig. 60 shows the convergence behavior of a stationary simulation with the RRKL 6007 and grease 1 to 500 rpm. Similar to a speed stage test, there was a gradual approach to the maximum speed of 500 rpm. Satisfactory convergence was achieved at each intermediate step.

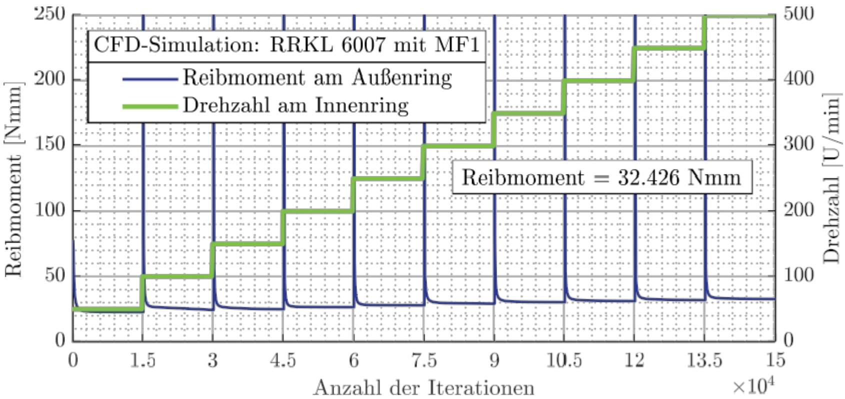


Fig. 60: 3D CFD simulation of rolling bearings: Convergence behavior of a stationary simulation with RRKL 6007 and grease at 2 to 500 rpm

The friction torque reached 32.4 Nmm at 500 rpm. In repeat tests on the rotational tribometer (see Fig. 23), this was 26 Nmm with a dispersion of +/- 8. In principle, the calculation of the torque on the outer ring surface seems plausible.

5.3.6 Stationary simulation with the RRKL and the SKL

In stationary simulation, a time-independent solution to a problem is sought. This requires that a finite state be established. In a first step, the Simple (too dt. simple) and Coupled (dt. coupled), both of which are considered robust. Convergence was very slow and fluid movement was very low. The pseudo-transient option can be selected to accelerate convergence. This is a form of implicit under-relaxation that has proven to be effective for stationary rolling bearing simulation. In addition, the rolling bearing temperatures derived in the tests were used for the lubricant temperatures at initiation, which can be seen in Table 23 and taken from the repeat tests of the rotation tribometer (Fig. 23).

Table 23**:** Initial grease temperatures in °C for the CFD simulation at 500 rpm

	Fat 1	Fat 2
RRKL	31	32
SKL	28	33

Fig. 61 shows simulation results for the RRKL 6007 with the same boundary conditions for grease 1 (left) and grease 2 (right). A certain amount of lubricant accumulates in front of the rolling element on both the inner and outer rings. Behind the contact area, however, the lubricant spreads in the form of tracks and flows back to the accumulated amount of grease (boundary condition in Fig. 57).

In addition, there are areas in the roller bearing that hardly move at all after their initial distribution. In particular, there are accumulated amounts of grease on the cage surface that hardly change in terms of quantity and position. In the area of the rolling element-raceway contact (rolling contact for short), there are greater differences in the amount of lubricating grease. Here, grease 2 is distributed more evenly across both the outer and inner rings. Overall, grease 2 appears to be distributed more widely and evenly than grease 1. Grease 2 produces fewer small fragments. This means that high pressures occur less frequently in the simulation, which may be due to its comparatively better flowability.

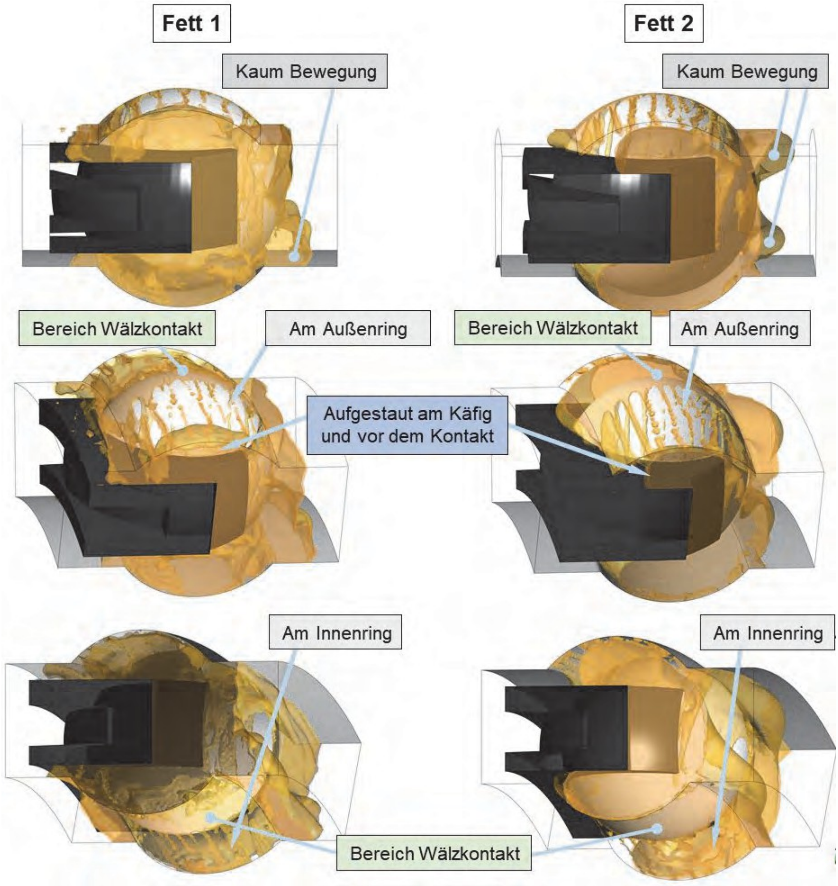


Fig. 61: Stationary simulation of grease distribution with RRKL 6007 and grease 1 (left) and grease 2 (right) at a speed of 500 rpm

The SKL shows very similar distribution characteristics. Fig. 62 clearly shows the distribution along the contact angle of 40° around the rolling element. Simulation results for speeds of 500 rpm (left) and 1,000 rpm (right) are shown. At 500 rpm, there is also a distribution along the outer ring in the form of tracks. Similar images were shown in the high-speed recordings through the transparent outer ring in the special tribometer tests (4.5.2).

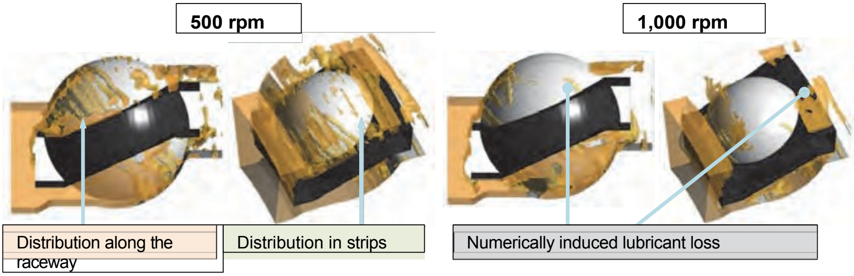


Fig. 62: Stationary simulation of grease distribution with SKL 7007 and grease 1 at left: 500 rpm and right: 1,000 rpm

However, as the speed increased, numerically induced mass losses of the grease occurred. Minor errors accumulated over the entire simulation period. The mass loss was observed exclusively in the stationary simulations.

Unlike in transient simulation, fluid movement occurs at each iteration in steady-state simulation. Constant redistribution can cause the phase boundary to become inaccurate. When using a VOF model, all phase volume fractions between 0 and 1 can occur at the interface. However, this can also blur the boundary between the lubricating grease and air phases, which can impair the interpretability of the results. For this purpose, conventional fluid mechanics software products such as Fluent have compression algorithms that continuously sharpen the phase boundary. These algorithms prevent the phase boundary from blurring. In particular, at the phase boundary, relatively small numerical errors in the mass conservation of the grease occurred during each iteration, which could not be detected using the classic residual – mass conservation. In the future, this effect must be reduced either by source terms (adding lubricant to compensate for losses) or improved solution settings.

This error can be seen starting at SKL in Fig. 62 between the speed stages shown. Up to 500 rpm, the largest error was less than 4% relative to the original mass. This corresponds to a loss of less than 84 mg for a total mass of 2.1 g. For further investigations at the AZRL, therefore, only the much more computationally intensive transient simulation was used in order to avoid these problems.

5.3.7 Unsteady simulation by AZRL

The transient simulation of roller bearings was performed with the AZRL 81212 at 500 rpm with both greases. The mesh of the fluid domain has fully conforming elements, which is why it has the best mesh quality compared to the other two types of rolling bearings. At the same time, the fluid domain could be meshed with a comparatively small number of elements due to the type of meshing (Table 22). This makes this mesh particularly suitable for simulating a ramp-up. Nevertheless, the time step size was only 2E-06 seconds. In tests with ball bearings, the time step sizes were even smaller. Nevertheless, simulating a ramp-up with the AZRL to 500 rpm took several weeks of computing time, which illustrates the effort required to produce the results.

Furthermore, the calculation of the friction torque from shear heating, which was explained in section 5.2.2, was also implemented here. The fluid domain has the temperature determined from the tests on the special tribometer as its initial solution (see Fig. 39, for grease 2: 38 °C and for grease 1: 33 °C).

Fig. 63 shows two views of a transient simulation with grease 2 when reaching a speed of 500 rpm. On the left, the grease distribution is shown by the phase volume fraction from 0 to 1, and on the right, the phase boundary is shown. This type of result representation clearly shows various basic characteristics of the grease distribution. The grease collar is clearly visible based on the phase volume fraction. This forms at the front ends of the rolling element and adheres to the rotating shaft disc. Between these is the raceway of the rolling element, which is provided with very small amounts of lubricant. A considerable amount of grease is distributed between the cage and the rolling element in the cage pocket. Lubricating grease is also visible on the cage pocket near the shaft disc. The semi-transparent representation of the phase boundary (Fig. 63, right) helps to see into the depth of the model. Here, the distribution on the upper cage surface (towards the shaft disc) can be seen even better. There is also an accumulation of lubricant, especially in front of, but also behind, the rolling element.

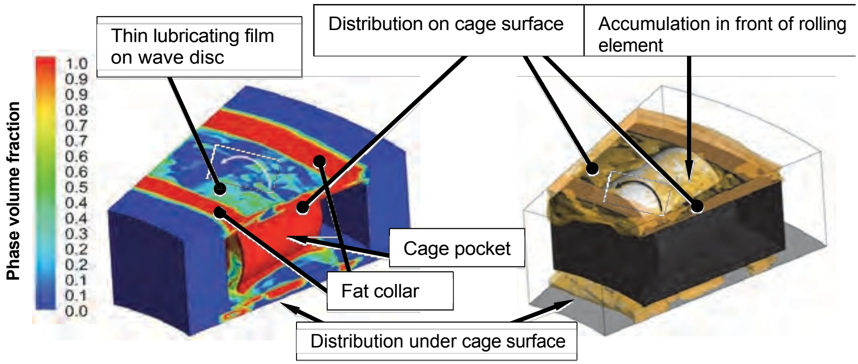


Fig. 63: Transient simulation of AZRL 81212 with grease 2 until a speed of 500 rpm is reached, left: Phase volume fraction for grease phase 100% in red (VF= 1) and for air phase with 100% air in blue (VF= 0), right: Phase boundary (transparent orange) for view of upper cage surface and rolling element

In summary, grease distribution can also be simulated using the transient (time-dependent) solution. Whether this calculated lubricating grease distribution is realistic will be discussed in the following chapter based on a comparison with the optical images. The results for both greases will also be presented here, and the evaluation of the friction torques from CFD and the experiment will be combined.

6 Discussion of the CFD results

This chapter discusses the results from the experimental investigations and the CFD simulations. To this end, both results are compared with each other. In addition, the consequences of kneading are discussed with regard to the flow behavior of greases. Finally, the most important findings from the rolling bearing tests are classified in the context of widely used calculation methods and typical practical applications.

6.1 Comparison between experiment and simulation

6.1.1 Model validation with rheometer tests

To validate the analytical grease model (5.1) and the cone-plate model (5.2), the results from the simulation and from the test based on the standardized shear test on the rheometer (see 4.1.1) are now compared. In the test, the moment is the measured variable, which is converted into viscosity using Eq. (*24)* and Eq. (*23)*. In the simulation, two methods for calculating the friction moment were developed (see 5.2.2):

1. Method: Calculation from shear stresses according to Eq. *(28)*
2. Method: Friction torque from power losses according to Eq. *(31)*.

Since there is currently no evidence for these calculation methods, they are also shown in the following figures.

Fig. 64 shows the results for Fat 2 at 20 °C. The moment is plotted on the first y-axis (left). The moment from the experiment is shown as a black line. This is followed by the results of the simulation using the calculation methods described. The first method is shown in red and the second method in blue. The relative deviation from the standard test is shown on the second y-axis (right). These deviations are shown in green for each calculation method in relation to the test. The test time on the x-axis is based on the actual test duration according to the standard test (see 4.1.1, Phase IV of the standard test). During this period'the cone is accelerated at a constant rate until it reaches a maximum shear rate of 1,000 s$(^{-1)}$.

The results in Fig. 64 show that the same friction torque is determined using both calculation methods. There is hardly any deviation between the two. This can be seen from the respective deviation from the results of the standard test (green in each case). The deviations from the standard test are less than 4% for large parts of the test. The uneven deviation curve is due to the relatively small amount of data from the standard test. Only one value is available for each second. Overall, the deviations are relatively small.

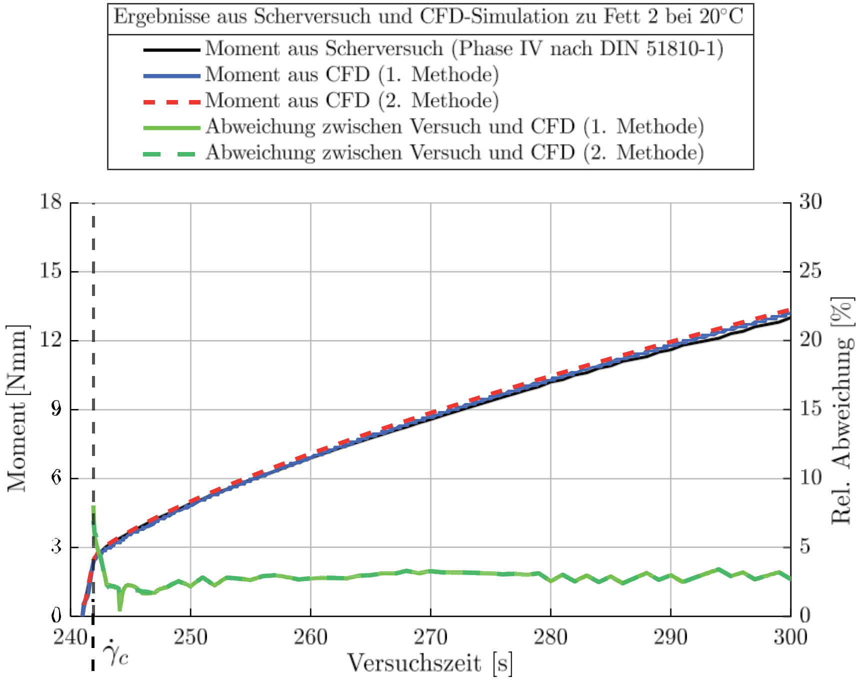


Fig. 64: Simulation of phase IV of the cone-plate standard test (see 4.1.1): Comparison of calculation methods for determining the friction torque in the CFD simulation using method 1 according to Eq. *(28)* and Eq. *(31)* and the respective deviations from the test for fat 2 at 20 °C

Fig. 65 shows the results for fat 1 at 80 °C. When regressing the rheometer data, the worst coefficient of determination was achieved with the data set for fat 1 at 80 °C (see Fig. 46, top left). Fig. 65 shows that after shearing begins, the deviations fall below 5%. Thus, a very good representation of the flow behavior has been achieved here as well. Such a time-dependent simulation of the shear test requires one week of computing time on a computing cluster using over 30 cores.

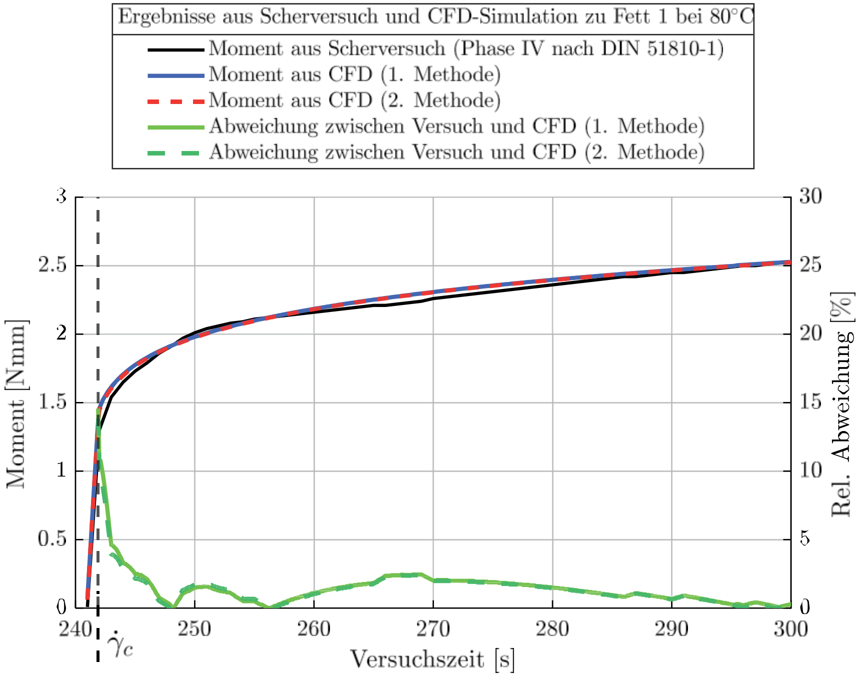


Fig. 65: Simulation of phase IV of the cone-plate standard test (see 4.1.1): Comparison of calculation methods for determining the friction torque in the CFD simulation using method 1 according to Eq. *(28)* and Eq.
(31) and the respective deviations from the test for grease 1 at 80 °C.

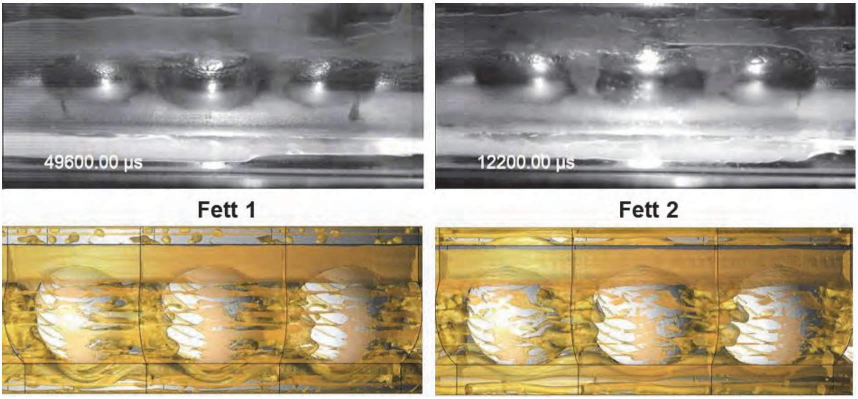
The validation of the analytical grease model showed that the CFD simulations for the shear test only deviated slightly from the actual shear test. The actual measured variable—the moment—was used for this purpose. This proved the successful implementation of the flow model in the CFD software. Furthermore, the calculation of the moment via the shear stress on the cone surface leads to the same result as the calculation of the moment via the power loss (see calculation methods in 5.2.2). This demonstrated that both calculation methods are valid.

6.1.2 e of grease distribution and friction torque in rolling bearings

The investigations on the special tribometer provide high-speed recordings of the grease distribution at 500 rpm for both the SKL and the AZRL. CFD simulation results are also available for these operating points. The grease distributions from the experiment and from the CFD are compared below.

Fig. 66 shows the results from the SKL. The test results for both greases are shown at the top and the simulation results at the bottom. Previously, only the simulated segment of the CFD model of a rolling bearing had been displayed. For better visualization of the simulation results, the segment was duplicated according to the rolling bearing boundary conditions and rotated around the bearing rotation axis (Fig. 66 below).

Fig. 66: Comparison of high-speed recordings with the SKL 7007 and a stationary distribution



simulation at 500 rpm, left: grease 1, right: grease 2

Numerous effects can be seen in the test image, which are also reflected in the CFD. A grease collar forms at the top in each case. The upper grease collar is slightly narrower and more visible in test 1 than in test 2. In contrast, the upper grease collar in test 2 has been pushed further upward and is wider overall than in test 1. This is also evident in a similar form in the simulation. The contact area can also be seen. In the area between the grease collars, a thin lubricating film layer forms for both lubricants in the real test. The grease located in the area of the cage is also visible in the simulation.

Fig. 67 shows the results with AZRL 81212. The optical measurements are now compared with the results of the 3D CFD simulation in the same way. The images were taken with a high-speed camera at the point of steady state (after three hours). Numerous similarities become apparent, which are explained below.

In the experiment, the outer edge of the grease collar is interrupted in grease 1, which was also evident in the CFD simulation. It is also relatively narrow compared to grease 2. In the CFD, the collar was also wider on the outside. There are there are further similarities. In the experiment with grease 1, the surface of the cage was often still visible. With grease 2, on the other hand, this can only be seen in a few places. Overall, grease 2 forms a much more uniform layer thickness on the cage surface. This characteristic was also evident in the CFD. First, the areas before and after the rolling element are analyzed. In the test, lubricant had built up on the cage surface before and after the rolling element. With grease 2, this protrudes further into the contact. This property can also be seen on both sides in the CFD.

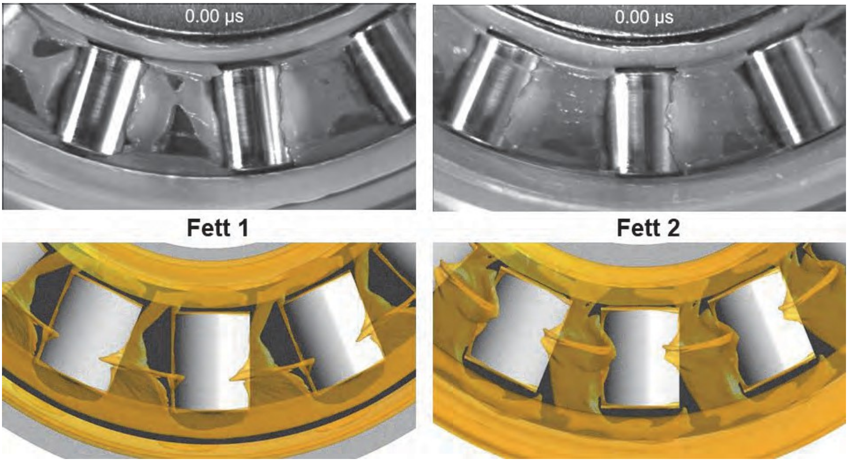


Fig. 67: Comparison of high-speed images with the AZRL 81212 and a transient distribution simulation at 500 rpm each, left: grease 1, right: grease 2

In the CFD simulation, friction torques can be determined across relevant areas, as demonstrated using the cone-plate rheometer as an example (Fig. 64). Table 24 compares friction moments from the experiment and from the CFD simulation. Here, the CFD calculates the friction moment for the AZRL using the shear stress on the housing disc and for the SKL and RRKL using the shear stress on the outer ring (method 1, see 5.2.2). The results show that most deviations are in the double-digit percentage range (Table 24, right column). The calculation of friction from shear heating was implemented for the AZRL (method 2, see 5.2.2). However, this evaluation led to significantly larger deviations, which were 100 times greater than the values presented in Table 24. Possible reasons for the deviations of this calculation method could be identified in post-processing. Here, the rolling contact area dominated the

Shear heating was evident throughout the entire fluid area. Incorrect, excessive shear heating could explain the deviations from this calculation method. However, it was not possible to conclusively determine why the shear heating was too high locally.

The AZRL was calculated as unsteady and the two ball bearings as steady. The results show that plausible values are determined in both cases. This confirms that distribution simulations with the developed grease model are possible in principle, both steady and unsteady.

Table 24: Comparison of friction torque between rolling bearing test and CFD simulation from the evaluation of shear stresses on the housing disc for the AZRL and from the outer ring for the SKL and RRKL

Rolling bearing type	Lubricating grease	Rotational speed	Test	CFD	Deviation
		[rpm]	[Nmm]	[Nmm]	[
AZRL 81212 (unsteady)	Fat 1	50	60	83	+38
	Fat 2		150	110	-25
SKL 7007 (stationary)	Fat 1	50	40	49	+23
	Fat 2		64	38	-40
RRKL 6007 (stationary)	Fat 1	500	26	32	+25
	Fat 2		29	94	+224

Overall, many aspects of grease distribution in numerical fluid mechanics can be modeled using the newly developed grease models and the methods developed. The calculation of the friction torque via the wall shear stress on the housing disc (AZRL) or the outer ring (SKL) provides values that are comparable to measured friction torques.

6.2 Final considerations on the flow behavior of

In the case of the friction torque in the rolling bearing, deviations in the two to three-digit percentage range occurred in the comparison between the experiment and the CFD simulation (6.1.2). In contrast, these deviations were in the single-digit percentage range in the simulation of the shear test (6.1.1). Finally, considerations on the flow behavior of the test greases are presented. These concern:

- the change in flow behavior as a result of the stress in the rolling bearing and
- the flow behavior at maximum shear rates.

In the first indent, a grease is measured again in the rheometer after the speed stage test in order to evaluate the consequences of the stress. The flow behavior at the highest shear rates presents measurement results at shear rates up to 71,000 s^{-1} on the MCR 702e (see 3.3.2). Each of these aspects is relevant to the flow behavior of greases and thus to the friction torque in rolling bearings.

6.2.1 Change in flow behavior as a result of the " " stress

One reason for differences in friction torque may be the stress on the lubricants. For the comparison between CFD and the test, AZRL 81212 recorded the friction torques measured in the test after a stress period of three hours. It is possible that the stress altered the fluidity of the grease. Below are some references that have addressed this problem in different ways. At the end of the article, there are four experiments that are intended to illustrate a change in flow behavior that is as realistic as possible.

GONCALVES was able to determine the extent to which the flowability of different fats changes under stress and quantify this based on the change in the individual Herschel-Bulkley coefficients [GONC15]. To do this, fats were subjected to stress at 80 °C in an apparatus similar to a centrifuge with an integrated sieve (ASTM-D4425). This test setup is similar to the oil separation test according to DIN 51817 (see 4 .1.3). In his investigation of aging, the author focuses clearly on the aspect of oil depletion. Furthermore, DORNHÖFER thermally aged large quantities of greases in oven tests. During the oven test, a small sample was taken at regular intervals and analyzed in a rheometer. Here, too, a change in flow behavior was observed [DORN16]. These analyses focus purely on thermo-oxidative stress.

KUHN uses a shear test to explain how the measured shear stress continuously decreases over hours of low but constant shear. The author describes that after some time, a state of equilibrium is reached [KUHN17]. The focus of the aging study is on the aspect of shear, i.e., purely mechanical stress.

ZHOU investigated the equilibrium state for some lithium greases on the basis of shear tests [ZHOU17] and later provided a master curve that is supposed to apply to all lithium-based greases [ZHOU19]. The curve describes the change in yield point in correlation with the amount of energy applied, based on mechanical stress. Later, his colleague CHATRA partially refuted this statement for roller bearing greases. To this end, numerous studies were conducted with combined stress, described as thermo-mechanical stress. Tests were carried out either in the shear roller [CHATR21] or in the roller bearing [CHATR23-2]. The kneading action in rolling bearings leads to a permanent structural change in the greases, which was confirmed by AFM images for both groups of greases. The author concludes that the stress on greases in rolling bearings causes different changes compared to those observed in a rheometer.

Building on these ideas, further experiments were conducted with the roller bearing greases used here. The aim here is to quantify the changes in flow behavior resulting from the roller bearing tests carried out here. For this purpose, the greases were analyzed again in a shear test following the roller bearing tests. For this purpose, tests were carried out on the high-speed test bench with the RRKL 6007 and both greases at speeds of up to 3,000 rpm. On the one hand, tests were carried out at the persistence temperature and then examined at 40 °C in the rheometer. On the other hand, the same tests were carried out at 80 °C and then analyzed in the rheometer at 80 °C. The test procedure corresponds to the speed step test used throughout on the rotational tribometer (see Table 11). This is intended to represent a load that is representative of the majority of the rolling bearing tests presented in this work.

The RRKL 6007 was selected for these representative stress tests. The snap cage of the RRKL makes it easier to take grease samples after the roller bearing test than with the window cage of the SKL 7007. In addition, the high-speed test bench was used, as two roller bearings are always tested simultaneously here. This ensured that sufficient lubricant was available for the planned shear test sufficient lubricant would be available for the planned shear test. Fig. 68 shows the fresh grease analyses in blue and the analyses resulting from the stress in the rolling bearing in red. The shear stress of both test greases decreased significantly across the entire shear rate range as a result of the stress. This results in a permanently reduced viscosity.

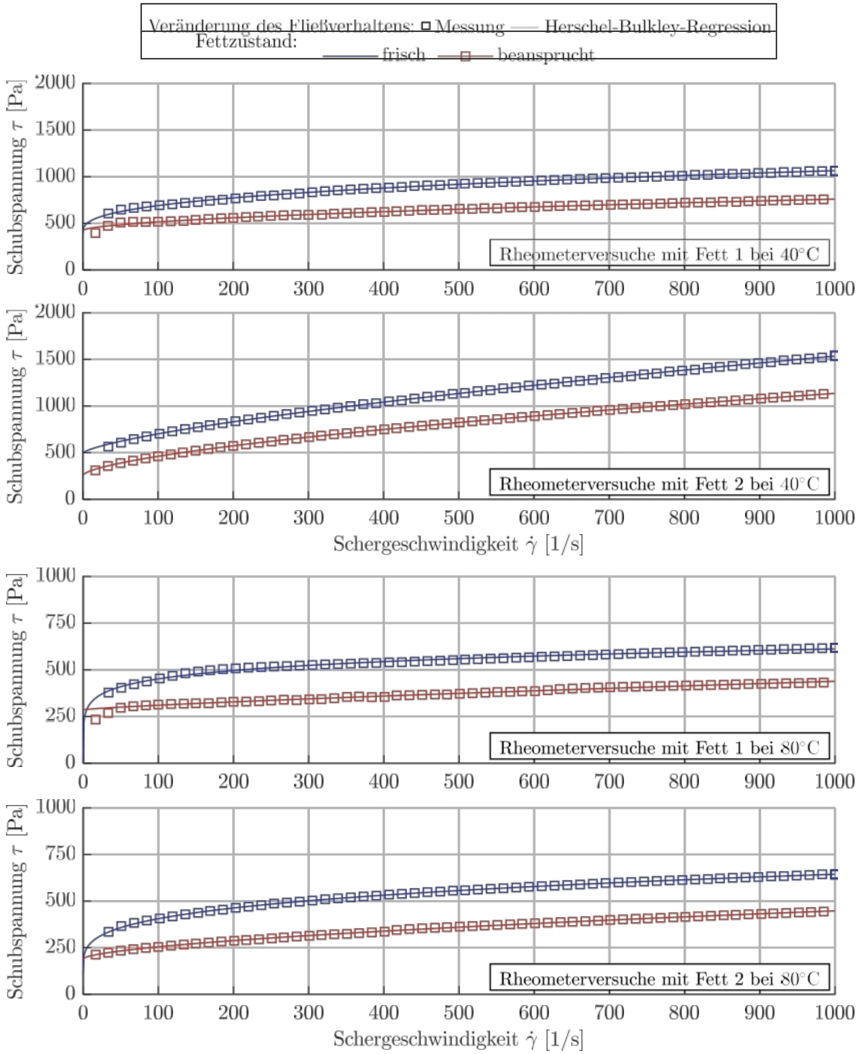


Fig. 68: Flow behavior of the test greases at 40 and 80°C: Fresh grease (blue) and altered flow behavior as a result of stress in the roller bearing (red): Measurement on the MCR 301 rheometer in accordance with DIN 51810-1 from phase IV as tiles and Herschel-Bulkley regressions as a line

Since as much of the grease as possible was removed for the downstream shear test in the roller bearing tests, the results of the shear tests must be understood as an integral of the change. After all, it cannot be assumed that all of the grease in the roller bearing was subjected to the same stress. It can therefore be assumed that there are areas of grease in a roller bearing that show virtually no change, while other areas show even greater changes than could be shown here. Following the example of GONCALVES, Herschel-Bulkley regressions were added to the measurements of the fresh and stressed grease. The coefficients derived from the regressions are used below to quantify the changes in flow behavior.

Table 25 illustrates the results of the regressions shown in Fig. 68. It contains the Herschel-Bulkley coefficients of both greases and states for 40 °C and 80 °C. The right-hand columns also show the percentage deviations of the model parameters from the fresh to the stressed grease. Once again, it is clear that a significant and permanent reduction in viscosity can be detected.

Table 25: Herschel-Bulkley coefficients according to Eq. *(10)* for fresh grease and grease stressed in the rolling bearing and their deviations from each other

Test temperature Rheometer	Grease	Condition of the grease	Model parameters			Deviation from		
			τ_0	k	n	τ_0	k	n
[°C]			[Pa]	[Pa·s]	[-]	[%]	[%]	[%]
40 (after test with RRKL at steady-state temperature)	Fat 1	fresh	394.9	57.41	0.356	-	-	-
		claimed	421.5	7.19	0.557	**7**	**-87**	**57**
	fat 2	fresh	494	8.21	0.701	-	-	-
		claimed	255.9	11.04	0.634	**-48**	**34**	**-10**
80 (after test with RRKL at 80 °C)	Fat 1	fresh	735.4	919.6	0.055	-	-	-
		claimed	286.2	0.619	0.797	**-61**	**-100**	**1349**
	Fat 2	fresh	112.8	88.68	0.259	-	-	-
		claimed	189.6	4.11	0.599	**65**	**-95**	**131**

The present results provide a possible reason why the CFD simulation of the shear test showed good agreement with regard to the friction torque, whereas in the case of the roller bearing, this evaluation measure was subject to deviations (6.1). The data obtained in the shear test at 40 °C and 80 °C are not sufficient to construct a comprehensive flow model as in 5.1. In addition, there is currently no way to predict the change in grease and thus implement it in the CFD simulation. A variety of operating parameters could influence the change in flow behavior, such as the maximum speed in the test (here 3,000 rpm) or the kinematics of the roller bearing used (here RRKL).

could influence the change in flow behavior, such as the maximum speed in the test (here 3,000 rpm) or the kinematics of the roller bearing used (here RRKL with plastic snap cage).

6.2.2 Shear tests at maximum shear rates

In section 4.1.1, the shear test according to DIN 51810-1 with fresh greases on the MCR 301 rheometer up to a shear rate of 1,000 s^{-1}was explained. This test forms the basis for the developed grease model (see 5.1). Now, two specific measurements are presented using the MCR 702e rheometer, which can measure shear rates up to 100,000 s($^{-1)}$
^{1}. The aim of the individual investigations is to compare the actual measured values at the highest shear rates with the developed lubricant model. The aim is to check how large the deviations are. On the one hand, this allows the data basis of the standard test (here: maximum shear rate of 1,000 s($^{-1)}$ $^{)}$and, on the other hand, the regression model (here: the Herschel-Bulkley model) to be questioned.

The MCR 702e Multidrive is equipped with an additional drive that drives the plate in the opposite direction to the rotation of the cone (see 3.3.2). A smaller measuring cone with an angle of 0.5° instead of 1.0° at the same drive speed also made it possible to double the shear rate and reduce the risk of gap emptying. Furthermore, an optical temperature measurement is integrated into the plate, which allows in-situ measurement in the lubrication gap. If shear heating occurs, it can thus be measured in the lubricant. In the present measurements, measurements without interference effects could be carried out up to a shear rate of 71,000 s($^{-1)}$. Interference effects only occurred above this shear rate, which will be discussed in the following results. Since it was already known that lubricating greases change as a result of stress, the test greases were pre-sheared in the rheometer up to the maximum speed before the measurements shown.

In Fig. 69 and Fig. 70, the results for fat 1 and fat 2 at 60 °C from the MCR 301 and MCR 702e are compared (top and middle). The Herschel-Bulkley flow model according to Eq. *(10)* was added, which is based on the measurement data from the MCR 301. The top diagram shows the viscosity versus shear rate up to 1,000 s($^{-1)}$.

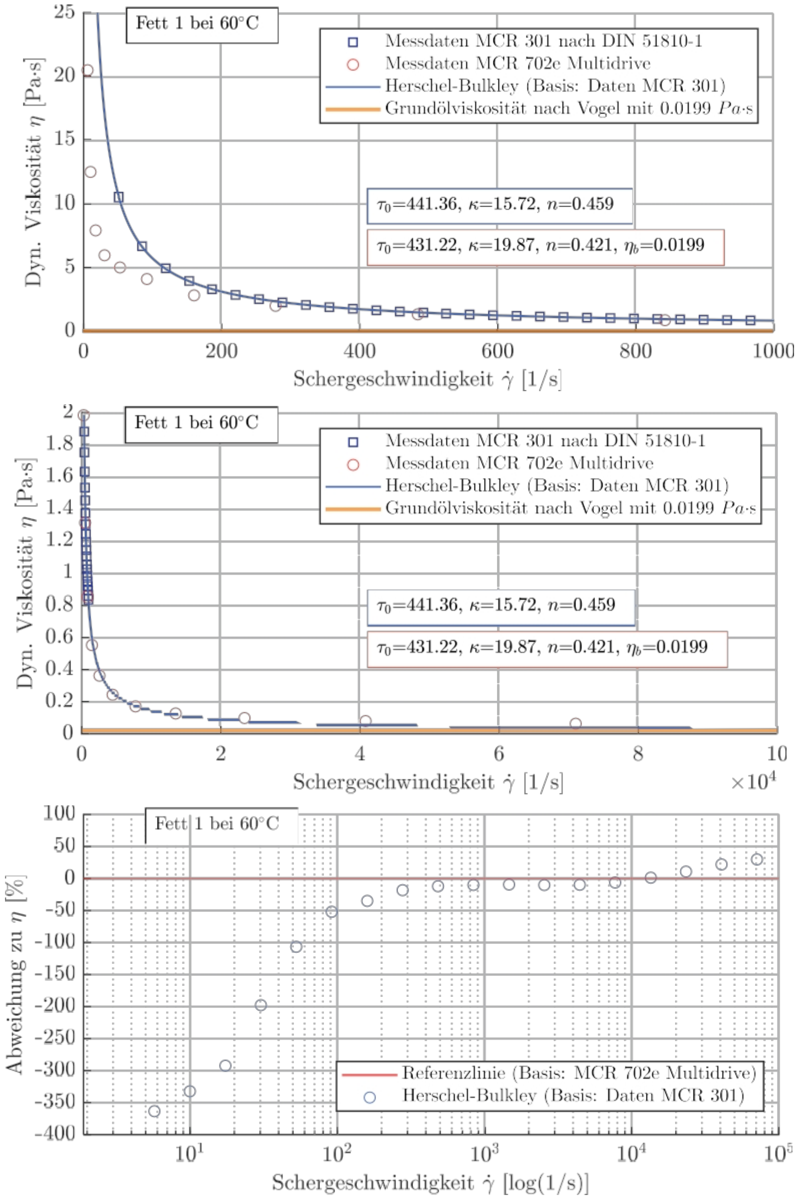


Fig. 69: Grease 1 at 60 °C: Data from the MCR 702e up to 71,000 s^{-1} and data from the MCR 301 up to 1,000 s^{-1} as well as the Herschel-Bulkley flow model

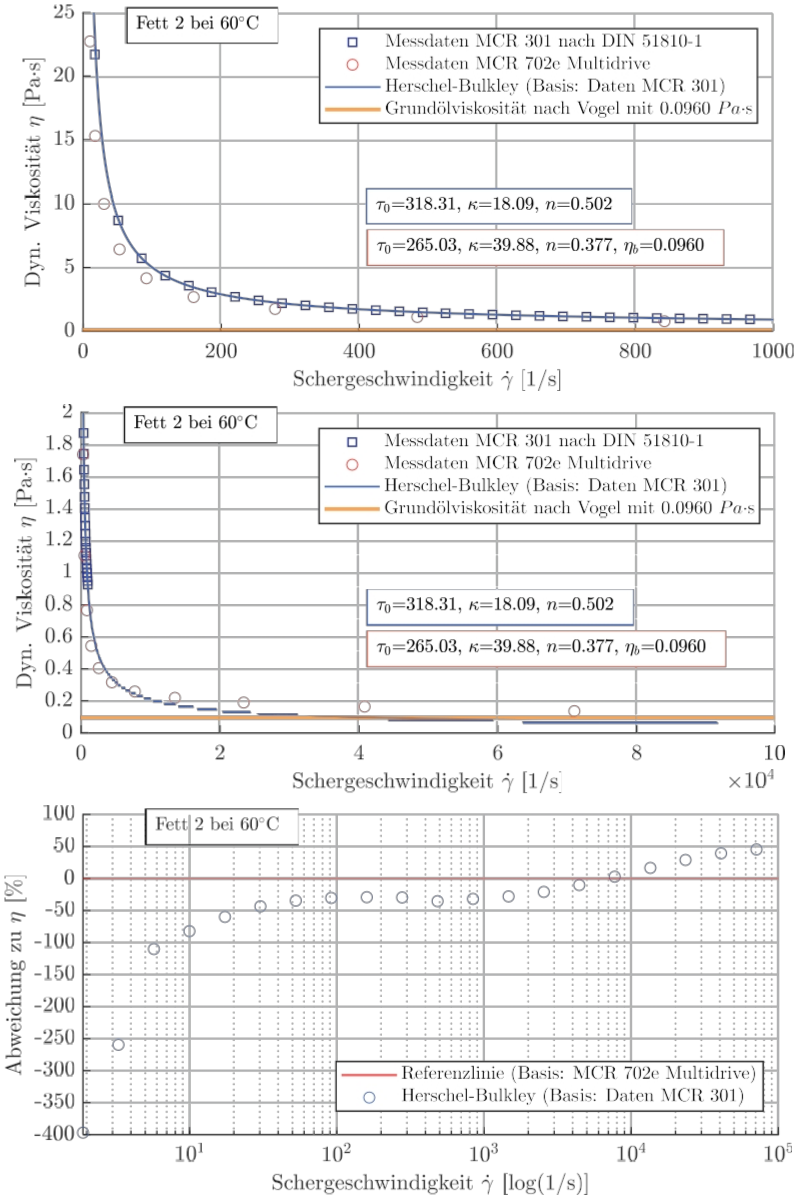


Fig. 70: Grease 2 at 60 °C: Data from the MCR 702e up to 71,000 s^{-1} and data from the MCR 301 up to 1,000 s^{-1} as well as the Herschel-Bulkley flow model

It becomes clear that the pre-shear leads to a greater discrepancy between the two measurements in the range up to 300 s^{-1}. Up to a shear rate of 1,000 $s^{(-1)}$, the differences become smaller again. Similarly, the flow model follows the fresh fat measurement data equally. In the middle diagram, the viscosity over the shear rate is shown up to

100,000 s^{-1} . There, it can be seen that the regression model also produces small deviations from the new measurement data in the high shear rate range.

These deviations are summarized in the bottom diagram. The percentage deviation from the dynamic viscosity η is shown there over the logarithmically represented shear rate. For this purpose, an ideal reference line was created from the new

Experimental data formed. The deviations from the regression model, which is still based on the fresh fat measurements, are shown at the discrete data points on this reference line. It is clear that both flow models up to a shear rate of

10,000 s^{-1} have an identical curve. Up to a shear rate of 71,000 $s(^{-1)}$, the first differences between the model approaches begin to emerge. This is due to the approximation of the base oil viscosity. It should be noted once again that the lubricating grease model implemented in the CFD cannot fall below the base oil viscosity (see 5.1.4). Basically, it can be deduced from the new measurements that, as a result of pre-shearing, there are two very different data sets for the test greases at 60 °C. The largest deviations are in the low shear rate range. This means that the stress in the rheometer differs from the stress in the rolling bearing (see 6.2.1). The two regression methods show the same discrepancy in this regard. The measurements from the MCR 702e are therefore once again a strong indication that the influence of the stress on the simulation is significant. A final conclusion that can be drawn from this is that the deviations at the highest shear rates are to be considered rather small compared to the deviations at low shear rates.

Finally, we will discuss the interference effects mentioned at the beginning, which occurred above the 71,000 s^{-1}shown here. These effects can be classified as shear thickening (increasing viscosity). Such effects are known from the rheology of dispersions [WILLE13]. WILLEMBACHER describes the formation of dispersion particles into a gap-encompassing cluster – a so-called hydrocluster. Whether clusters also form with this measuring system and the substrate "lubricating grease" could not be clarified here. According to the device manufacturer, however, further analyses could investigate such effects, which in principle should also make it possible to measure shear rates of 100,000 $s(^{-1)}$ and more with greases.

6.3 Classification of the results

In addition to comparing optical distribution images and CFD simulations, tests were also carried out at speeds of up to 18,000 rpm and 80 °C. Fig. 71 compares selected tests using two analytical methods for calculating the friction torque according to Eqs. *(1)* and *(2)*. The result bars for the measurements are light blue and those for the calculation methods are either green or dark blue. For the calculation methods, the respective base oil viscosity of the greases at 80 °C was calculated using the coefficients of Vogel's equation given in Table 20. This results in a kinematic viscosity of 12.54 mm²/s at 80 °C for grease 1 , while grease 2 has a value of 55.49 mm²/s.

The upper diagram shows results with grease 1 from the test bench and the SKL. The SKF method (Eq. *(2)*) provides relatively good agreement with the tests up to high speeds. The Schaeffler method (Eq. *(1)*) significantly underestimates the friction torque above 7,500 rpm. The situation is different for grease 2 (Fig. 71 below). Here, the method shown in green exceeds the friction torque from 5,000 rpm. However, there are also operating points (up to 3,000 rpm and at 18,000 rpm) where grease lubrication causes a significantly higher friction torque in the test than can be predicted. At low speeds, the measured friction is always higher than the result of both calculation methods.

Both methods equate grease lubrication with oil mist lubrication. An oil mist prevents splashing losses. It follows that work losses are also ignored and friction in rolling contact is assumed to be very good under lubrication conditions. Since both methods show good agreement with the experiment at many operating points, this is an indication that work is actually present at only a few operating points. As already demonstrated on the rotational tribometer, temperature plays a major role in this. At 80 °C, it was significantly more difficult to quantify work, as the differences in friction torque due to variations in grease quantity were very small in absolute terms. Similarly, the grease distribution run can minimize the occurrence of work losses. Finally, the tests were also accompanied by grease leakage, which was always less than 10% in the two tests shown here (see Table 17).

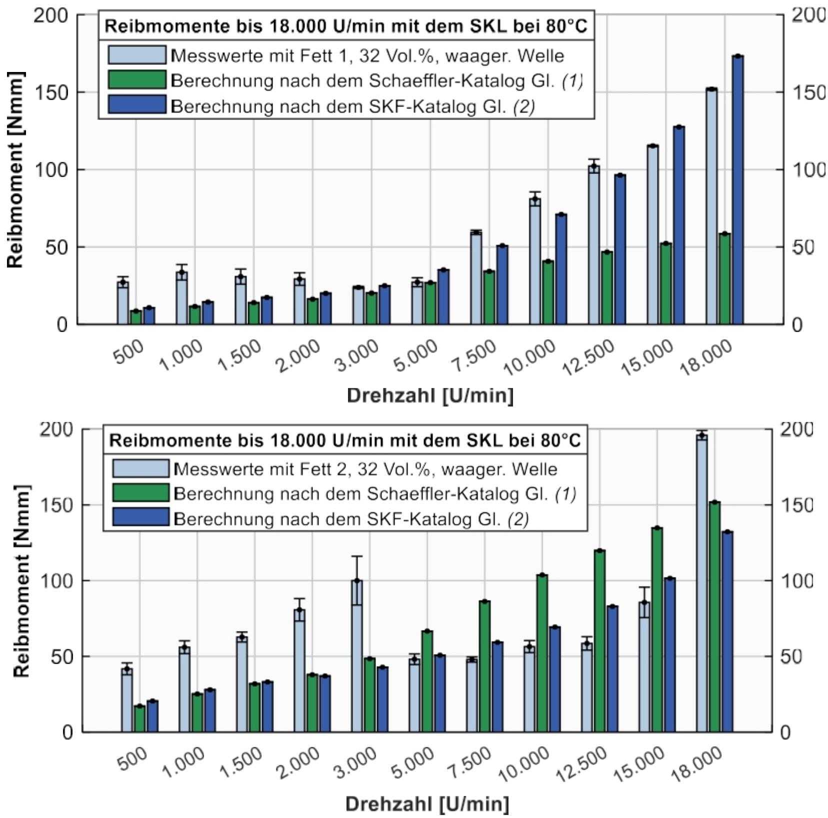


Fig. 71: Comparison between selected measurements from the high-speed test bench and analytical calculation methods of two roller bearing manufacturers, top: grease 1 and bottom: grease 2

Fig. 72 compares the results from the rotational tribometer with the same catalog method. For the comparison, tests without temperature control (4.3.1) were selected with the RRKL, both greases, and their base oils. The base oil viscosity is required for the catalog method. The measured temperature at each speed level was used to calculate the base oil viscosity (see Fig. 23). The measured values of the grease tests (light blue) differ significantly in some cases from the oil test (yellow). These differences are attributed to the kneading action. From a speed of 1,500 rpm, kneading is the dominant friction influence in the grease tests. For both grease 1 and grease 2, this increase in friction torque is underestimated by both catalog methods. Although both methods equate grease lubrication with oil mist lubrication, the friction torque of the oil test is overestimated. This illustrates that an improvement of the methods is necessary in the long term, as many aspects such as the degree of filling and the work of kneading are not represented.

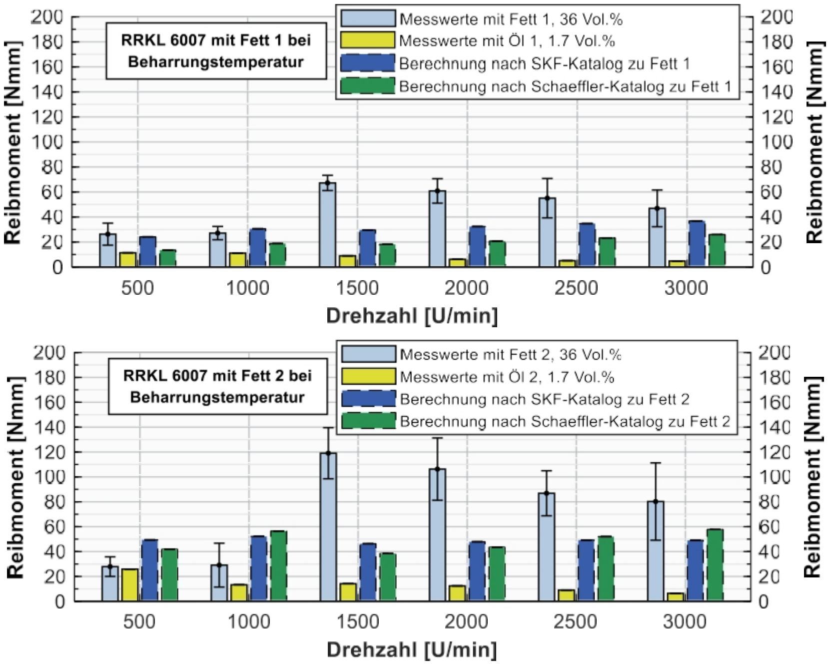


Fig. 72: Comparison between selected measurements from the rotational tribometer and analytical calculation methods of two roller bearing manufacturers, top: grease 1, bottom: grease 2

These tests up to a speed of 3,000 rpm correspond to the speed characteristic value 145,500 n·dm. Targeted tests on the drastic increase in friction torque were able to narrow down the onset of the effect to the speed range between 1,300 and 1,400 rpm (see 4.3.1.4). Applied to the example of a passenger car wheel bearing, this would correspond to a speed range between 136 km/h and 146 km/h for a commonly used tire (205/55 R17). These findings and the test method developed here are therefore relevant for a widespread application.

Furthermore, it was explained in Chapter 2.3 that no clear value for the ideal amount of grease can be determined for a rolling bearing. For the initial greasing of rolling bearings, a grease quantity of, for example, 90% of the free bearing volume is used for low speeds. This is recommended by a lubricant manufacturer, among others. Below 200,000 n·dm, even over 90% possible [KLÜB20]. However, it was shown, for example, with the RRKL and a fill level of 36%, that the scouring work can dominate the measured friction torque. As a result of the scouring work, the friction torque triples in some configurations during the speed step test (see 4.3.1.4). Furthermore, the tests demonstrated that a slight reduction in the amount of grease leads to a significant reduction in work of kneading. A reduction of 4 vol.% is sufficient for this (see 4.3.1.2). These results underline the added value of the findings from the rolling bearing tests and the test method developed.

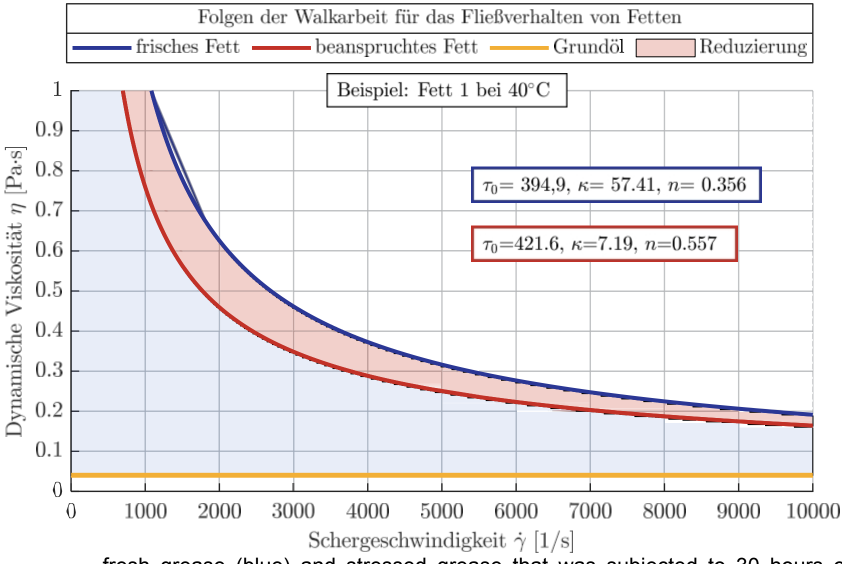
The use of transparent rolling bearing components on the special tribometer made it possible to observe and evaluate the grease distribution in the rolling bearing in detail. This deepened the understanding of grease distribution in the rolling bearing for the greases used here. Although such methods are state of the art in the field of research, they nevertheless enabled the CFD rolling bearing simulation to be visually validated.

Based on a standardized shear test, a lubricating grease model was developed for numerical simulation. This model reproduces the database with sufficient accuracy, which was also confirmed by a CFD simulation of the same test. The simulation results for roller bearings of different designs provide a good initial approximation of the actual grease distributions from the special tribometer and the measured friction torques. However, it was not possible to dispense with the rolling bearing temperature as an initial condition in the simulation model. This means that the simulation model does not meet the requirement of independence. Experimental investigations on rolling bearings are therefore still necessary. Furthermore, no comprehensive mesh study was carried out for the rolling bearing models, which is why the error from the meshes used was not quantified. Finally, possible causes were identified for the deviations between the friction moments from the rolling bearing simulation on the one hand and the rolling bearing tests on the other. These include the flow behavior at high shear rates and the change in flow behavior as a result of the work done in the rolling bearing.

There has already been some research on the topic of work rate in the context of grease-lubricated rolling bearings, which is explained in detail in 1.2 and 6.2.1. Based on HORTH, CHATRA describes that greases have poor work rate behavior when they exhibit high work rates for a period of more than three hours [HORTH71, CHATR23-2]. Based on the experiments presented here, it was shown that work is present in the rolling bearing tests and that this is maintained for a period of significantly more than three hours. According to HORTH and CHATRA, the test greases examined for this work (see Table 2) can therefore be categorized as greases with poor workability. This is evident not only from the measurement of friction torque and temperature in the component tests (see 3.3), but also from the changes in flow behavior (6.2.1). Using grease 1 as an example, a suggestion for taking this aging aspect into account in the 3D CFD simulation is now provided.

Fig. 73 is based on the measurements from the rheometer presented in Fig. 68. It shows a result at 40 °C for the dynamic viscosity. In addition, only the regressions based on the Herschel-Bulkley model for the fresh and stressed fat have been indicated here by the coefficients. The base oil viscosity was also added as a comparative value. The apparent viscosity and the base oil viscosity differ significantly in this area. The red-colored area also shows that the stress of only 30 hours in the roller bearing has significantly and permanently changed the flow behavior.

Fig. 73: Effects of kneading using the example of grease 1 at 40 °C: HB regression to



fresh grease (blue) and stressed grease that was subjected to 30 hours of rolling bearing testing (red), as well as the base oil viscosity (yellow) and the reduction in viscosity of the fresh grease as a result of stress in the rolling bearing test.

In order to draw conclusions about the change in viscosity directly from a fresh grease measurement, there is currently still a lack of understanding about how these changes arise. It is conceivable that the extent of the change depends on classic test parameters such as roller bearing type, friction torque, proportion of kneading work, ambient temperature, maximum speed, and others. The literature refers to a quasi-stationary final state [KUHN20, ZHOU19]. A formula for greases is already available for this purpose, which includes a calculation of the yield point τ_f according to DIN 51810-2 [CHATR23-2]. However, as explained in 5.1.2,

that yield points τ_f from the oscillation test [DIN 51810-2] and τ_0 from the Shear test [DIN 51810-1] differ significantly from each other. The two methods cannot be combined. Furthermore, knowledge of a yield point would not be sufficient to describe the flow behavior. Ultimately, all factors in the Herschel-Bulkley equation change, as shown in Table 25. This means that the current state of the art is not sufficient to predict the new flow behavior.

Without knowing the exact mechanisms for the change in flow behavior or being able to predict them, one possible approach to describing this change would be to use correction factors. In this way, stressed greases from rolling bearing tests, which raise questions for numerical flow simulation, could be reduced in the same way by means of a shear test. For the shear test, a test temperature is then specifically set which is known from the fresh grease and corresponds as closely as possible to the rolling bearing temperature. In this way, the grease model presented in 5.1 would continue to be applicable in this temperature range.

Based on the current status, an extension of the previous flow model according to Eq. *(27)* could look as follows:

$$\tau(\dot{\gamma}, \vartheta, K) = K_\tau \tau_{(0)}(\vartheta) + K_\kappa \cdot \kappa(\vartheta) \cdot \dot{\gamma}^{K_n \cdot n(\mathrm{P})} \qquad (33)$$

For the case shown in Fig. 73, the following correction factors can be used to convert from the curve for fresh fat (blue) to the curve for stressed fat (red): K_{τ_0} = 1.07; K_κ = 0.13; K_n = 1.57. The factors were derived from Table 25.

7 Summary

In this study, grease-related bearing losses were investigated on three different test benches with regard to various aspects. Two representative grease-lubricated rolling bearings relevant to practical applications were used: the radial deep groove ball bearing 6007 and the angular contact ball bearing 7007 (Table 1). In addition, two rolling bearing greases were used that are assigned to the same NLGI class and have the same chemical base (Table 2). Due to different base oil viscosities and thickener proportions, the rheological properties of the greases differ significantly.

In order to investigate work loss experimentally, a test methodology was developed using a widely used test system, the rotational tribometer (4.3.1.2). Based on this methodology, comprehensive investigations were carried out with ball bearings and different types of lubrication. These included grease lubrication, with both a typical amount of grease and various reduced amounts of grease, as well as oil lubrication, with the base oils of the respective greases. The friction torque curves for these different types of lubrication were statistically verified. By calculating the difference between the friction torque curves, it was possible to quantify the work done by the lubricating grease (4.3.1.2). Based on the quantification and a wide range of test conditions, the following factors influencing the work done were experimentally verified:

- Lubricant rheology,
- Temperature,
- lubricant quantity,
- Rotational speed, and
- Bearing type.

Furthermore, a CFD simulation model was developed for three different types of rolling bearings (5.3). The central aspect of the simulations was the distribution of grease in the rolling bearing. This in turn requires knowledge of the flow behavior of the test greases, which was described using a model known from the literature and established in numerical fluid mechanics (2.6). The plastic-structural viscous flow behavior was derived from the data obtained in a standardized rheological shear test (4.1.1). Initially, the model only includes the dependence of the shear stress on the shear rate (5.1.1). Through data analysis, it was possible to extend it to include temperature dependence (5.1.3). To validate the lubricating grease model developed, a 3D simulation model of the shear test was created, taking into account the test standard for numerical fluid mechanics (5.2). This allowed the temperature and shear rate dependence to be checked and validated at a comprehensible level (6.1.1).

Using the validated lubricating grease model, 3D CFD simulations were performed with roller bearings of different designs (5.1). This presented various challenges: the interconnection of the roller bearings with areas of very high and very low shear rates, the simulation of a two-phase flow taking into account reasonable calculation times, and the evaluation of the friction torque. Solutions to these problems were identified. The simulations were used to derive the grease distribution and the friction torque caused by pure fluid friction (without mixed friction). In the CFD simulations of roller bearings of different designs, it has not yet been possible to map speeds above 500 rpm. At higher speeds, numerical problems arose that could not be solved within the scope of this work.

Using transparent rolling bearing components, distribution properties in rolling bearing types could be observed on a special tribometer. For this purpose, various recordings were made using a high-speed camera in speed step tests on the axial cylindrical roller bearing 81212 (4.5.1) and the angular contact ball bearing 7007 (4.5.2). Since 3D CFD simulations were performed for both rolling bearing types, the grease distribution could be compared with the visual recordings (6.1.2). This revealed a good correlation between the simulated and experimentally determined grease distribution.

In addition, friction torques from the rolling bearing test and from the CFD simulation of rolling bearings were compared with each other. The friction moments derived from the rolling bearing simulation provide plausible values for all three rolling bearing types (6.1.2). Thus, both distribution aspects and friction moments can be derived from the CFD using the lubricating grease model. The absolute deviations from the measured friction moment are in the double-digit percentage range in almost all cases.

On a high-speed test bench, the SKL 7007 was also used to vary lubricant quantities up to a speed of 18,000 rpm. In contrast to the other test benches, a grease distribution run was carried out before the actual speed step test. It can be assumed that this grease distribution run equalized differences in the friction torque curves resulting from the variations in grease quantity. In addition, lubricant leakage of up to 20% made it difficult to determine the load capacity of the friction torque curves for the lubricant quantities used. In tests on the other roller bearing test benches, these were 0% in most cases and a maximum of 5% in some cases. For these reasons, it is questionable whether the walking losses can be clearly derived from the measured friction torque curves on the high-speed test bench.

8 Outlook

Based on the results, three focal points should be set for further research work:

1. Rolling bearing tests:
   Now that a test methodology for investigating work loss has been developed and various dependencies in real bearings have been demonstrated using this methodology, the following aspects should be investigated in the future.
   a. Lubricating grease rheology: The results have demonstrated that not only rheology but also, for example, rolling bearing kinematics can influence kneading behavior. Since only two test greases were considered here, which differ both in their thickener content and in their base oil viscosity, it is still unclear which specific rheological property is decisive for the detectable kneading behavior. In order to better address this issue, more greases need to be examined. These primarily include test greases with a finer gradation of rheology, such as the thickener content and base oil viscosity. Furthermore, practical greases from which the test greases used here were derived are also of interest. Finally, greases with a different thickener chemistry, such as polyurea greases, are equally important.
   b. Rolling bearing kinematics: Since the operating pressure angle differs greatly between the tested bearings (SKL 40° to RRKL 9°), a finer gradation of the operating pressure angle by spindle bearings is conceivable. This could allow critical parameters to be derived, as the delivery behavior increases with increasing pressure angle. Significant improvements in flexing behavior may only be achieved if greases are tailored to highly restricted bearing groups (more customized solutions).
   c. Free volume in the rolling bearing (sealing concept, grease quantity, and cage design): In this study, the grease quantities in rolling bearings were varied extensively. In this way, it was possible to show in some configurations that a small change in the grease quantity can have a significant effect on the rolling work (Fig. 24). This is a strong indication that the work of the seal is significantly influenced by the free volume. It was therefore necessary to examine the seal concept and cage design. For tapered roller bearings, one rolling bearing manufacturer has specific rules

for seal design [SCHAE13, p. 107]. There are no fixed rules for other bearing types. In order to test how sensitive the flexing behavior is to the seal design or the free volume, the height of the seal shoulder used could be varied (see Fig. 20, blue area). The volume of steel sheet, brass, or plastic cages also differs significantly. This aspect would be easy to investigate for the RRKL 6007 used here (versions with plastic or steel sheet cages are available).

2. 3D CFD simulation:

   Approaches for improving rolling bearing simulation are explained below.

   a. Lubricant model: It is not yet clear whether other flow models other than the Herschel-Bulkley model, which has been extended here, are more suitable for the application of rolling bearing greases. It is also important to consider models that assume base oil viscosity at high shear rates in order to ensure a smooth transition from grease viscosity to base oil viscosity. This aspect could lead to an improvement.
   b. Network quality: Finally, improvements should be made to the simulation model of the ball bearings. These primarily concern the networking. It was shown how the friction torque can be derived from the shear heating in the entire simulation area (5.2.2). In the rolling bearing simulation, excessive shear heating was postulated, resulting in an excessive friction torque being derived using this method. For this reason, only isothermal simulations were performed, using the persistence temperature determined in the experiment as the input variable. The preliminary assumption is that the network quality is the cause of this.
   c. Computing time: A final aspect must be improvements in computing time. Many simulations took several weeks to complete on a cluster with more than 40 cores. However, high speeds could not be achieved. To save time, a comprehensive comparison between stationary and transient simulations should therefore be carried out first.

3. Grease aging:

   a. Mechanical stress: In this work, a change in flow behavior as a result of stress in the rolling bearing was experimentally proven (see 6.2.1). In the topic complex

In recent years, there have been numerous publications by CHATRA and LUGT on grease aging, particularly on the consequences of kneading (see 1.2). Chapter 5 explained why this literature is not sufficient to describe new flow behavior in CFD. An initial proposal was made to correct the lubricating grease model according to Eq. *(27)*. Initial 3D CFD simulations were performed on this basis. They take into account the correction factors set out in Eq. (*33)* (referred to as K factors in Fig. 74). In both cases, an improvement in terms of friction torque was achieved, which could not be fully verified for all types of rolling bearings. This merely shows that the friction torque that can be derived from the CFD simulation shows a positive trend when the aspect of stressed greases is taken into account:

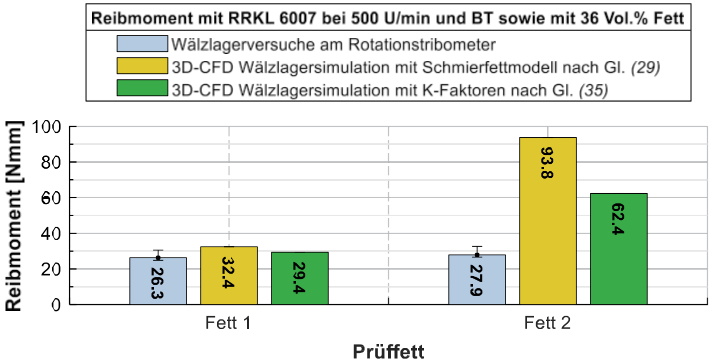


Fig. 74: Comparison between measured friction torques from the rotational tribometer and friction torques derived from the CFD simulation with and without correction factors according to Eq. *(33)* for RRKL 6007 at 500 rpm with both greases.

b. <u>Regenerative capacity:</u> It remains unclear how regenerative capacity (e.g., through start-stop operation) and time dependence (continuous reduction of shear stress) should be implemented. All these aspects are fundamentally relevant for CFD simulation.

c. <u>Heterogeneous aging state in the rolling bearing</u>: An important aspect that has not yet been addressed in the literature is the highly inhomogeneous state of the grease in the rolling bearing. Many areas of the grease are no longer sheared after the grease distribution run until the end of the grease service life. Nevertheless, this grease also fulfills the purpose of supplying oil to tribological contacts. How these different stress states can be represented in the CFD simulation should also be worked out in the future.

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Curriculum

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Occupation

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2020-2024	Influence of kinematics and load on grease aging in rolling bearings (FVA project no. 913 I)
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